

Development of novel RF coils for ^{23}Na imaging at ultra-high field MRI

Von der Fakultät für Mathematik, Informatik und Naturwissenschaften der
RWTH Aachen University zur Erlangung des akademischen Grades eines
Doktors der Ingenieurwissenschaften genehmigte Dissertation

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Tag der mündlichen Prüfung: 12. November 2019

DECLARATION

I, Yonghyun Ha, declare that this dissertation titled, “Development of novel RF coils for ^{23}Na imaging at ultra-high field MRI”, and the work presented in it is my own. Studies discussed in the part of Chapter 1, Chapter 4, Chapter 5, and Chapter 6 have been either published or accepted in the following citations and I am the author of the works.

- Apart from the main projects of this thesis, the author was involved in the development of a simple multinuclear MRI system, which has also been published in Journal of Magnetic Resonance (C.-H. Choi, Y. Ha, P. Veeraiah, J. Felder, K. Möllenhoff, N.J. Shah, *Design and implementation of a simple multinuclear MRI system for ultra-high field imaging of animals*, J. Mag. Reson., 273, 28-32, 2016, doi: 10.1016/j.jmr.2016.10.007). Figures and text fragments of this paper are reused in the Chapter 1 and Chapter 5 with permission of the publisher Elsevier.
- The work presented in the Chapter 4 and the Chapter 6 was previously published in the journal IEEE transaction on Medical Imaging (Y. Ha, C.-H. Choi, N.J. Shah, *Development and implementation of a PIN-diode controlled, quadrature-enhanced, double-tuned RF coil for sodium MRI*, IEEE Trans. Med. Imaging, 2017, doi: 10.1109/TMI.2019.2786466). Figures, tables, and text fragments of this paper are reproduced in this manuscript with kind permission of the publisher IEEE.
- The development of a folded four-ring birdcage coil has also been published as a full article in the Journal of Magnetic Resonance (Y. Ha, C.-H. Choi, W. A. Worthoff, A. Shymanskaya, M. Schöneck, A. Willuweit, J. Felder, N.J. Shah, *Design and use of a folded four-ring double-tuned birdcage coil for rat brain sodium imaging at 9.4 T*, J. Mag. Reson., 286, 110-114, 2018, doi: 10.1016/j.jmr.2017.12.003). Figures, tables, and text fragments of this paper are reproduced in the Chapter 5 and the Chapter 6 with kind permission of the publisher Elsevier.
- Other results of this thesis have been presented to the MR community at international scientific meetings organized by the ISMRM (International Society for Magnetic Resonance in Medicine) and ESMRMB (European Society for Magnetic Resonance in Medicine and Biology) between October 2015 and October 2017. The conference abstracts are cited throughout the manuscript and can be found in the Bibliography as well as in the LIST OF PUBLICATIONS at the end of the document.

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ABSTRACT

Sodium (^{23}Na) magnetic resonance imaging (MRI) provides valuable cellular and metabolic information. Since the Larmor frequency of the ^{23}Na nucleus is different from that of protons (^1H), a radiofrequency (RF) coil tuned to the ^{23}Na Larmor frequency is required for sodium MR imaging. However, the MR signal of ^{23}Na is much smaller in amplitude than that of ^1H due to its low intrinsic MR sensitivity and lower natural abundance and ^1H images are very often required in conjunction with the ^{23}Na images. For this reason, $^{23}\text{Na}/^1\text{H}$ double-tuned RF coils are often used for ^{23}Na studies. In this case the proton signal is used for localisation and B_0 shimming. Double-tuned RF coils are often designed by including additional lossy components such as LC-traps or PIN-diodes. Thus, the sensitivity of the coil – which is proportional to the quality factor of the resonant circuit – is reduced at both resonance frequencies, when compared to that of a single-tuned coil. The aim of this thesis is to develop both surface and volume double-tuned RF coils for ^{23}Na brain imaging with high efficiency. Of special interest is the quality factor at the ^{23}Na Larmor frequency in order to maintain the best possible signal-to-noise ratio (SNR) for the lower MR sensitive X-nucleus.

For a heterogeneous resonator, a double-tuned surface coil can be designed using PIN-diodes with series capacitors in parallel with the tuning capacitor. When the PIN-diode is forward biased, a capacitor is added in parallel to the tuning capacitor. The resonance frequency of the coil is then shifted to the lower frequency (for example to the ^{23}Na Larmor frequency). While not simultaneously double-resonant, such networks allow rapid switching between two resonance frequencies under electronic control. However, the sensitivity at the lower frequency can be significantly decreased due to the residual RF resistance of the PIN-diode. A PIN-diode controlled, quadrature enhanced, double-tuned RF coil is developed for an animal scanner to enhance the SNR of the ^{23}Na image. A single-tuned ^{23}Na butterfly coil is added on the PIN-diode controlled coil to drive this coil in quadrature. The coil is connected to the scanner using a home-built transmit/receive (T/R) switch. Either forward current or reverse voltage is applied simultaneously to the individual PIN-diodes using a home-built PIN-diode control unit.

In another approach, for a homogeneous resonator, the four-ring birdcage coil is comparable to a single-tuned birdcage coil terms of efficiency and homogeneity. However, it requires a physical addition in length for the two outer end-rings and this limits its use for brain studies due to the presence of shoulders forming a natural restriction of overall coil dimensions. In order to resolve this issue, a three-dimensional (3D) spatially folded four-ring birdcage coil is developed that mitigates space limitations.

For the suggested surface coil, the SNR values of gradient echo (GRE) images were compared to that of single-tuned coils under the same condition. It was found that ^{23}Na SNR is higher than that of single-tuned coil, for both the loop and the butterfly arrangements. The performance of the folded four-ring birdcage coil was also compared with a single-tuned and a larger four-ring birdcage coil on a phantom. It is shown that both B_1 field homogeneity and image SNR of ^{23}Na are comparable to that of the single-tuned birdcage coil. In vivo healthy rat brain images at 9.4 T using the novel folded four-ring birdcage coil are presented to validate the suitability of the novel design for brain imaging.

It was demonstrated that using of the quadrature-enhanced surface coil and folded four-ring birdcage coil has a distinct benefit for ^{23}Na imaging. It is also possible to combine both coils as a transmit-only/receive-only RF system to generate homogeneous B_1^+ field using the folded four-ring birdcage coil and for the higher SNR using the quadrature-enhanced surface coil.

ZUSAMMENFASSUNG

Natrium (^{23}Na) Magnetresonanztomographie (MRT) liefert sowohl wertvolle zelluläre als auch metabolische Informationen. Da die Larmorfrequenz von Natrium (^{23}Na) anders ist als diejenige von Protonen (^1H) ist es für die Natrium Bildgebung erforderlich, die Hochfrequenz (HF)-Spule auf die ^{23}Na Larmorfrequenz abzustimmen. Außerdem hat das MR-Signal von ^{23}Na aufgrund der geringen intrinsischen MR Sensitivität und der geringeren Konzentration von Natrium Kernen in biologischem Gewebe eine kleinere Amplitude als dasjenige von Protonen. Daher werden zur Untersuchung von ^{23}Na häufig HF-Spulen verwendet, welche auf die Resonanzfrequenzen von ^{23}Na und ^1H abgestimmt sind; sogenannte $^{23}\text{Na}/^1\text{H}$ doppelt abgestimmte (double-tuned) Spulen. Das ^1H -Signal wird dann zumeist für anatomische Übersichtsaufnahmen und B_0 -Shimmen eingesetzt. Doppelt abgestimmte HF-Spulen werden oft mittels zusätzlichen verlustbehafteter Komponenten wie Sperrkreisen oder PIN-Dioden realisiert. Daher ist die Empfindlichkeit dieser Spule für beide Resonanzfrequenzen im Gegensatz zu einer monofrequenten Spule reduziert. Das Ziel dieser Arbeit ist es, sowohl eine Flächen- als auch eine Volumenspule zur Natrium Bildgebung des Gehirns mit möglichst hohem Wirkungsgrad bei der ^{23}Na Larmorfrequenz zu entwickeln.

Eine doppelt abgestimmte Spule kann mithilfe von PIN Dioden realisiert werden. Dazu wird die PIN Diode in Serie mit einem Kondensator geschaltet und parallel zu dem Abstimmkondensator eingebaut. Wenn die PIN Diode HF leitend ist (forward bias) liegt ein zusätzlicher Kondensator parallel zum Abstimmkondensator. Dies reduziert die Resonanzfrequenz der Spule und verschiebt sie auf die Larmorfrequenz des Natriumkerns. Obwohl die Spule in dieser Realisierung nicht zeitgleich doppelt resonant ist, ermöglichen PIN Dioden basierte Netzwerke die schnelle Umschaltung zwischen zwei Resonanzfrequenzen unter elektronischer Kontrolle. Nachteil dieser Anordnung ist, dass die Sensitivität der Spule – sie ist proportional zur Güte des Schwingkreises – auf der X-Kern Frequenz durch den residuellen Widerstand der PIN Diode deutlich verringert werden kann. Eine PIN Dioden gesteuerte, doppelt abgestimmte HF-Spule mit zirkularer Feldpolarisation für einen Tierscanner wird entwickelt, welche das Signal-zu-Rausch Verhältnis (SNR) der ^{23}Na Bilder optimiert und eine Kompression des Sendesignals vermeidet. Sie besteht aus einer einzeln abgestimmten ^{23}Na Schmetterlings Spule in Kombination mit einer PIN Dioden gesteuerten, doppelt resonanten Spule. Sie wird über einen dedizierten Send-/Empfangs Umschalter mit dem Scanner verbunden. Eine ebenfalls neu entwickelte PIN Dioden Steuereinheit versorgt die individuellen PIN Dioden des Spulenarrays mit den benötigten Bias-Spannungen um Kompression und Intermodulation des Sendepulses zu vermeiden.

Im Hinblick auf Effizienz und Homogenität ist die Birdcage-Spule mit vier Endringen vergleichbar

mit einer monoresonanten Birdcage-Spule, besitzt jedoch die Eigenschaft auf zwei homogene Moden mit verschiedenen Resonanzfrequenzen erzeugen zu können. Allerdings machen die zusätzlichen Endringe die Spule länger, was eine Einschränkung für die Verwendung in der Gehirn Bildgebung darstellt: die Schultern von Mensch und Tier beschränken die maximale Länge der Anordnung physikalisch. Folglich wurde eine in der dritten Dimension „gefaltete“ Birdcage Spule mit vier Endringen konzipiert.

Das SNR der neu entwickelten Spule wird anhand von Gradientenecho (GRE) Bildern mit demjenigen einer einzeln abgestimmten Spule unter den gleichen Bedingungen verglichen. Es wird festgestellt, dass der SNR-Wert für ^{23}Na höher ist als derjenige der einzeln abgestimmten Spule, sowohl für das Schleifen-, als auch für das Schmetterlings-Arrangement. Es wird gezeigt, dass sowohl die B_1 -Homogenität als auch das SNR von ^{23}Na Bildern der neu entwickelten Anordnung vergleichbar mit demjenigen von monoresonanten Birdcage Spulen sind. In einer ersten Anwendung werden in vivo Bilder von gesunden Rattenhirnen bei 9.4 T mit der gefalteten, doppelt resonanten Birdcage Spule aufgenommen. Dies demonstriert, dass die neue Anordnung auch in der Gehirn Bildgebung sinnvoll angewendet werden kann.

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LIST OF ABBREVIATIONS

1D	One-dimensional	NEMA	National electrical manufacturers association
2D	Two-dimensional	NMR	Nuclear magnetic resonance
3D	Three-dimensional	PE	Phase encoding
CP	Circularly polarised	PIN	Positive-intrinsic-negative
CT	Computer tomography	pTx	Parallel transmit
DC	Direct current	Q-factor	Quality factor
EMF	Electromotive force	RF	Radio frequency
FA	Flip angle	ROI	Region of interest
FID	Free induction decay	RU	Related uniformity
FLASH	Fast low angle shot	Rx	Receive
FOW	Field of view	SAR	Specific absorption rate
GRE	Gradient echo	SE	Spin echo
KVL	Kirchhoff's voltage law	SI	Signal intensity
LP	Linear polarised	SNR	Signal to noise ratio
LNA	Low noise preamplifier	S-parameters	Scattering parameters
MEMS	Microelectromechanical system	T/R	Transmit and receive
MR	Magnetic resonance	TE	Echo time
MRI	Magnetic resonance imaging	TR	Repetition time
MTL	Micro-strip transmission line	Tx	Transmit

LIST OF SYMBOLS

I	Spin	B₁	RF field
μ	Magnetic moment	B_{1x}	x-component of the RF field
J	Angular momentum	B_{1y}	y-component of the RF field
γ	Gyromagnetic ratio	B_{1z}	z-component of the RF field
$\hbar$	Planck's constant over 2π	ω_1	Resonance frequency of the RF field
B₀	External magnetic field	t	Time
E	Energy	M_{xy}	Transverse magnetisation
m	Magnetic quantum number	T_1	Spin-lattice relaxation time
ν	Frequency	T_2	Spin-spin relaxation time
h	Plank's constant	f_c	Centre frequency
ω	Angular frequency	Δf	Bandwidth
$N_{-1/2}$	Population of the spin-down energy level	Δz	Slice thickness
$N_{+1/2}$	Population of the spin-up energy level	$\bar{\gamma}$	Gyromagnetic ratio over 2π
k_B	Boltzmann constant	G_x	x-directional magnetic gradient
T	Absolute temperature	G_y	y-directional magnetic gradient
M_0	Net magnetisation	G_z	z-directional magnetic gradient
N	Total number of spins	S_{RF}	RF sensitivity factor
M	Net magnetisation vector	μ_0	Permeability of free space
B	Magnetic field vector	Q	Quality-factor
M_x	x-component of the magnetisation vector	V_s	Volume of the coil
M_y	y-component of the magnetisation vector	L	Inductance
M_z	z-component of the magnetisation vector	R	Resistance
B_x	x-component of the magnetic field vector	C	Capacitance
B_y	y-component of the magnetic field vector	ρ	Specific resistance
B_z	z-component of the magnetic field vector	ε	Induced electromotive force
ω_0	Precessing frequency		

Chapter 1

INTRODUCTION

Magnetic resonance imaging (MRI) is a non-invasive imaging technology used primarily in medical diagnosis for imaging soft tissues inside of the human body. MRI makes use of the nuclear magnetic resonance (NMR) phenomenon [1, 2], by using a strong static magnetic field, radio frequency (RF) waves, and field gradients to spatially localise the signal [3, 4]. MRI is often compared to computed tomography (CT), which uses ionising radiation, since both are tomography methods but it is very different from CT. Although MRI is also not effective for materials with little water content, it has many positive characteristics. MRI provides high quality images with excellent soft tissue contrast and contrast agents are not always required to produce detailed images of blood vessels. In contrast with CT, MR images can be obtained from oblique slices.

In theory, all nuclei with non-zero spin can be used for magnetic resonance imaging and/or spectroscopy. X-nuclei (i.e. non-proton) MRI provides supplementary information to protons and thus is of interest. Studies using X-nuclei MRI have been of great interest over recent years due to the availability of higher magnetic field strengths, improved gradient hardware and novel sequence developments [5-7]. The image quality of the X-nuclei image can be further improved by increasing the number of receive channels and enhanced reconstruction methods [8, 9].

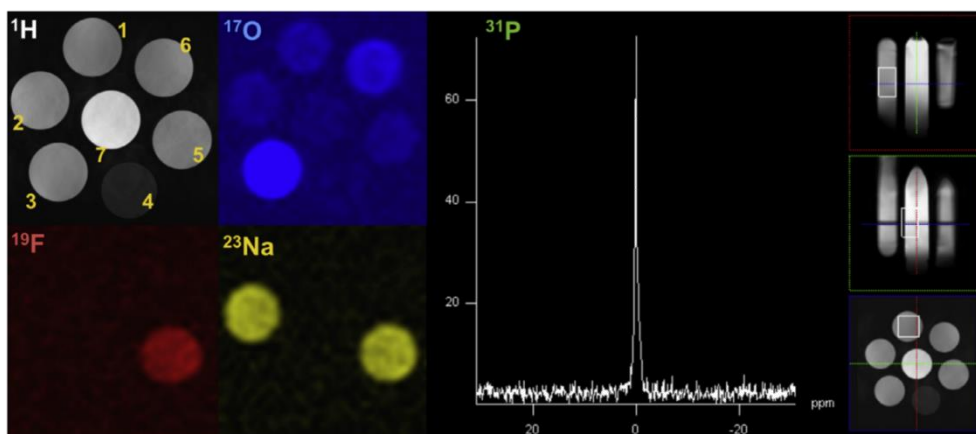


Figure 1-1 Multinuclear MR results obtained using a phantom comprising of (1) 10 mM KH_2PO_4 , (2) 100 mM NaCl, (3) 0.15% ^{17}O , (4) 4% agarose gel, (5) 100 mM NaF, (6) 0.05% ^{17}O , (7) 1 mM CuSO_4 doped distilled water [6].

Figure 1-1 shows selective multinuclear MR images and a spectrum of the phantom obtained using single-tuned birdcage coils which are tuned to the Larmor frequency of each nucleus at 9.4 T [6]. As can be visually assessed in Figure 1-1, SNR of the X-nuclei is smaller than that of the proton due to their lower natural abundance and less MR sensitivity [10]. Localiser image and B_0 shimming with X-nuclei signal are also problematic due to its low signal, therefore coupled acquisition of proton imaging is advantageous [9].

Most X-nuclei MR imaging studies are performed on sodium because the sodium ions are the most abundant MR sensitive X-nuclei in the human body [11, 12]. Because sodium ions play key roles in biological processes, metabolism, and homeostasis, ^{23}Na MRI can provide direct insights into the metabolic activity and cellular integrity of tissues [5, 13]. Imaging ^{23}Na nuclei is often performed using $^{23}\text{Na}/^1\text{H}$ double-tuned coils [9, 14-16]. ^1H signal is then used for scout images and B_0 shimming as mentioned above.

Many different techniques are used to create a double-resonant RF coil, for example, modifying a single-tuned coil using traps or PIN-diodes. A RF coil can be set to resonate at two different frequencies by adding a LC parallel trap circuit [17]. However, the sensitivity of the double-tuned coil is usually less than that of a geometrically identical single-tuned coil [18]. The sensitivity of the trapped coil at two resonance frequencies depends on the component values of the trap [19]. When the inductance of the trap is bigger, smaller trap capacitance is required to tune the coil [20] and the sensitivity at ^1H Larmor frequency is lower [19]. On the other hand the smaller the trap inductance used, the higher the sensitivity becomes at the X-nuclei Larmor frequency.

When PIN-diode switches with capacitors are added in the coil to actively switch the resonance frequencies [21, 22], the sensitivity at the lower frequency (usually Larmor frequency of the X-nuclei) can be significantly compromised [22] due to the resistance of the forward biased PIN-diode (resistance of the forward biased PIN-diode is typically in the range of $0.1\text{-}1\ \Omega$ at 100 MHz) [23]. Traps and PIN-diode switches can be also added to the birdcage coil for the same purpose [19, 22, 24-26]. Instead of adding traps or PIN-diodes, two additional end-rings can be inserted to the birdcage coil, creating a double-resonating birdcage resonator [27, 28]. This four-ring approach has an advantage in terms of efficiency and homogeneity for both frequencies when it meets its optimised inner and outer leg length ratio [29]. However, the additional length required when adding the outer end-rings restricts the accessible space, in particular for brain studies [30].

The main objective of this work is the development of both heterogeneous and homogeneous double-tuned RF resonators for brain ^{23}Na imaging with highest possible sensitivity at the ^{23}Na Larmor frequency. A four-ring birdcage coil, which maintains the efficiency and sensitivity at both

frequencies, is modified to meet in vivo imaging constraints. For the heterogeneous resonator, a quadrature coil in combination with a PIN-diode switched loop coil is designed for SNR enhancement at the ^{23}Na frequency compared to the geometrically identical, single-tuned coil.

In Chapter 2, the fundamental principles of MRI are summarized. Starting with the physical principles of NMR these are extended to the imaging case. An overview of the required MR hardware completes this chapter.

Chapter 3 provides an overview of widely used RF coil configurations and the RF chain of a MR system. The properties of different RF coils are discussed and the operation of RF coil within the RF chain is described. The methods to create a double-resonant RF coil are introduced.

The original contribution of this work starts in Chapter 4. There, a PIN-diode controlled, quadrature-enhanced, double-tuned RF coil to improve the ^{23}Na image SNR is presented. To acquire ^{23}Na signals in quadrature and ^1H signals in linear modes, a T/R switch with specific layout using ^{23}Na hybrid coupler is designed. A PIN-diode control unit is also introduced and used to equally apply either forward current or reverse voltage to the individual PIN-diodes.

Chapter 5 introduces the folded four-ring birdcage coil which is designed to overcome the length restriction issue for in vivo brain imaging is introduced. Concepts and bench measurement result are presented.

Chapter 6 provides all the MR experiments result. For the PIN-diode controlled, quadrature-enhanced, double-tuned RF surface coil, both ^{23}Na and ^1H image SNRs were then compared with those of a geometrically identical, single-tuned coil. However, higher RF transmit voltage was required than that of single-tuned coil although applied RF transmit voltage was lower than system reverse voltage for PIN-diode. For the folded four-ring birdcage coil, the SNR and the B_1 uniformity are compared with a single-tuned and a larger four-ring birdcage coil using phantom measurements. In vivo healthy rat brain images at 9.4 T using the folded four-ring birdcage are also presented.

Finally, in Chapter 7, the results are discussed and the most important conclusions of this dissertation are drawn.

Chapter 2

MAGNETIC RESONANCE IMAGING

2.1 Nuclear magnetic resonance

NMR is a physical phenomenon in which nuclei with non-zero spins placed in a magnetic field absorb and re-emit radio frequency energy [31]. The nucleus consists of protons and neutrons. All nuclides with even numbers of both protons and neutrons have a net spin of zero and hence, they are MR insensitive. All the other molecules have an intrinsic magnetic moment and angular momentum. Table 2-1 shows the general rules for determination of nuclear spin quantum numbers [32]. For example, ^1H has one proton so spin=1/2 and ^{23}Na contains an odd number of neutrons, resulting in a net spin ($I=3/2$) and permitting their use in NMR [33].

Table 2-1 General rules for determination of nuclear spin quantum numbers

Mass Number	Number of Protons	Number of Neutrons	Spin (I)	Examples
Even	Even	Even	0	^{16}O , ^{12}C
	Odd	Odd	Integer (1, 2, ...)	^2H , ^6Li
Odd	Even	Odd	Half-integer (1/2, 3/2, ...)	^{13}C , ^{15}O , ^{17}O
	Odd	Even	Half-integer (1/2, 3/2, ...)	^{23}Na , ^{31}P , ^{19}F

2.1.1 Spins in a magnetic field

A magnetic moment $\boldsymbol{\mu}$ of an atomic nucleus arises from a non-zero spin. The nuclear magnetic moment and an angular momentum $\mathbf{J}$ are related by the following formula

$$\boldsymbol{\mu} = \gamma \mathbf{J}, \quad 2-1$$

where γ is nuclear gyromagnetic ratio, which is a nucleus-dependent constant and is 42.56 MHz/T for the hydrogen nucleus and 11.262 MHz/T for the sodium. The angular momentum is proportional to the intrinsic spin, $\mathbf{I}$, of the nucleus.

$$\mathbf{J} = \hbar \mathbf{I}, \quad 2-2$$

where $\hbar$ is Planck's constant over 2π ($1.054 \times 10^{-34} \text{ m}^2 \text{ kg/s}$).

When there is no external magnetic field, the nuclear magnetic moments in a sample are randomly oriented, and thus the net magnetisation is zero. If, however, an external magnetic field $\mathbf{B}_0$ is presented to the nuclei, the magnetic moments align either parallel (low energy) or anti-parallel (high energy) with the applied field. For the ^1H nucleus, $I = 1/2$ and therefore there are only two energy levels. The high energy level is occupied by spins which are aligned antiparallel to $\mathbf{B}_0$ and the low energy level is occupied by spins aligned with $\mathbf{B}_0$. Each energy level is given by

$$E = -m\hbar\gamma\mathbf{B}_0, \quad 2-3$$

where m is the magnetic quantum number which is $-I, -I + 1, \dots, I - 1, I$, in this case $\pm 1/2$. Figure 2-1 (a) shows how the energy levels are arranged for a spin 1/2 nucleus.

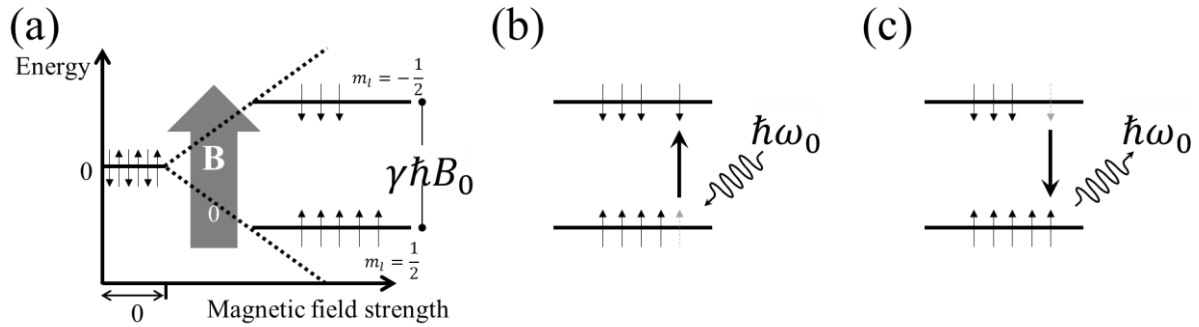


Figure 2-1 (a) In the presence of an external magnetic field, energy level difference for a system of $I = 1/2$. (b) Absorption of a photon of energy $E = h\nu$ to promote a transition between nuclear energy levels, called a spin flip. (c) The photon is emitted with the spin moving from higher energy level to its lower energy level.

The energy of a photon is proportional to its frequency ν .

$$E = h\nu = \hbar\omega, \quad 2-4$$

where h is Planck's constant. When a proton with energy $\hbar\omega = \Delta E$ is absorbed, transition between the two energy levels occurs as shown in Figure 2-1 (b). The required frequency for absorption to occur is given as

$$\begin{cases} \nu_0 = \frac{\gamma B_0}{2\pi} \\ \omega_0 = \gamma B_0 \end{cases} \quad 2-5$$

which is known as the Larmor frequency. For most clinical MRI machines, B_0 is on the order of Tesla (T) while γ is on the order of 10^7 . Consequently, the electromagnetic radiation required is on the order of dozens or even hundreds of MHz (radio frequency, RF). As soon as the electromagnetic radiation is off, the excited nuclei (i.e. nuclei in the higher energy level) start to return to their thermodynamically stable equilibrium state. This phenomenon is called relaxation. The relaxation is the opposite process of the excitation where the energy absorbed during the excitation is released (see Figure 2-1 (c)).

The population difference of the spins between the two energy levels is determined by the Boltzmann distribution, which can be written as

$$\frac{N_{-1/2}}{N_{+1/2}} = e^{-\Delta E/k_B T}, \quad 2-6$$

where $N_{-1/2}$ and $N_{+1/2}$ are the population of the spin-down (lower) and spin-up (higher) energy levels, respectively, $k_B = 1.38 \times 10^{-23} \text{ J/K}$ is the Boltzmann constant and T is the absolute temperature of the spin system. The net magnetisation is proportional to the population difference of the spins between the lower and the higher energy state $\Delta N = N_{-1/2} - N_{+1/2}$. The net magnetisation M_0 under the high temperature assumption ($k_B T \gg \Delta E$) can be computed as

$$M_0 = \frac{N \gamma^2 \hbar^2 I(I+1)}{3k_B T} B_0, \quad 2-7$$

where $N = N_{-1/2} + N_{+1/2}$ is the total number of spins. Thus the net magnetisation is proportional to the applied $\mathbf{B}_0$ field. In this way, the whole spin system can be characterized in terms of the net magnetisation, which can be described by classical physics instead of quantum mechanics.

2.1.2 Bloch equations

The Bloch equations are phenomenological equations that are used to calculate the nuclear magnetisation as a function of time [34]. In this section, the precession, excitation, and relaxation are described using Bloch equations.

2.1.2.1 Precession

If the system is at thermal equilibrium, the net magnetisation $\mathbf{M}$ and external magnetic field $\mathbf{B}$ are

pointing in the same direction. If the thermal equilibrium is disturbed and $\mathbf{M}$ and $\mathbf{B}$ do not point in the same direction, a torque on $\mathbf{M}$ proportional to the rate of change of the angular momentum $\mathbf{M}/\gamma$ will be produced by $\mathbf{B}$.

$$\frac{d\mathbf{M}}{dt} = \gamma \mathbf{M} \times \mathbf{B}. \quad 2-8$$

For the static magnetic field $\mathbf{B}_0$ along the z-axis ($B_x = 0, B_y = 0, B_z = B_0$), the bulk magnetisation vector can be analysed with x-, y-, and z-components and the motion equation for the individual components of $\mathbf{M}$ are

$$\frac{dM_x}{dt} = \gamma M_y B_0, \quad \frac{dM_y}{dt} = -\gamma M_x B_0, \quad \frac{dM_z}{dt} = 0. \quad 2-9$$

The solution describes a free precession of the magnetisation vector with the Larmor frequency ω_0 .

$$\begin{aligned} M_x(t) &= M_{x0} \cos(\omega_0 t) + M_{y0} \sin(\omega_0 t) \\ M_y(t) &= -M_{x0} \sin(\omega_0 t) + M_{y0} \cos(\omega_0 t) \\ M_z(t) &= M_{z0}. \end{aligned} \quad 2-10$$

As shown in Figure 2-2 (a), $\mathbf{M}$ is precesses clockwise around z-axis at frequency ω_0 .

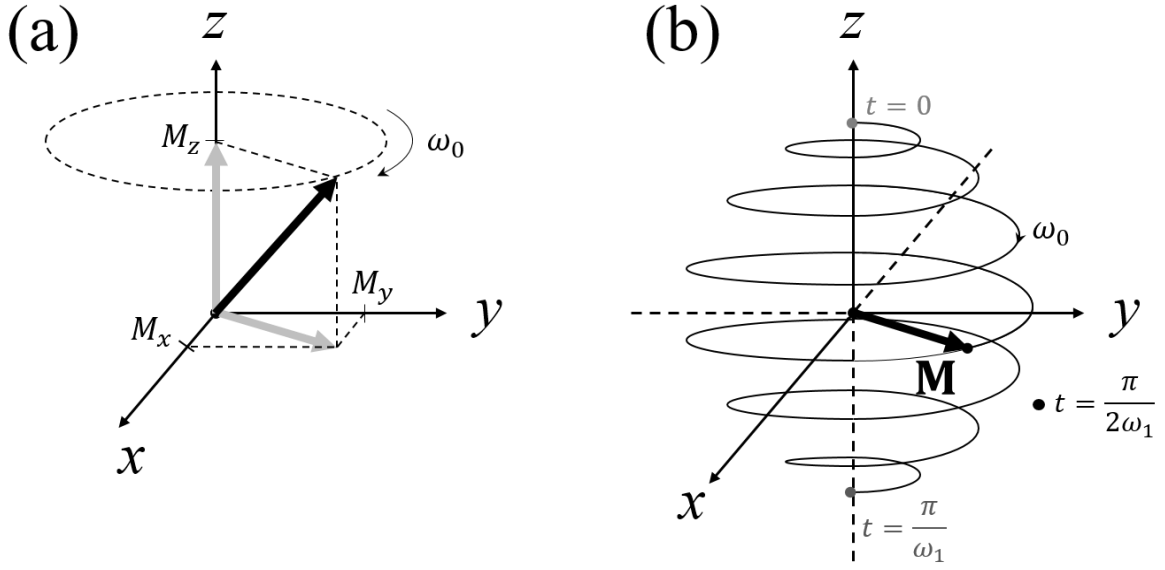


Figure 2-2 The behaviour of the net magnetisation $\mathbf{M}$ in the presence of a static magnetic field $\mathbf{B}_0$ and in the absence of the relaxation as seen from laboratory frame. (a) $\mathbf{M}$ is precesses around z -axis. (b) $\mathbf{M}$ is flipped by an applied oscillating magnetic field $\mathbf{B}_1$.

2.1.2.2 Excitation

If an RF field $\mathbf{B}_1$ which is perpendicular to the $\mathbf{B}_0$ and rotating with frequency ω_0 is applied in addition to the $\mathbf{B}_0$, the Eq. 2-8 can be written as

$$\frac{dM_x}{dt} = \gamma(M_y B_0 - M_z B_{1y})$$

$$\frac{dM_y}{dt} = \gamma(-M_x B_0 + M_z B_{1x}) \quad 2-11$$

$$\frac{dM_z}{dt} = \gamma(M_x B_{1y} - M_y B_{1x}),$$

with $B_{1x} = B_1 \cos(\omega_0 t)$ and $B_{1y} = -B_1 \sin(\omega_0 t)$. The solutions are given by

$$M_x(t) = M_0 \sin(\omega_1 t) \cos(\omega_0 t)$$

$$M_y(t) = -M_0 \sin(\omega_1 t) \sin(\omega_0 t) \quad 2-12$$

$$M_z(t) = M_0 \cos(\omega_1 t),$$

where $\omega_1 = \gamma B_1$. At equilibrium ($t = 0$) $\mathbf{M}$ is aligned with $\mathbf{B}_0$ so that $\mathbf{M}$ has only z-component, at $t = \pi/(2\omega_1)$ the magnetisation vector lies in the xy-plane ($M_z = 0$) (Figure 2-2 (b)). A RF pulse with $t = \pi/(2\gamma B_1)$ is called a “90° pulse” and one with $t = \pi/(\gamma B_1)$ is known as the “180° pulse”.

2.1.2.3 Relaxation

As shown in Figure 2-3, the magnetization can be decomposed in a longitudinal and a transverse component. Relaxation causes the longitudinal magnetisation M_z to recover while the transverse magnetisation $M_{xy} = \sqrt{M_x^2 + M_y^2}$ is decaying. These two processes will be discussed in the following sections.

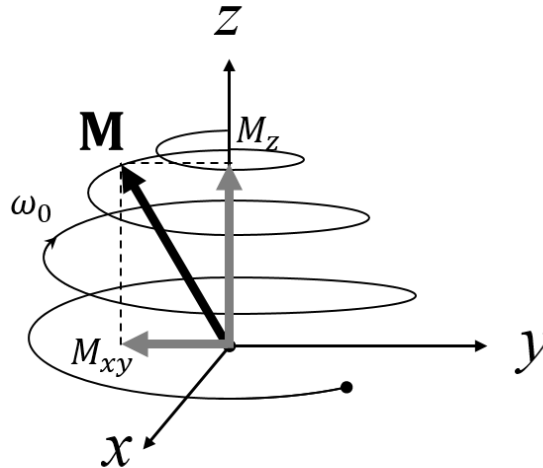


Figure 2-3 Trajectory of the net magnetisation during the relaxation.

2.1.2.3.1 Spin-lattice relaxation

During relaxation, the longitudinal magnetisation energy is released to the molecular lattice and this process is called longitudinal or spin-lattice relaxation. Its behaviour is mathematically described by [34]

$$\frac{dM_z}{dt} = -\frac{M_z - M_0}{T_1}. \quad 2-13$$

where T_1 is the longitudinal (spin-lattice) relaxation time which is the time when M_z has reached approximately 63% of M_0 ($M_z(T_1) \cong 0.63M_0$) (Figure 2-4 (a)). The solution of this differential equation after 90° pulse ($M_z(0) = 0, M_{xy}(0) = M_0$) can be written as

$$M_z(t) = M_0 \left(1 - e^{-t/T_1}\right), \quad 2-14$$

2.1.2.3.2 Spin-spin relaxation

Simultaneously with the longitudinal relaxation, the transverse magnetisation decays because of interaction between the spins and this process is called transverse or spin-spin relaxation. This behaviour is characterised by the time constant T_2

$$\frac{dM_{xy}}{dt} = -\frac{M_{xy}}{T_2}. \quad 2-15$$

The solution is given as an exponential decay, e.g. after a 90 degree pulse

$$M_{xy}(t) = M_0 e^{-t/T_2},$$

2-16

where T_2 is the transverse (spin-spin) relaxation time which is the time, when M_{xy} has decayed by a 90 degree pulse to approximately 37% of M_0 ($M_{xy}(T_2) \cong 0.37M_0$) (Figure 2-4 (b)).

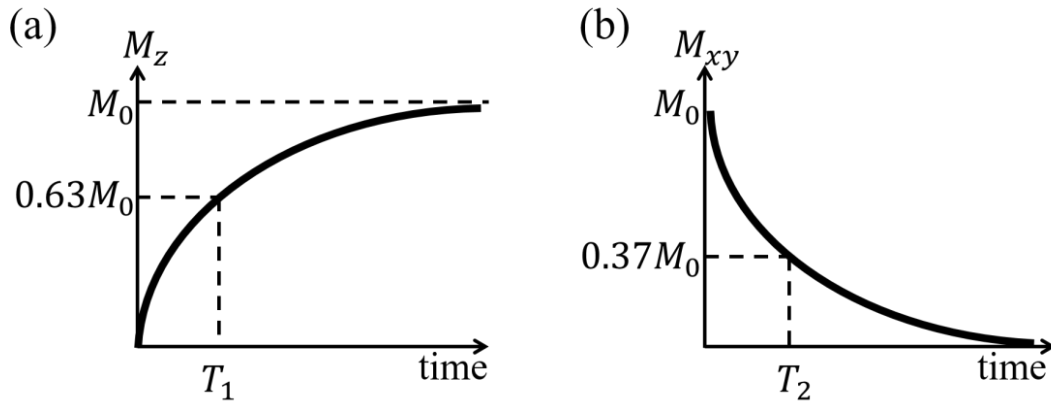


Figure 2-4 Relaxation curves of (a) longitudinal and (b) transverse magnetisation.

In reality, the transverse magnetisation decays faster than expressed by T_2 because inhomogeneity in the $\mathbf{B}_0$ produce variations in the resonance frequency according to Eq. 2-5. Due to that, the transverse magnetisation decay is given by T_2^* .

$$\frac{1}{T_2^*} = \frac{1}{T_2} + \frac{1}{T_2'},$$

2-17

where T_2' accounts for $\mathbf{B}_0$ inhomogeneity which can come from magnet design imperfections, susceptibility differences in susceptibility between the different sample tissues. The combination of Eq. 2-8, 2-13, and 2-15 is known as the Bloch equation.

$$\frac{d\mathbf{M}}{dt} = \gamma \mathbf{M} \times \mathbf{B} - \frac{(M_x \hat{x} + M_y \hat{y})}{T_2} - \frac{(M_z - M_0)}{T_1} \hat{z}, \quad 2-18$$

$$M_{xy}(t) = M_0 e^{-t/T_2^*}. \quad 2-19$$

where $\hat{x}$, $\hat{y}$, and $\hat{z}$ represents the unit vectors along the x-, y-, and z-axis, respectively.

2.1.3 Free induction decay

The rotating magnetisation in the transverse plane can be detected as an induced voltage by an RF receive coil placed with its longitudinal axis perpendicular to the $\mathbf{B}_0$. Details about RF coil will be discussed in chapter 3. However, the measured NMR signal based on Eq. 2-19 is known as the free induction decay (FID) which is shown in Figure 2-5.

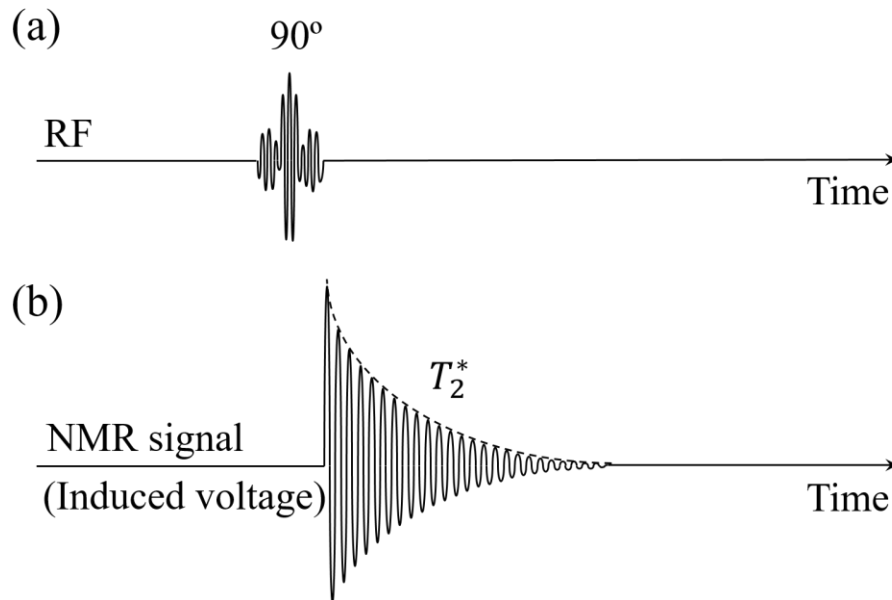


Figure 2-5 (a) A (sinc-shaped) RF excitation pulse of 90° flip angle. (b) A FID signal with a T_2^* envelope.

The FID signal oscillates at the Larmor frequency but is damped by the T_2^* decay. The FID signal

can be created by an RF pulse with not only 90° but also with any other flip angles except 180° because the 180° pulse only inverts the longitudinal magnetisation and does not generate any transverse components.

2.1.4 Spin-echo

Since the transverse magnetisation decays with T_2^* , the FID signal is damped rapidly as shown in Figure 2-5. However, T_2' decay can be reversed by generating a spin echo (SE) [35]. The SE signal is acquired by applying a 180° pulse, sometime τ after the 90° pulse [2, 36]. Figure 2-6 depicts the SE sequence and the effect of the pulse.

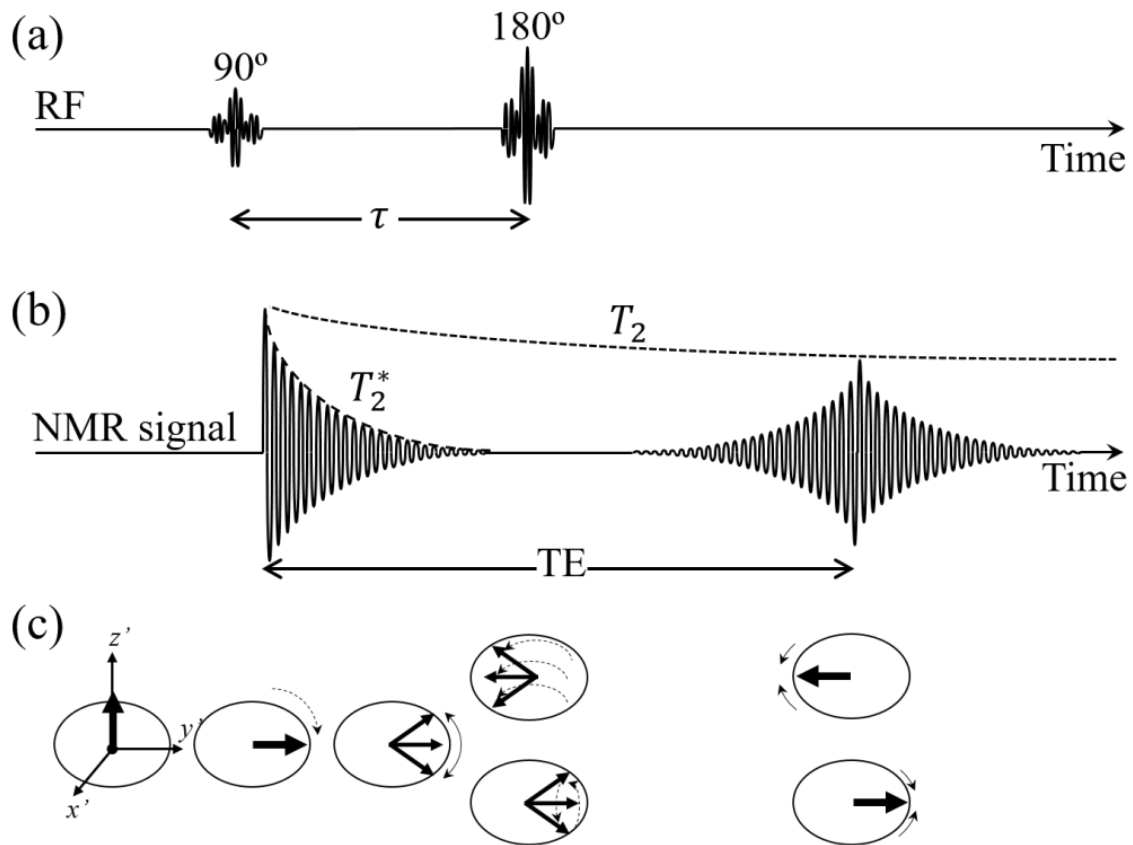


Figure 2-6 (a) RF pulse used in the spin-echo sequence and (b) the measured signal in the laboratory frame of reference. (c) From left to right the figure shows: spin before, and after excitation, dephasing, application of 180° pulse along the x' -axis (top) or y' -axis (bottom) from a rotating frame of reference at the frequency ω_0 , and refocused spin [35].

After excitation by 90° pulse, transverse components of magnetisation M_{xy} dephase due to the T_2^* effect. When the 180° pulse is applied, a time of τ after the 90° pulse, magnetisation vectors are inverted. M_{xy} is then rephased at echo time $TE = 2\tau$. As shown in Figure 2-6 (c), 180° pulse can be applied with either in phase [2] or 90° out of phase with respect to the initial 90° pulse [36]. Pulse-related errors can be reduced by applying 90° phase shifted 180° pulse [35].

2.2 Magnetic resonance imaging

In this section, the principles of the spatial localisation of a given NMR property are briefly discussed based on the two-dimensional (2D) Cartesian grid. Spatial localisation is necessary to create an image using the NMR signal. The resonance frequency is determined by the main magnetic field strength (Eq. 2-5). This resonance frequency changes when a gradient is applied. The gradient generates a small additional magnetic field variation, which depends on the spatial position. Thus, when the gradient is on, the resonance frequency is determined by the combination of the static magnetic field strength and the additional field created by the gradient, which is a function of the position in space [37].

2.2.1 Slice selection

When the magnetic gradient is applied along the perpendicular direction of the slice, for example in z -direction, the spins at each position rotate slightly faster or slower than ω_0 as shown in Figure 2-7 (a).

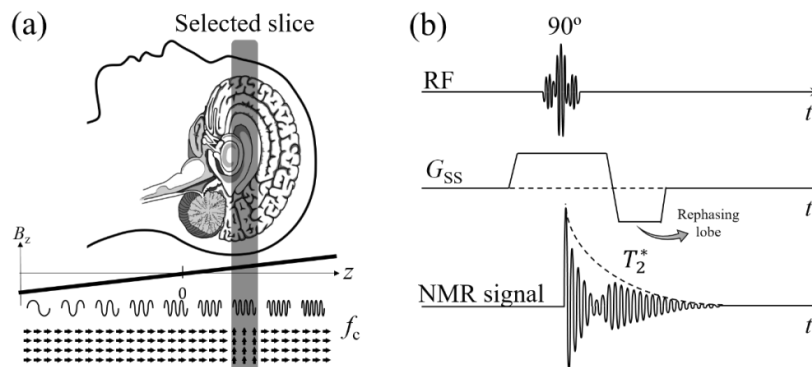


Figure 2-7 (a) The use of a magnetic gradient to tune a specific slice so that it can be selectively excited by an RF pulse. (b) The pulse sequence used for slice selected excitation.

A narrow-band RF pulse is simultaneously applied where its frequency band is designed to match the frequency band of the slice to be excited. As a result, only the spins in the slice are excited. The centre frequency (f_c) and bandwidth (Δf) of the RF pulse can be selected by the position (z) and thickness (Δz) of the slice as equation below.

$$\begin{cases} \omega_c = \omega_0 + \gamma G_z z \\ \Delta \omega = \gamma G_z \Delta z \end{cases}, \quad 2-20$$

where $G_z = B_z/dz$. The phase of transverse magnetisation is dispersed after excitation due to the applied magnetic gradient. An opposite sign of gradient, so-called rephasing lobe (Figure 2-7 (b)) is normally applied to rephase the magnetisation and the area under the rephasing lobe is only half the area under the main portion of the slice-select gradient [37, 38].

2.2.2 Frequency encoding

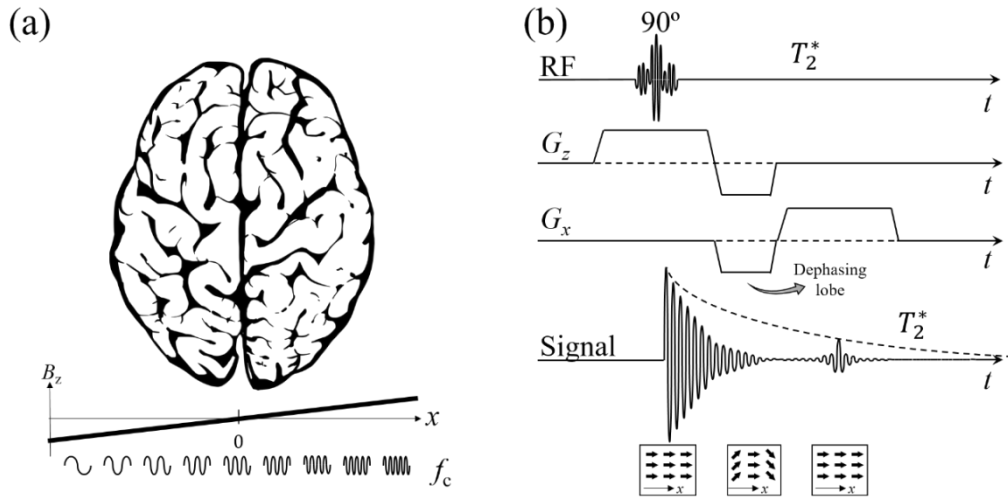


Figure 2-8 (a) A linearly increasing frequency-encoding gradient ($\frac{\partial B_z}{\partial x}$) applied along the horizontal axis (x). (b) The pulse sequence in order to acquire 1D projection data of 2D plane.

When the magnetic gradient is applied along the readout direction, for example x -direction, the precession frequencies are lower and higher than ω_0 at the $x < 0$ and $x > 0$, respectively (Figure 2-8(a)).

$$\Delta\omega(x) = \gamma G_x x, \quad 2-21$$

where $G_x = B_z/dx$. The Larmor frequency varies with x-position due to the applied x-gradient.

$$\omega(x) = \omega_0 + \Delta\omega(x). \quad 2-22$$

The solution to the Bloch equation is

$$M_{xy}(x, t) = M_0(x) e^{-t/T_2^*} e^{-i\gamma \int_0^t G(\tau) \cdot x d\tau}. \quad 2-23$$

The received signal is then a mixture of the signal with different frequencies. The effects of T_2 relaxation are ignored in the following equation.

$$s(t) = \int_x \rho(x) e^{-i2\pi k_x(t) \cdot x} dx. \quad 2-24$$

where $\rho(x)$ represents the spin density in the object and proportional to the initial magnetisation according to Eq. 2-7 and

$$k_x(t) = \frac{\gamma}{2\pi} \int_0^t G_x(\tau) d\tau, \quad 2-25$$

the time integrals of the gradient waveforms. Since the one-dimensional (1D) Fourier transform of $\rho(x)$ is given as

$$P(k_x) = \int_x \rho(x) e^{-i2\pi k_x x} dx. \quad 2-26$$

The signal contribution of each frequency component can be calculated by inverse Fourier transformation. Since the frequencies within a column in Figure 2-8 (a) are the same, one-dimensional (1D) projection data of 2D plane can be obtained using G_z and G_x . The spatial frequency space is called k -space, where k represents the spatial frequency variable. The dephasing lobe is applied before applying frequency encoding gradient to acquire a single k -space line from the $k_{x\min}$ to the $k_{x\max}$; the minimum and maximum k_x is determined by the given imaging parameters (i.e. field of view (FOV), matrix size). Here, the echo signal is expected to be sampled at the centre of the frequency encoding gradient.

2.2.3 Phase encoding

The phase information can also be used for spatial encoding using linear magnetic gradient. The phase encoding (PE) gradient G_{phase} is normally applied between slice selection gradient and frequency encoding (readout) gradient [38]. While the G_{phase} is applied, typically for a short time, the spins at a different location in the PE direction will precess with different frequencies. After turning off the G_{phase} , all spins then again precess with the same frequency but with different phase offset according to the position in the PE direction as shown in Figure 2-9. The phase encoding process is repeated to fully encode an image with a different magnitude of the G_{phase} .

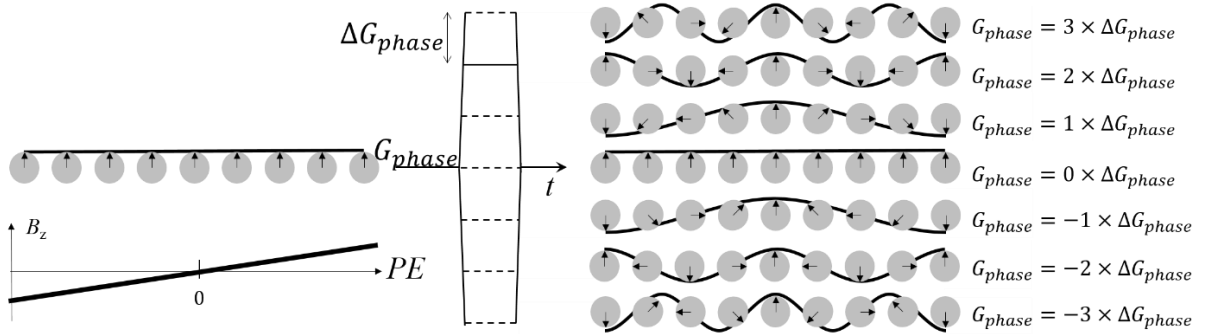


Figure 2-9 Magnetisation vector diagrams before and after applying phase encoding gradient.

2.2.4 k-space

Similar to the Eq. 2-24 and 2-26, from the signal equation with x- and y-gradient,

$$s(t) = \int_x \int_y \rho(x) e^{-i2\pi[k_x(t) \cdot x + k_y(t) \cdot y]} dx dy, \quad 2-27$$

the 2D Fourier transform of $\rho(x, y)$ is given as:

$$P(k_x(t), k_y(t)) = \int_x \int_y \rho(x) e^{-i2\pi(k_x(t) \cdot x + k_y(t) \cdot y)} dx dy. \quad 2-28$$

The relationship between signal and the 2D Fourier transform of $\rho(x, y)$ is expressed as

$$s(t) = P(k_x(t), k_y(t)), \quad 2-29$$

At any given time t , $s(t)$ equals the value of the 2D Fourier transform of $\rho(x, y)$ at a specific spatial frequency. The receive signal $s(t)$ therefore maps to 2D spatial frequency space (k -space) as determined by the time integrals of the applied gradient waveforms $G_x(t)$ and $G_y(t)$. Figure 2-10 shows an example of gradient echo (GRE) sequence to obtain the entire horizontal line. A slice selection gradient G_z is applied simultaneously with the RF excitation pulse. The negative lobe G_x and phase encoding gradient G_y set the start of the k -space trajectory to the upper left corner of k -space. After that, only the frequency encoding gradient G_x is applied causing a sweep of $\mathbf{k}$ along k_x (from left to right) and the horizontal line on the k -space is acquired. Only two phase-encoding steps are depicted in Figure 2-10 but other k -space lines should be acquired by repeating the sequence (a) with a different magnitude of phase encoding gradients.

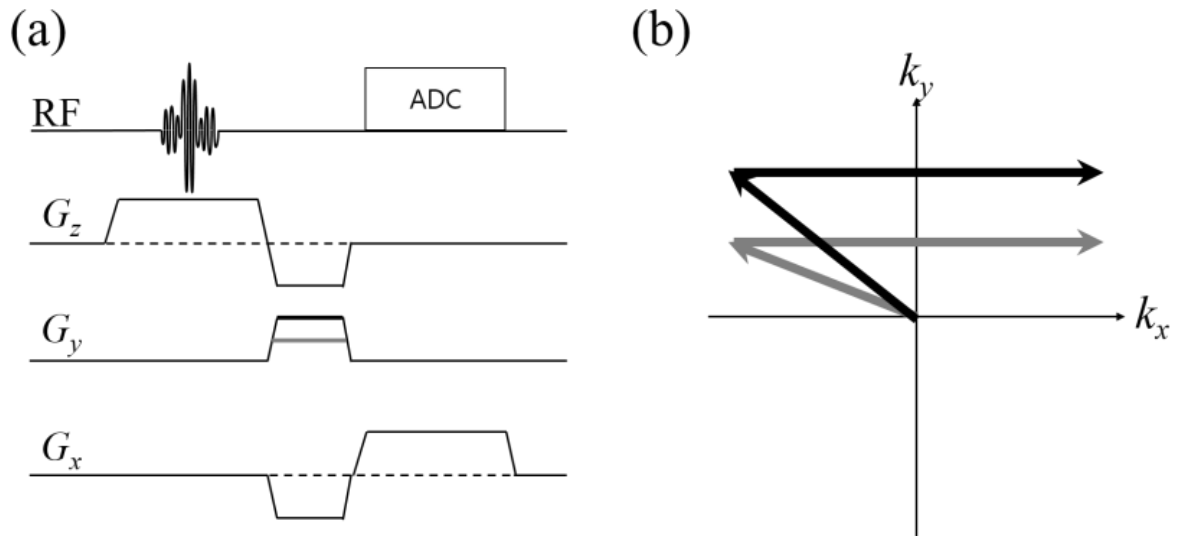


Figure 2-10 (a) An example of GRE sequence and (b) k -space trajectory.

2.2.5 Basic imaging sequences

Figure 2-11 shows the pulse sequence diagrams of the SE and GRE with three directional imaging gradients.

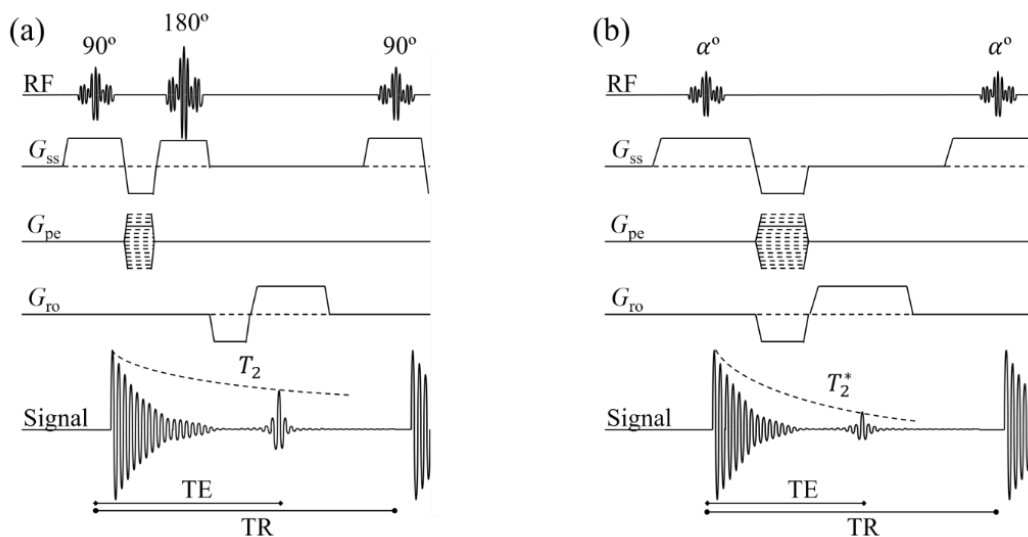


Figure 2-11 (a) A spin echo imaging sequence and (b) a gradient echo imaging sequence.

A SE is produced by using 180° refocusing RF pulse, while GRE occurs during the TE using gradient reversal. Because only one RF pulse is applied in GRE, the echo can be recorded much more quickly when compared to standard SE. When low flip angles are used, repetition time (TR) can be also shorter. Shorter TR and TE values allow for very rapid signal acquisition, therefore GRE is preferred for rapid imaging techniques. GRE gives T_2^* contrast since T_2' effects due to susceptibility or field inhomogeneity are not refocused, whereas image contrast of SE is dictated by T_2 relaxation. For this reason, GRE sequences are more frequently affected by susceptibility and chemical shift artefacts.

2.2.6 MRI hardware

2.2.6.1 MRI hardware overview

Figure 2-12 shows the schematic representation of the major systems on an MRI and a few of the major interconnections.

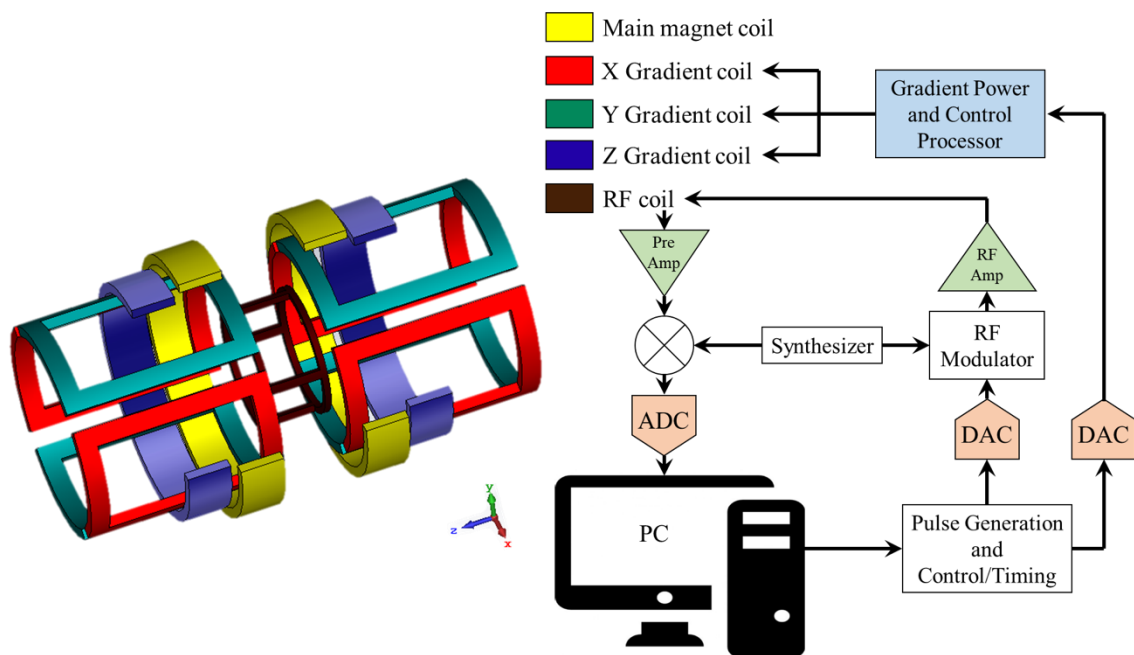


Figure 2-12 Schematic diagram of an MRI hardware system with cross section of main magnet and z gradient coil.

An MRI system consists of three main components, which are the main magnet, the gradient systems, and the RF system. A superconducting magnet generates the strong static magnetic field $\mathbf{B}_0$, which is required to induce the nuclear polarisation. The gradient coils generate the required linear magnetic fields which are used to spatially encode the MR signal. Moreover, these gradient coils are used as first-order shimming by applying offset bias [39]. The RF coil generates the alternating magnetic field $\mathbf{B}_1$, which is perpendicular to $\mathbf{B}_0$, and used to rotate the spins by 90° , 180° or any other value selected by the pulse sequence. The RF coil also detects the weak MR signal being returned from the tissue. More details about the RF system are described in the Chapter 3. A number of computers are used to generate the digital representations of the gradient and the RF pulse, to reconstruct the image, and also to display the image on a graphical user interface.

2.2.6.2 Gradient coils

Three sets of gradient coils are used in almost every MR system. Each of x-, y-, and z-gradient coils is driven by an independent power amplifier and these coils can create a gradient field whose z-component varies linearly along the chosen axis. The Maxwell coil (Figure 2-13 (a)) can be used for z-gradient coil while the Golay coils (Figure 2-13 (b)) can be used for transverse (x- and y-) gradient coil [40, 41]. A Maxwell gradient coil consists of two parallel circular coils as shown in Figure 2-13 (a) [42]. When the opposite direction of current with the same magnitude is applied to the two loops, the Maxwell coil produces a linear magnetic gradient [43]. When the loop radius a and the distance between the loops satisfy $d = \sqrt{3}a$, the Maxwell coil generate linear z-gradient [44]. Figure 2-13 (d) shows the Golay coil which consists of two pairs of saddle coils [45]. When z_0 and z_r fulfil the equation below for a given loop radius a , linear x-gradient is generated by the Golay coil [46].

$$\begin{cases} z_0 = 0.39a \\ z_r = 2.57a \end{cases} \quad 2-30$$

When the x-gradient coil is rotated by 90° around the z-axis, the coil is used for the y-gradient coil. However, modern gradient coils are much more complex than described here [40, 47].

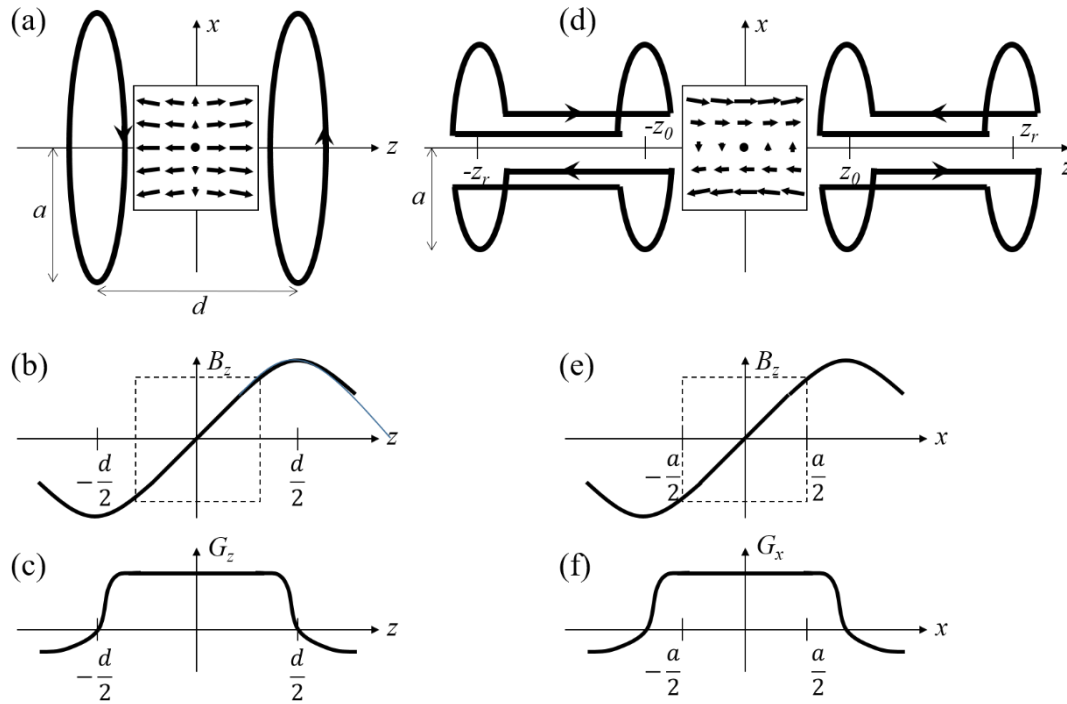


Figure 2-13 The gradient coils. (a) A Maxwell coil (used as z -gradient coil) and its magnetic field on the xz -plane. (b) the z component of magnetic field B_z generated by the Maxwell coil along the z -axis. (c) The z -gradient G_z of the Maxwell coil along the z -axis. (d) A golay coil (used as x -gradient coil) and its magnetic field on the xz -plane. (e) B_z generated by Golay coil along the x -axis. (f) The x -gradient G_x of the Golay coil along the x -axis.

Chapter 3

RADIOFREQUENCY COIL AND CHAIN

The RF coil is one of the main components of a scanner. During the transmission phase, an oscillating/rotating magnetic field ($\mathbf{B}_1$), which is orthogonal to $\mathbf{B}_0$, is generated by the RF coil and it is used for excitation of the nuclear spins. When used as receiver, the net magnetic flux from the excited spin system can be detected by the coil as an induced electric voltage. The RF coils can be used as transmit (Tx)-only, receive (Rx)-only, and transceiver. This chapter provides a description of the RF coils, including some of the RF chain components commonly used with RF coils. The RF coils are grouped into two main categories; heterogeneous resonators and homogeneous. The RF coils can be also tuned at two or more different frequencies to acquire non-proton (X-nuclei) signal.

3.1 Heterogeneous resonators

As the name suggests, a surface coil rests on the surface of the object being imaged. Since the surface coil can be smaller and located closer to the object (possibly higher filling factor η), it picks up signal and noise only from a small region not the entire sample. Thus, it provides higher RF sensitivity factor S_{RF} at a given frequency ω .

$$S_{RF} = \sqrt{\frac{\mu_0 \omega \eta Q}{4K_B T_c V_s}}, \quad 3-1$$

where μ_0 , Q , T_c , and V_s are the permeability of free space, quality-factor, temperature, and volume of the coil, respectively [48, 49]. These coils are commonly much smaller than the homogeneous resonators. Since smaller surface coils have less resistance and they receive noise only from nearby regions, they have higher signal to noise ratio (SNR) [49].

3.1.1 Loop coil

Figure 3-1(a) and (b) show the illustration of simple circular and rectangular surface loop coil made of conductive wires with a capacitor.

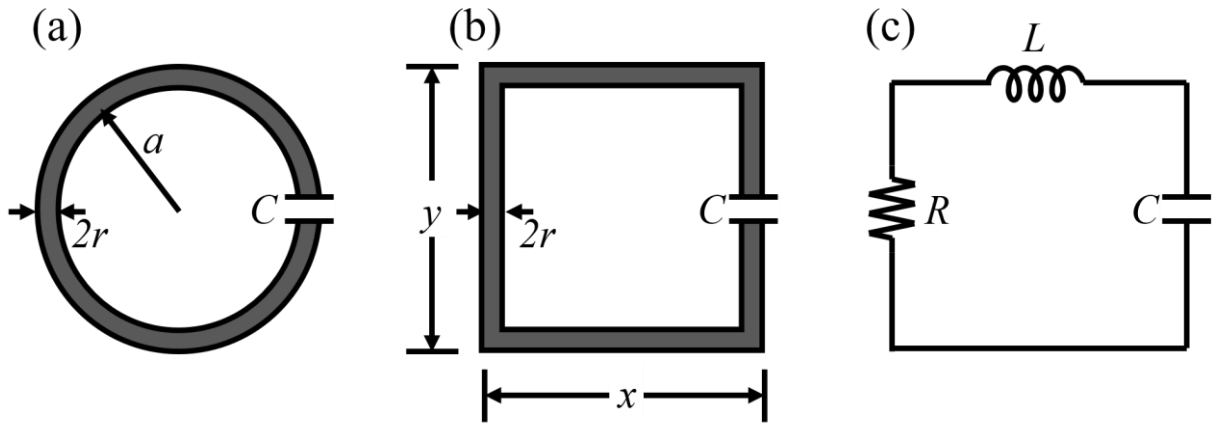


Figure 3-1 Illustration of simple surface loop coils. (a) A circular wire loop, (b) a rectangular wire loop, and (c) equivalent circuit of the surface loop coil.

These surface loops can be analysed using Maxwell's equations. However, if the size of RF coil is relatively small compared to the wavelength, RF coils can be analysed using an equivalent circuit using Kirchhoff's voltage and current laws and Biot-Savart's law [46]. Figure 3-1(c) shows the equivalent circuit diagram of the surface loop coil. The value of inductance (L) and resistance (R) can be calculated from the radius of the wire and the coil size and shape. The appropriate formulas for calculating the inductance and resistance are given in Eq. 3-2 and Eq. 3-3, respectively, for a circular wire loop [50]:

$$L = \mu_0 a \left[\ln \left(\frac{8a}{r} \right) - 1.75 \right], \quad 3-2$$

where a and r denote, respectively, the loop radius and radius of a circular wire, and

$$R = \frac{2a}{r^2} \rho, \quad 3-3$$

where ρ is specific resistance of the material, and in Eq. 3-4 and Eq. 3-5 for rectangle of a conductor with circular cross section [50],

$$L = \frac{\mu_0}{\pi} \left[x \ln \left(\frac{2x}{r} \right) + y \ln \left(\frac{2y}{r} \right) + 2\sqrt{x^2 + y^2} - x \sinh^{-1} \left(\frac{x}{y} \right) - y \sinh^{-1} \left(\frac{y}{x} \right) - 1.75(x + y) \right], \quad 3-4$$

where x and y denote width and height of the loop as shown in Figure 3-1(b), and

$$R = \frac{2(x + y)}{\pi r^2} \rho. \quad 3-5$$

From Eq. 3-3 and Eq. 3-5, R is only considered coil resistance. However, RF resistance might become higher due to the skin effect and radiation loss [51, 52]. The total impedance of the loop coil is then written as

$$Z = R + i \left(\omega L - \frac{1}{\omega C} \right). \quad 3-6$$

Since resonance condition is reached when the imaginary part of the impedance is zero, resonance frequency can be written as

$$\omega = \frac{1}{\sqrt{LC}} \quad 3-7$$

The magnetic field generated by a loop coil can be easily calculated using Biot-Savart's law.

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0 I}{4\pi} \int_C \frac{d\mathbf{l} \times \mathbf{R}}{R^3}, \quad 3-8$$

where μ_0 , I , $d\mathbf{l}$, and $\mathbf{R}$ are the magnetic constant, current passed through the wire of the coil, vector differential in the direction of current, and the full displacement vector from the wire element to the point at which the field is being computed, respectively. Figure 3-2 show the circular and square loop coils in the x - z plane. The magnetic field generated by the circular loop along the y -axis has only a y -component, whose magnitude is given by

$$\mathbf{B}(y) = \frac{\mu_0 I a^2}{2(y^2 + a^2)^{3/2}} \hat{y}, \quad 3-9$$

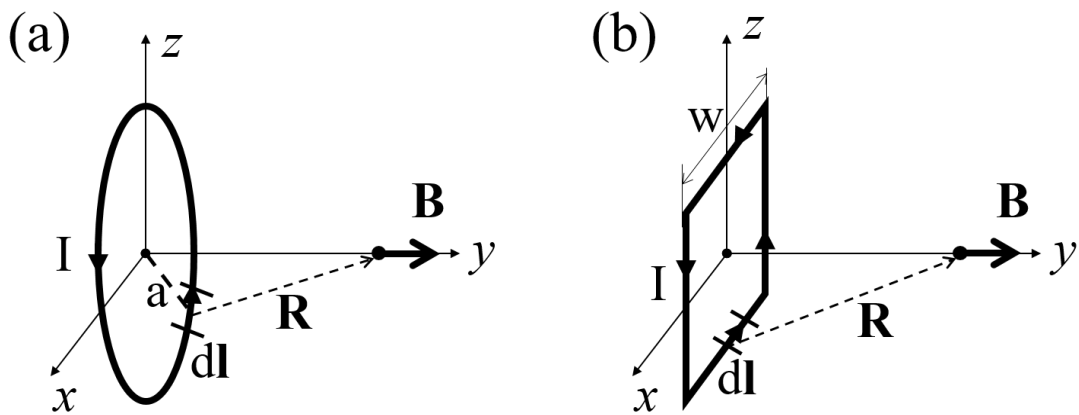


Figure 3-2 (a) A circular loop and (b) a square loop.

The magnetic field generated by the square loop of width w along the y -axis also has only a y -component and the magnitude of the magnetic flux density is given as [46]

$$\mathbf{B}(y) = \frac{\mu_0 I w^2}{2\pi \left[y^2 + \left(\frac{w}{2} \right)^2 \right]^{3/2}} \hat{y}. \quad 3-10$$

3.1.2 Butterfly coil

Figure 3-3 shows two different shapes of butterfly surface coils [53, 54]. The equivalent circuit of the butterfly coil is same as the loop coil but the butterfly coil generates a magnetic field with an orthogonal direction compared to that of the loop coil and it provides good RF isolation from surface coil when the loop coil is located at the centre of the butterfly coil [53].

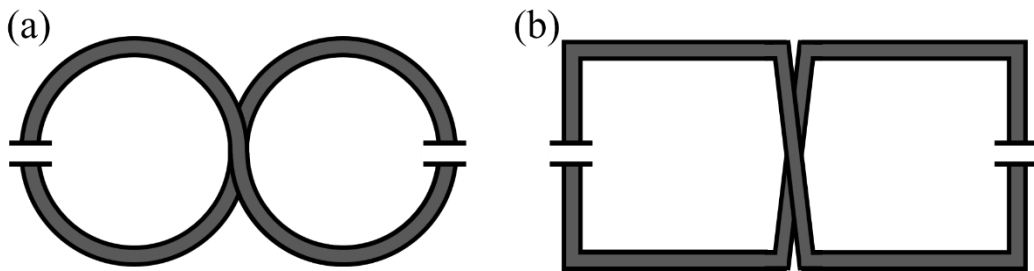


Figure 3-3 Illustration of butterfly coils. (a) Circular butterfly coil. (b) Rectangular butterfly coil.

When the circular or rectangular butterfly coil is arranged in x - z plane as shown in Figure 3-4, the butterfly coil generates x -directional magnetic field at P on the y -axis, unlike the loop coil.

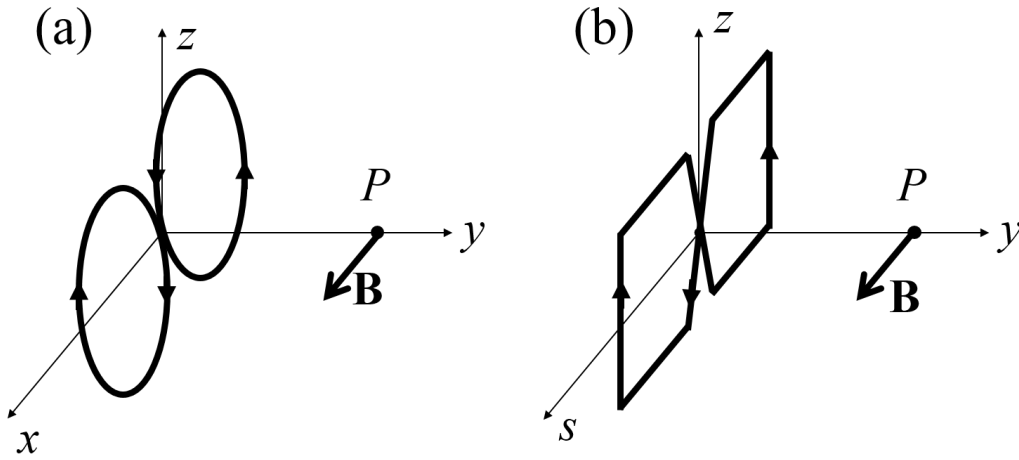


Figure 3-4 (a) Circular and (b) rectangular butterfly coils and magnetic field at P on the y -axis.

3.1.3 Array coil

The array coil is a group of coils operating independently. Recently, most MRI scanners have the ability to use a multi-channel phased-array coil for receive and some scanner have the ability even for transmit. A multi-channel transmit array coil is used for parallel RF transmission (pTx) which has been proposed as a solution to improve flip angle uniformity and specific absorption rate (SAR) reduction at ultra-high field [55, 56]. On the other hand, a receive phased-array coil is widely used to improve SNR [57] and reduce the scan time [58-60].

3.1.3.1 Decoupling

When the two or more coils are placed in close vicinity, the time-varying magnetic flux generated by one can be coupled to the others. The mutual coupling between the coils causes splitting of the resonance frequencies [61]. The coil is then detuned and mismatched, causing reduction in SNR. Therefore, decoupling is very important in array coil design and it also reduces the amount of correlated noise. There are many approaches to reduce the coupling effect [62-65]. Among them, geometric decoupling is introduced in this section.

3.1.3.1.1 Coupling effect

Induced electromotive force (EMF) ε can be calculated using Faraday's induction law.

$$\varepsilon = \iint_{\Sigma} -\frac{\partial \mathbf{B}(\mathbf{r}, t)}{\partial t} \cdot d\mathbf{A}, \quad 3-11$$

where $\mathbf{B}$ and $d\mathbf{A}$ are the magnetic field generated by one element and the surface area of the other element, respectively. This EMF produces a coupled current on another coil, which can be modelled as a circuit with mutual inductance (M). Figure 3-5 shows an example of the coupled coils (two coils are considered).

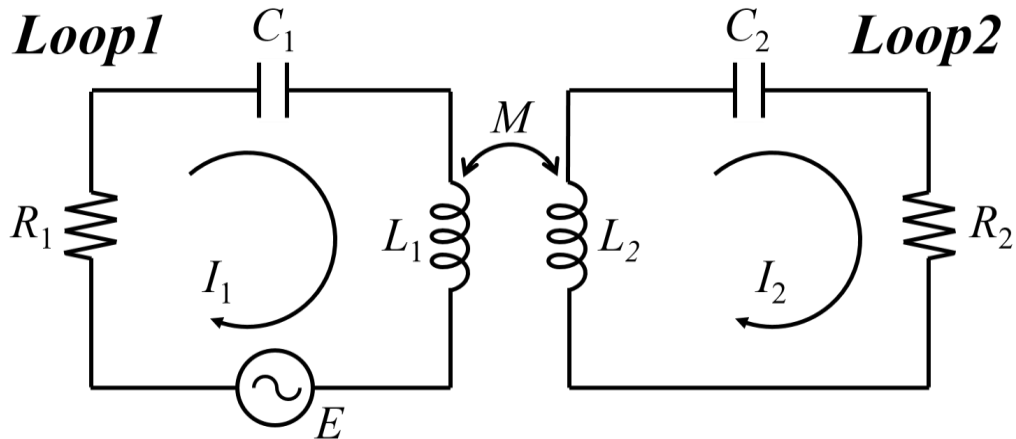


Figure 3-5 Equivalent circuit for two coupled coils.

Applying Kirchhoff's voltage law (KVL), the loop equations can be written.

$$E = I_1(R_1 + jX_{C1} + jX_{L1}) + I_2jX_M, \quad 3-12$$

$$0 = I_1 jX_M + I_2(R_2 + jX_{C2} + jX_{L2}). \quad 3-13$$

Solving Eq. 3-13 for I_2 ,

$$I_2 = -\frac{I_1 jX_M}{R_2 + jX_{C2} + jX_{L2}}. \quad 3-14$$

and eliminating I_2 from Eq. 3-12,

$$E = I_1(R_1 + jX_{C1} + jX_{L1}) + I_1 \frac{X_M^2}{R_2 + jX_{C2} + jX_{L2}}. \quad 3-15$$

The Loop2 in Figure 3-5 can be modelled as an impedance that is added in series with the primary impedance. The resonance frequency of Loop1 may be shifted through the added impedance and the efficiency of Loop1 will be reduced as a result of the coupling.

3.1.3.1.2 Geometric decoupling

As shown in Eq. 3-12, coils can be decoupled whenever X_M is equal to zero, X_M depends on the distance and the orientation of the coils. The coupling can thus be reduced by adjusting the distance between the coils and their orientation. From the Eq. 3-11, ε can be zero when the magnetic flux Φ_B through a surface, whose boundary is a wire loop, is zero.

3.1.3.1.2.1 Overlap decoupling

Figure 3-6 shows two circular loop coils which are on the same plane without overlapping. Once the RF signal is applied on Loop1, it generates a magnetic field. The magnitude of the generated magnetic field is higher near to Loop1 and lower further away from the Loop1. Thus, the mutual

coupling effect is stronger when Loop2 is closer to Loop1. Since the direction of the magnetic field inside Loop1 is opposite to that outside, the total magnetic flux inside Loop2 can be zero by overlapping the two loops as shown in Figure 3-7.

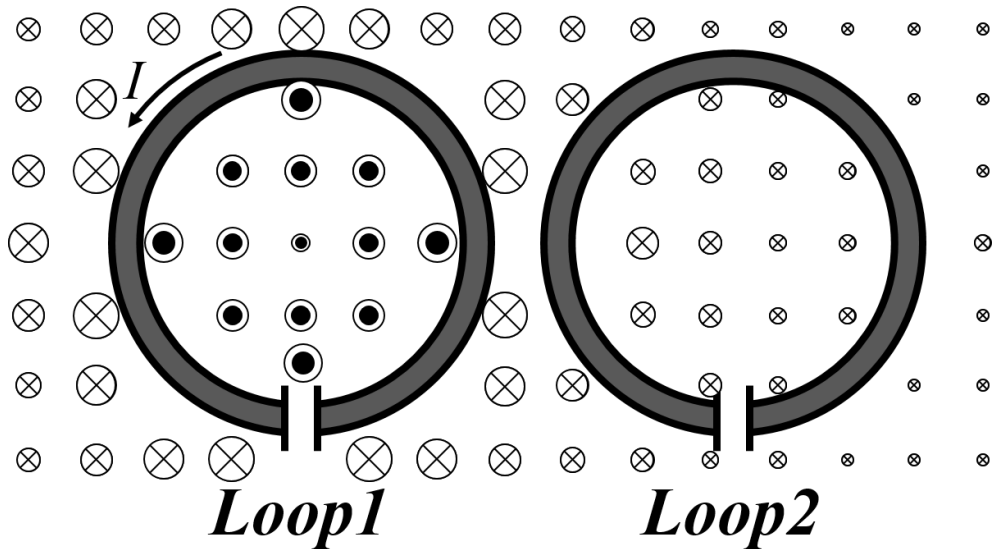


Figure 3-6 Illustration of two circular loop coils. Circles with a dot ($\odot$) and an X ($\otimes$) in Figure 3-6 shows the magnetic field generated by Loop1 and represent the magnetic field coming out of the page and going into the page, respectively. Size of the circle means magnitude of the magnetic field and larger circles indicate higher magnitude.

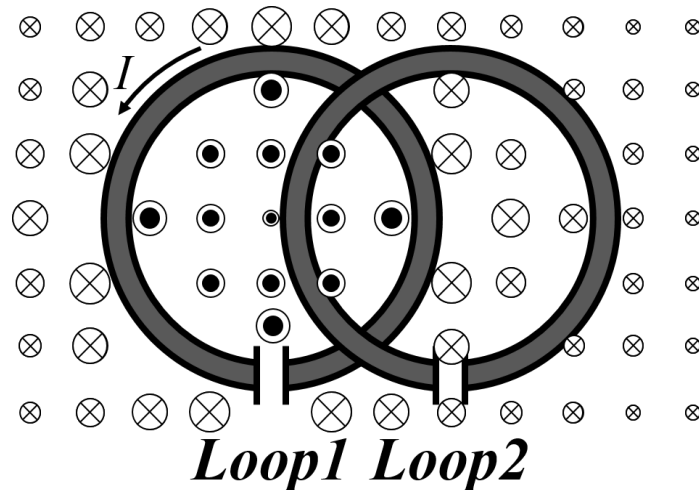


Figure 3-7 Illustration of two overlapped circular loop coils.

3.1.3.1.3 Perpendicular magnetic field decoupling

Each of the loops of the butterfly coil generates magnetic flux of equal magnitude but opposite direction. The magnetic field generated by a loop coil is perpendicular to that generated by a butterfly coil on the y-axis, as shown in Figure 3-2 and Figure 3-4.

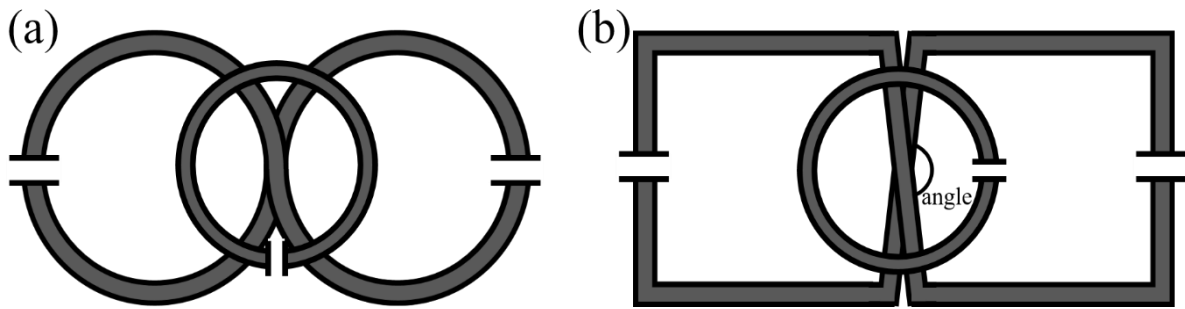


Figure 3-8 A circular loop coil with (a) a circular butterfly coil and (b) a rectangular butterfly coil.

When a loop coil is used with a butterfly coil as shown in Figure 3-8, the magnetic flux generated by one coil is cancelled out on the surface of the other coil. For a rectangular butterfly coil, the angle in Figure 3-8 (b) between the crossover elements at the centre of the coil can be 150° or larger for the high efficiency [53]. Instead of using a butterfly coil, other coils whose magnetic field is majorly perpendicular to that of the loop coil can be also used for decoupling, such as micro-strip transmission line (MTL) resonator [53], monopole antenna [66], and dipole antenna [67].

3.1.3.1.4 Preamplifier decoupling

When the array coil is used for only receiving MR signal, which is usually very low, each of loops should be connected to the low noise preamplifiers (LNAs). If the input impedance of the preamplifier (R_p) is close to a short circuit, the loop current that cause the inductive coupling can be reduced by single stage matching network [62]. The input noise matching network of such LNAs are named preamplifier decoupling. Figure 3-9 shows an example schematic diagram of a preamplifier decoupling circuit typically used for a receive-only coil. When the coil fulfils the three equations below, the coil is matched well to 50Ω and high impedance is present in series, with the loop at the

Larmor frequency.

$$X_{L_C} - X_{C_t} - X_{C_m} = 0, \quad 3-16$$

$$X_{L_m} - X_{C_m} = 0, \quad 3-17$$

$$X_{C_m} = \sqrt{50R_C}, \quad 3-18$$

where L_C , R_C , C_t , C_m , and L_m are the resistance, and the inductance of the loop, capacitance of the capacitor on the loop for the tuning, capacitance, and inductance of the matching components, respectively. The R_p is lower, the decoupling impedance of the loop is higher. The L_m is higher, the blocking impedance of the loop is also higher. However, for cases in which the R_C is very small, the value of L_m will be hard to realise and it also lead to reduce the blocking impedance. In this case, loop can be matched to 100 Ω to increase blocking impedance [68] or can be matched using more matching component to use realisable component [63].

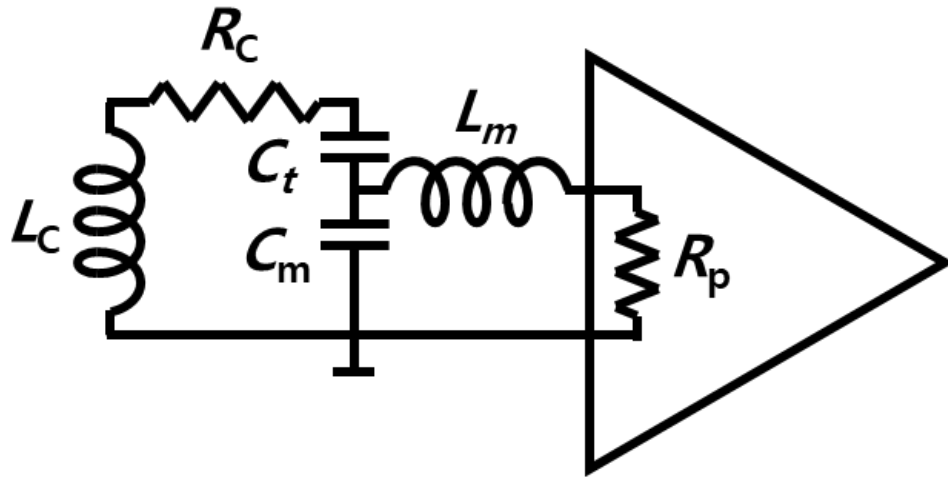


Figure 3-9 A schematic diagrams of the preamplifier decoupling circuits using an LC matching.

3.1.3.2 Quadrature surface coils

When the RF signal is driving a single loop coil, the coil generates a linearly polarised (LP) magnetic field as shown in Figure 3-10(a). The magnitude of the generated magnetic field can be controlled by adjusting the phase of the RF signal (Figure 3-10 (b)). When two coils generate two perpendicular magnetic fields of identical strength and the phase of the magnetic field generated by one coil is shifted by 90° , these coils generate a rotating magnetic field as shown in Figure 3-10 (c). This is known as circular polarisation (CP) or quadrature driving [69]. Since spins are excited by a rotating magnetic field during transmission, half the Tx power is required, compared to a linear polarised coil [70]. During reception, SNR is improved by a factor of $\sqrt{2}$, because random noise rotates both ways while spins only rotate one way [71]. The RF coils which are introduced in section 3.1.3.1.3 can be driven in quadrature.

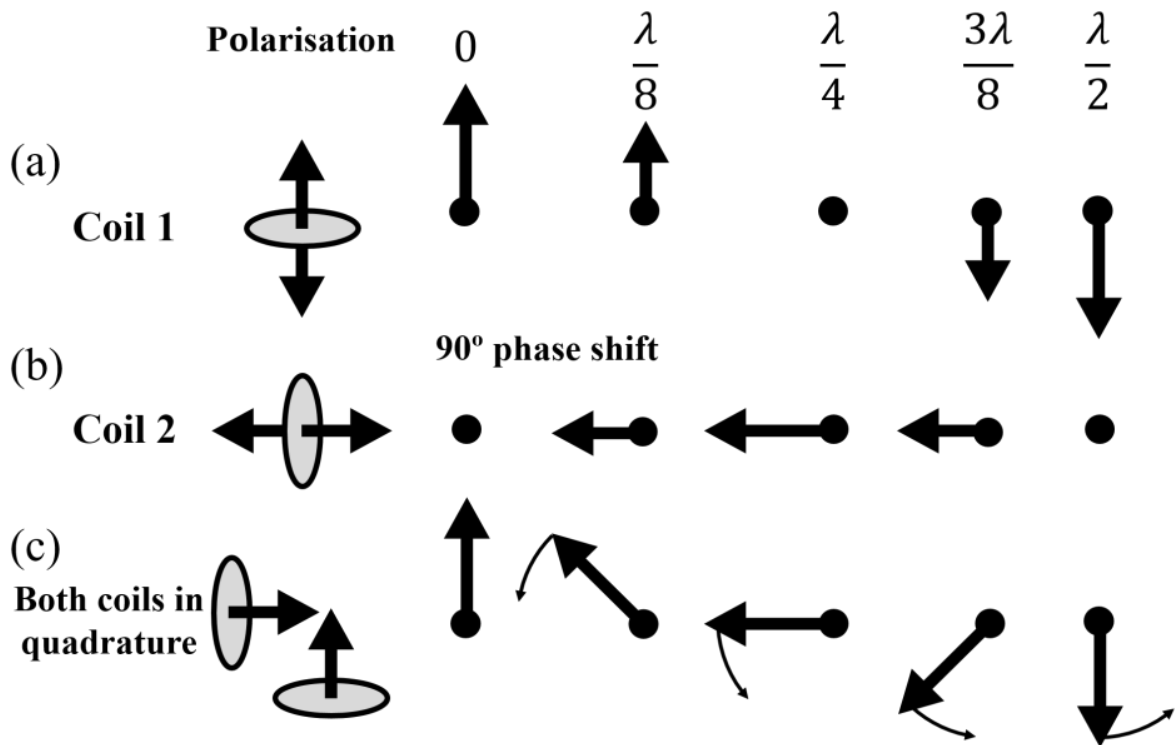


Figure 3-10 Vector diagram of linearly polarised coils placed in quadrature. (a) Linearly polarised coil 1 is driven by a sinusoidal current. (b) Linearly polarised coil 2 is orthogonal and 90° out of phase with coil 1. (c) Both coils driven in quadrature. [69]

3.2 Homogeneous resonators

Many different types of volume coils, such as paired saddle coils [72, 73], Alderman-Grant resonators [74], and birdcage [75] coils, are used for RF transmission and reception since these coils generate a homogeneous B_1 field (Figure 3-11). The homogeneous B_1 field is especially useful for Tx since spins in the region of interest (ROI) can be excited with same flip angle.

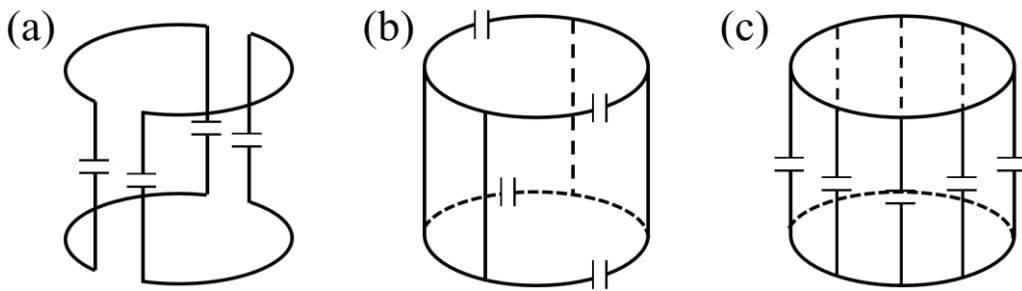


Figure 3-11 Illustration of homogeneous resonators. (a) Paired saddle coil. (b) Alderman-Grant resonator. (c) Birdcage coil.

Among them, the birdcage coil is widely used as Tx head and body coil in clinical MRI today since it provides the most homogeneous field of the realisations listed above [76]. As shown in Figure 3-11(c), the birdcage coil is composed of two circular conductive loops, called end-rings, connected by an even number of conductive straight elements, referred to as rungs or legs. The birdcage coils are grouped into three categories; high-pass, low-pass, and band-pass birdcage coils, containing capacitors in their end-rings, rungs, and both end-rings and rungs, respectively, as shown in Figure 3-12.

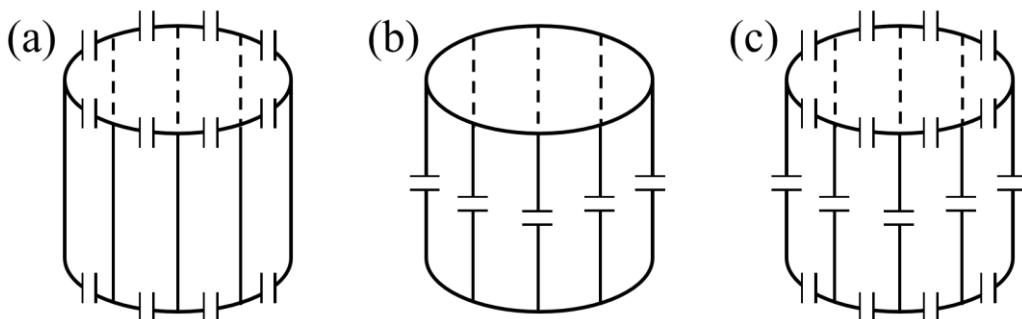


Figure 3-12 Illustration of birdcage coils. (a) High-pass birdcage. (b) Low-pass birdcage. (c) Band-pass birdcage.

Since the high-pass birdcage coil is designed with bigger capacitor values compared to the low-pass birdcage coil [77], high-pass birdcage coils are favoured for field strengths above 1.5T for the human head [30]. The birdcage coil can also easily be used with quadrature excitation and reception which increases SNR by a factor of $\sqrt{2}$ and decreases the required RF power by a factor of two [71].

3.2.1 High-pass birdcage coil

Symmetric N-rung high-pass birdcage coils can be modelled by a high-pass ladder network, a section of which is represented in Figure 3-13 [78]. In the Figure 3-13, C_1 denotes the capacitance of the capacitor connected between the legs, L_R denotes the self-inductance of the rung, and L_E denotes the self-inductance of the part of end-ring used to connect the capacitor.

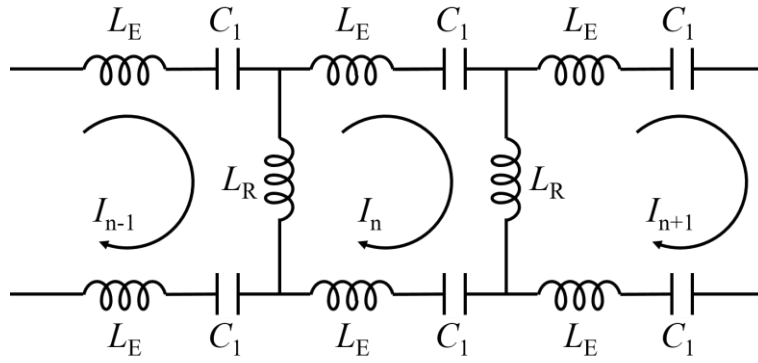


Figure 3-13 A section of the equivalent circuit for high-pass birdcage coil.

According to the Kirchhoff's voltage law, for the middle loop in the Figure 3-13, we have

$$i\omega L_R(I_n - I_{n+1}) + i\omega L_R(I_n - I_{n+1}) + 2i\omega L_E I_n - \frac{2i}{\omega C_1} I_n = 0, \quad 3-19$$

where I_n denotes the current in this loop. This equation can be simplified as

$$L_R(I_{n+1} + I_{n-1}) + 2\left(\frac{1}{\omega^2 C_1} - L_E - L_R\right)I_n = 0, \quad 3-20$$

where n take the integer values zero to $N-1$, and where all indices are modulo N due to the cylindrical symmetry. Consequently, the N linearly independent solutions (called modes) of Eq. 3-20 have the form

$$(I_n)_m = \begin{cases} \cos \frac{2\pi n m}{N}, & m = 0, 1, 2, \dots, \frac{N}{2}, \\ \sin \frac{2\pi n m}{N}, & m = 0, 1, 2, \dots, \frac{N}{2} - 1 \end{cases} \quad 3-21$$

The resonance frequency for each solution is determined from Eq. 3-20 and Eq. 3-21 as

$$\omega_m = \frac{1}{\sqrt{C_1 \left(L_E + 2L_R \sin^2 \frac{\pi m}{N} \right)}}, \quad m = 0, 1, 2, \dots, \frac{N}{2}. \quad 3-22$$

Analysing this result, the N -rung high-pass birdcage coil has $N/2+1$ modes and the highest resonance frequency occurs for the mode $m = 0$ which is the end-ring mode. The mode (called dominant mode) of $m=1$ has the second highest resonance frequency and this mode is capable of producing a uniform magnetic field [46].

3.2.2 Low-pass birdcage coil

A section of the equivalent circuit for a symmetric N -rung low-pass birdcage coil is shown in Figure 3-14. C_2 in Figure 3-14 denotes the capacitance of the capacitor connected between the end-rings.

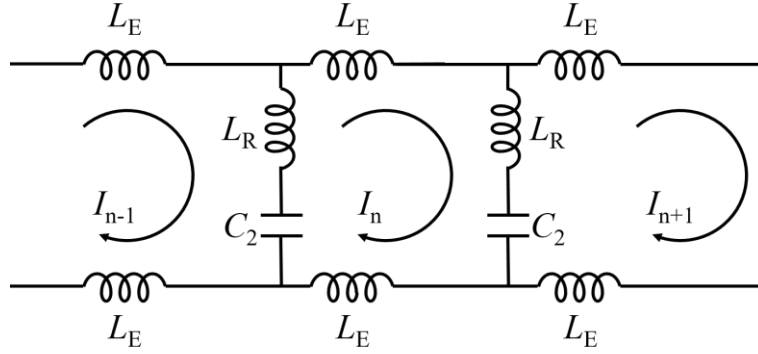


Figure 3-14 A section of the equivalent circuit for low-pass birdcage coil.

According to the Kirchhoff's voltage law, for the middle loop in the Figure 3-14, we have

$$\left(L_R - \frac{1}{\omega^2 C_2}\right)(I_{n+1} + I_{n-1}) + 2\left(\frac{1}{\omega^2 C_2} - L_E - L_R\right)I_n = 0. \quad 3-23$$

The resonance frequency for each solution is determined from Eq. 3-21 and Ep. 3-23 as

$$\omega_m = \frac{1}{\sqrt{C_1 \left(L_E + \frac{L_R}{2 \sin^2 \frac{\pi m}{N}} \right)}}, \quad m = 0, 1, 2, \dots, \frac{N}{2}. \quad 3-24$$

In this case, the resonance frequency for the end-ring resonant mode ($m = 0$) is zero and the dominant mode ($m = 1$) has the second lowest resonance frequency.

3.2.3 Band-pass birdcage coil

A section of the equivalent circuit for a symmetric N-rung band-pass birdcage coil is shown in

Figure 3-15. When $1/C_1$ is set to zero, this structure is the same as a low-pass birdcage coil. On the other hand, this structure is the same as a high-pass birdcage coil when $1/C_2$ is set to zero.

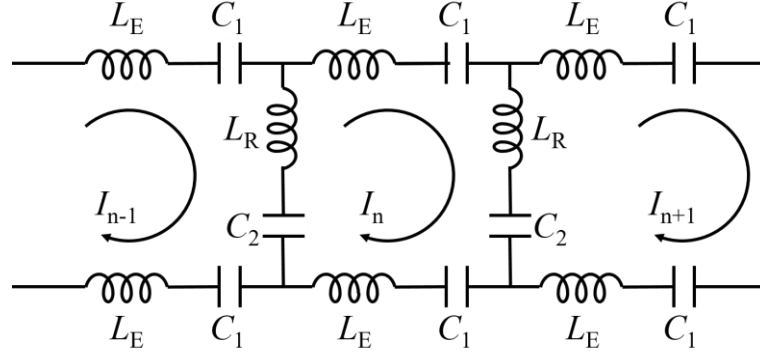


Figure 3-15 A section of the equivalent circuit for band-pass birdcage coil.

According to the Kirchhoff's voltage law, for the middle loop in the Figure 3-15, we have

$$\left(L_R - \frac{1}{\omega^2 C_2}\right)(I_{n+1} + I_{n-1}) + 2\left(\frac{1}{\omega^2 C_1} + \frac{1}{\omega^2 C_2} - L_E - L_R\right)I_n = 0. \quad 3-25$$

The resonance frequency for each solution is determined from Eq. 3-21 and Ep. 3-25 as

$$\omega_m = \sqrt{\frac{\frac{2}{C_2} \sin^2 \frac{\pi m}{N} + \frac{1}{C_1}}{2L_R \sin^2 \frac{\pi m}{N} + L_E}}, \quad m = 0, 1, 2, \dots, \frac{N}{2}. \quad 3-26$$

with the end-ring mode resonance frequency given by

$$\omega_0 = \frac{1}{\sqrt{L_E C_1}}. \quad 3-27$$

The resonance frequency for the dominant mode can be either the second lowest or the second highest.

3.3 Double-tuned resonators

As previously discussed, not only ^1H nuclei but also other nuclei (X-nuclei) with spin $I > 0$ can be used for MR signal detection, for example ^{23}Na , ^{31}P , ^{19}F , ^{13}C , ^{17}O , and so on. Recently, MR imaging with these nuclei has risen in clinical research because of their crucial role in cellular and metabolic processes [5, 79-84]. Since the resonance frequency of the X-nuclei is determined by their gyromagnetic ratio, as shown in Table 3-1 (the sign of the gyromagnetic ratio, γ , determines the sense of precession), a coil tuned at X-nuclei frequency is required for X-nuclei imaging/spectroscopy. However, X-nuclei MR signal is much smaller than that of ^1H due to the low sensitivity and/or natural abundance as shown in Table 3-1 [85].

Table 3-1 Characteristic of important isotopes.

Nuclei	γ (MHz/Tesla)	Natural abundance (%)	Sensitivity relative to ^1H
^1H	42.58	99.985	1
^{19}F	40.05	100	0.83
^{31}P	17.24	100	0.066
^{23}Na	11.26	100	0.093
^{13}C	10.71	1.108	0.016
^{17}O	-5.77	0.037	0.029

Due to the low SNR of the X-nuclei signal, long acquisition times are required and the resulting

image has usually poor spatial resolution. Consequently, an examination of the X-nuclei generally begins with ^1H images (B_0 shimming and localiser image) [6]. Double-tuned RF coil is thus widely used for X-nuclei imaging.

As shown in Eq. 3-7, the resonance frequency of the coil is related with inductance and capacitance of the coil. However, the coil can be resonating at more than one frequency by adding a trap or frequency switching circuit. A multiple-tuned coil can be also constructed using separate coils. In this case, decoupling is also very important since sensitivity of the coil is reduced especially at higher frequency due to the mutual coupling effect between the loops.

3.3.1 Double-tuned coil using lumped element traps

The loop coil can be resonating at two different frequencies when a trap circuit is added in series [17, 86]. The trap circuit can be designed in various ways using a combination of inductors and capacitors. The simplest form of the trap circuit is the LC parallel resonance circuit as shown in Figure 3-16 (a).

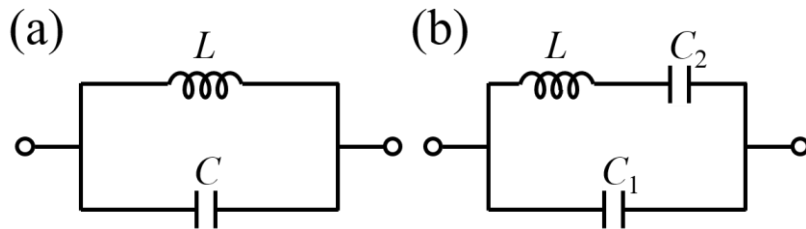


Figure 3-16 Illustration of the equivalent circuit for traps. (a) LC trap. (b) LCC trap.

Figure 3-16 (b) shows another kind of trap circuit, which is called LCC trap [20]. The LC and LCC trap's circuit properties are described below.

3.3.1.1 An LC trapped coil

Figure 3-17 shows the equivalent circuit diagram for an LC trapped loop coil whose impedance can

be calculated [86]:

$$Z = R_L + jX_{L_L} - jX_{C_L} + \frac{X_{L_{t1}}X_{C_{t1}} - jR_{t1}X_{C_{t1}}}{R_{t1} + j(X_{L_{t1}} - X_{C_{t1}})}, \quad 3-28$$

where R_L , L_L , C_L , L_{t1} , C_{t1} and R_{t1} are the resistance, and the inductance of the loop, capacitance of the capacitor on the loop, inductance and capacitance of the trap components, resistance of the L_{t1} , respectively.

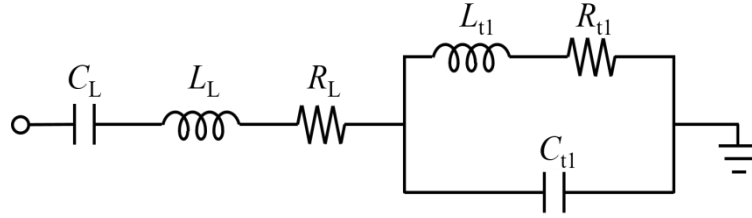


Figure 3-17 Equivalent circuit for an LC trapped, double-tuned, loop coil.

Since resonance condition is reached whenever the imaginary component of the impedance or admittance vanishes, resonance condition can be calculated by

$$X_{L_L} - X_{C_L} - \frac{R_{t1}^2 X_{C_{t1}} + X_{L_{t1}} X_{C_{t1}} (X_{L_{t1}} - X_{C_{t1}})}{R_{t1}^2 + (X_{L_{t1}} - X_{C_{t1}})^2} = 0 \text{ or } \infty. \quad 3-29$$

However, the resistance of the inductor (R_{t1}) can be set to zero for calculation of resonance frequencies since it is usually very small. From the Eq. 3-29, three different resonance frequencies can then be defined as:

$$\omega_1 = \sqrt{\frac{1}{2L_L C_L} + \frac{1}{2L_L C_{t1}} + \frac{1}{2L_{t1} C_{t1}} - \frac{\sqrt{(L_L C_L + L_{t1} C_L + L_{t1} C_{t1})^2 - 4L_L L_{t1} C_L C_{t1}}}{2L_L L_{t1} C_L C_{t1}}}, \quad 3-30$$

$$\omega_2 = \sqrt{\frac{1}{L_{t1} C_{t1}}}, \quad 3-31$$

$$\omega_3 = \sqrt{\frac{1}{2L_L C_L} + \frac{1}{2L_L C_{t1}} + \frac{1}{2L_{t1} C_{t1}} + \frac{\sqrt{(L_L C_L + L_{t1} C_L + L_{t1} C_{t1})^2 - 4L_L L_{t1} C_L C_{t1}}}{2L_L L_{t1} C_L C_{t1}}}. \quad 3-32$$

The impedance is a local minimum at ω_1 and ω_3 . At ω_2 , the impedance of the loop is maximum. A single loop double-tuned coil can be tuned at ω_1 and ω_3 . Since L_L and coil R_L are determined by coil geometry, C_L and C_{t1} can be written as function of ω_1 , ω_3 , and L_{t1} .

$$C_L = \frac{L_{t1} C_{t1} \omega_1^2 - 1}{\omega_1^2 (L_L L_{t1} C_{t1} \omega_1^2 - L_L - L_{t1})}, \quad 3-33$$

$$C_{t1} = \frac{L_L L_{t1} (\omega_3^4 - \omega_1^4) \pm \sqrt{[L_L L_{t1} (\omega_3^4 - \omega_1^4)]^2 - 4L_L L_{t1}^2 \omega_1^2 \omega_3^2 (L_L + L_{t1}) (\omega_3^2 - \omega_1^2)}}{2L_L L_{t1}^2 \omega_1^2 \omega_3^2 (\omega_3^2 - \omega_1^2)}. \quad 3-34$$

If both L_L and R_L are fixed, C_L and C_{t1} have two possible solutions for each value of L_{t1} as shown in Eq. 3-34. In practice, a smaller solution of C_{t1} is used for a LC trapped double-tuned coil because a LC trapped double-tuned coil with a bigger solution of C_{t1} (using plus sign in front of radical) works similarly to the higher frequency single-tuned coil (See Appendix A).

R_{t1} can be calculated from quality factor (Q -factor) and $X_{L_{t1}}$ using Eq. 3-35.

$$R_{t1} = \frac{X_{L_{t1}}}{Q}. \quad 3-35$$

R_{t1} increases with increasing frequency and also increases with increasing inductance value. Figure 3-18 shows the resistance of the air core inductor (Midi spring, Coilcraft, USA) as a function of inductance at 150 MHz as an example.

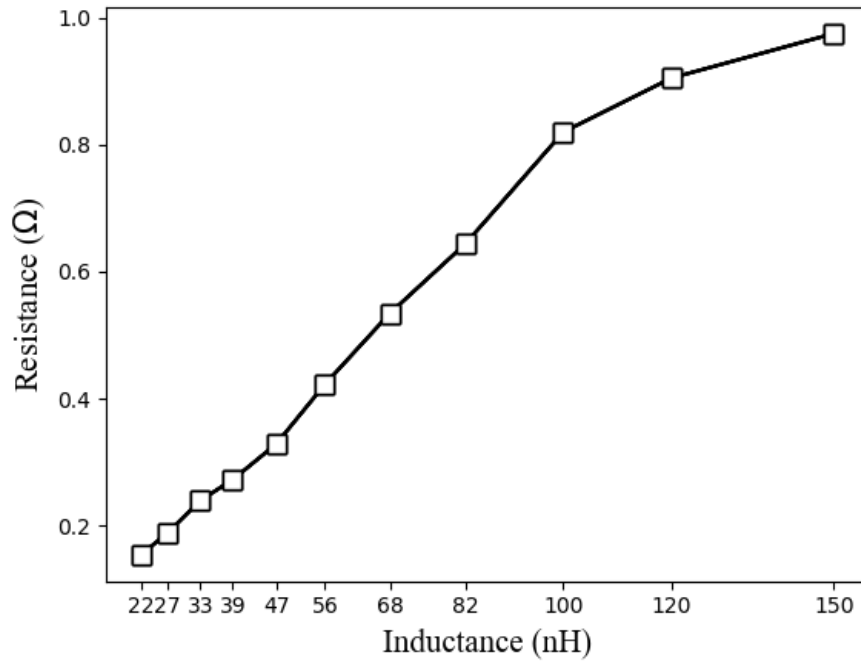


Figure 3-18 Resistance of the inductor as a function of its inductance.

The resistance term of the total impedance is given as

$$R_{tot} = R_L + \frac{R_t X_{C_{t1}}^2}{R_t^2 + (X_{L_{t1}} - X_{C_{t1}})^2}. \quad 3-36$$

As shown in Figure 3-19, the coil sensitivity at lower frequency is higher when lower value of trap inductance is used. The resistance of the coil at ω_1 is lower with lower L_{t1} . The resistance of the coil at ω_3 , on the other hand, is lower when higher value of L_{t1} is used [19].

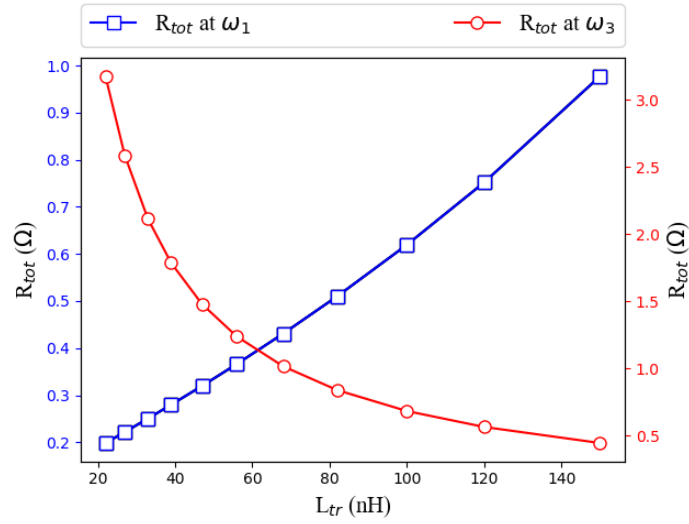


Figure 3-19 Total resistance of an LC trapped coil at two different frequencies as a function of trap inductance, where R_L and L_L are considered as 0.1Ω and 100nH , respectively.

3.3.1.2 An LCC trapped coil

Instead of using LC trap, an LCC trap, as shown in Figure 3-20, can also be used to split the resonance frequency of the coil. This has the advantage of adding one degree of freedom to the design equations.

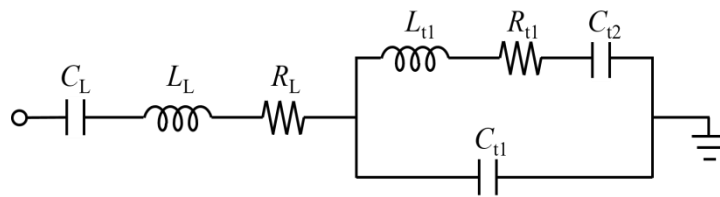


Figure 3-20 Equivalent circuit for an LCC trapped, double-tuned, loop coil.

When $1/C_{t2}$ is set to zero, this structure is the same as a LC trapped coil. From Eq. 3-28 and Eq. 3-29, ' $X_{L_{t1}}$ ' is replaced with ' $X_{L_{t1}} - X_{C_{t2}}$ '. The three different resonance frequencies of LCC trapped coil can be then defined as:

$$\omega_1 = \sqrt{\frac{C_{t1} + C_{t2}}{2L_{t1}C_{t1}C_{t2}} + \frac{C_L + C_{t1}}{2L_L C_L C_{t1}} - \frac{\sqrt{\{L_L C_L (C_{t1} + C_{t2}) + L_{t1} C_{t2} (C_L + C_{t1})\}^2 - 4L_L L_{t1} C_L C_{t1} C_{t2} (C_L + C_{t1} + C_{t2})}}{2L_L L_{t1} C_L C_{t1} C_{t2}}}, \quad 3-37$$

$$\omega_2 = \sqrt{\frac{C_{t1} + C_{t2}}{L_{t1} C_{t1} C_{t2}}}, \quad 3-38$$

$$\omega_3 = \sqrt{\frac{C_{t1} + C_{t2}}{2L_{t1}C_{t1}C_{t2}} + \frac{C_L + C_{t1}}{2L_L C_L C_{t1}} + \frac{\sqrt{\{L_L C_L (C_{t1} + C_{t2}) + L_{t1} C_{t2} (C_L + C_{t1})\}^2 - 4L_L L_{t1} C_L C_{t1} C_{t2} (C_L + C_{t1} + C_{t2})}}{2L_L L_{t1} C_L C_{t1} C_{t2}}}. \quad 3-39$$

Since *LCC* trapped coil should be tuned at ω_1 and ω_3 , C_L and C_{t1} can be written as function of ω_1 , ω_2 , L_{t1} , and C_{t2} .

$$C_L = \frac{L_{t1} C_{t1} C_{t2} \omega_1^2 - C_{t1} - C_{t2}}{\omega_1^2 (L_L L_{t1} C_{t1} C_{t2} \omega_1^2 - L_L C_{t1} - L_L C_{t2} - L_{t1} C_{t2}) + 1} \quad 3-40$$

$$C_{t1} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}, \text{ where}$$

$$a = L_L (\omega_3^2 - \omega_1^2) \{L_{t1}^2 C_{t2}^2 \omega_1^2 \omega_3^2 - L_L C_{t2} (\omega_1^2 + \omega_3^2) + 1\} \quad 3-41$$

$$b = C_{t2} (\omega_3^2 - \omega_1^2) \{2L_L - L_L L_{t1} C_{t2} (\omega_1^2 + \omega_3^2)\}$$

$$c = C_{t2}^2 (L_L + L_{t1}) (\omega_3^2 - \omega_1^2).$$

The resistance term of the total impedance is given as

$$R_{tot} = R_L + \frac{R_t X_{C_{t2}}^2}{R_t^2 + (X_{L_{t1}} - X_{C_{t1}} - X_{C_{t2}})^2}. \quad 3-42$$

As shown in Figure 3-21, the graphs of *LCC* trapped coil resistance are the graphs of *LC* trapped coil resistance shifted to the right and the shifting distance is determined by the choice of C_{t2} .

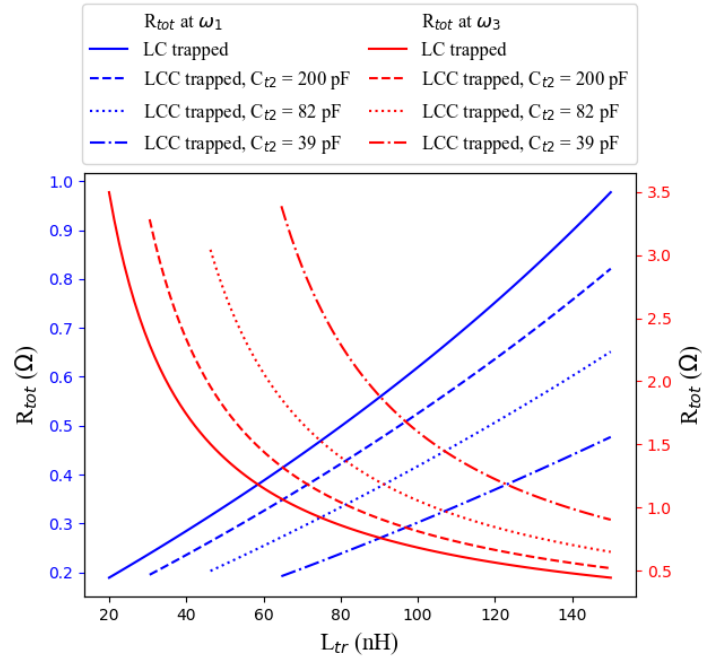


Figure 3-21 Total resistance of an *LCC* trapped coil at two different frequencies as a function of trap inductance, where R_L and L_L are considered as 0.1Ω and 100nH , respectively.

3.3.2 Double-tuned coil using PIN-diode switching unit

PIN-diodes are widely used as active tuning/detuning elements in coil combinations [87, 88] and for T/R switches [89], but can also be used to switch the resonance frequency of an RF coil [21, 22, 90, 91]. Figure 3-22 shows the schematic of the two different PIN-diode switched dual-tuned coils, where C_L , C_{t1} , C_{t2} , L_t , and D are capacitor on the loop, capacitor for tuning, capacitor, and inductor for frequency switching, and PIN-diode, respectively. The PIN-diode switched dual-tuned coil shown in Figure 3-22 has one PIN-diode controlled capacitor or inductor switching network. For the initial frequency operation, the PIN-diode is reverse biased, removing C_{t2} or L_t from the loop. Forward biasing D puts C_{t2} or L_t in parallel with C_{t1} , reducing or increasing the resonance frequency of the loop.

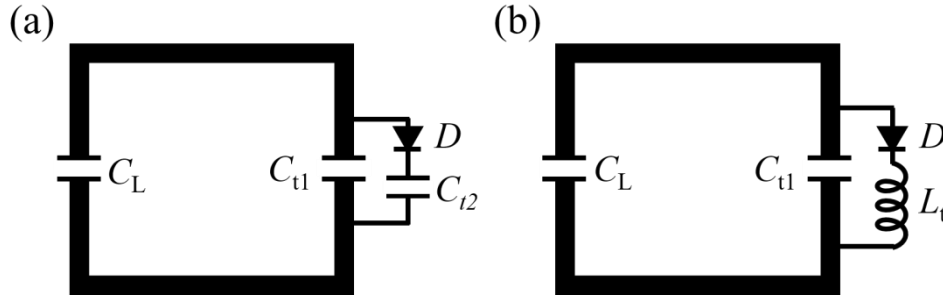


Figure 3-22 PIN-diode switched coils (a) using a capacitor, and (b) using an inductor

However, the PIN-diode appears as a small resistor when the PIN-diode is forward biased. When reverse bias is applied to the PIN-diode, the PIN-diode behaves as a small capacitor. For PIN-diodes used on MR coils, capacitance of the reverse biased PIN-diode is typically in the range 0.5-1.5pF. While not simultaneously resonant at both frequencies, such PIN-diode switched coils allow rapid switching between resonance frequencies under electronic control.

3.3.2.1 PIN-diode switched coil using a capacitor

Equivalent circuits for a PIN-diode switched coil using a capacitor when the PIN-diode is forward and reverse biased are shown in Figure 3-23. When the PIN-diode is forward biased, it presents resistance R_D with series bonding inductance. Since the bonding inductance is usually very small (on the order of 0.10 nH, it can be negligible at frequencies below 1 GHz [92]. Thus, only the forward resistance R_D is considered for below calculations. The higher the forward current is applied to the PIN-diode, the smaller the R_D becomes. The resonance frequency is obtained from Eq. 3-7 using L_L and total capacitance of the circuit. The total capacitance C_{tot} for forward biasing case is given as

$$C_{tot} = \frac{C_L(C_{t1} + C_{t2})}{C_L + C_{t1} + C_{t2}}. \quad 3-43$$

However, the PIN-diode becomes a small capacitance C_D with parallel resistance by applying reverse bias. Since the effect of the parallel resistance is lower than the resistance of a typical solder joint [93],

it is neglected for below calculations. C_{tot} for reverse biasing case is then given as

$$C_{tot} = \frac{C_L \left(C_{t1} + \frac{C_D C_{t2}}{C_D + C_{t2}} \right)}{C_L + C_{t1} + \frac{C_D C_{t2}}{C_D + C_{t2}}} \approx \frac{C_L C_{t1}}{C_L + C_{t1}}. \quad 3-44$$

Since the capacitance of the PIN-diode is small, combined capacitance of C_D and C_{t2} is small enough to ignore for most applications. When the PIN-diode is forward biased, the resonance frequency of the coil is shifted to the lower frequency as shown in the Eq. 3-7, Eq. 3-43 and Eq. 3-44.

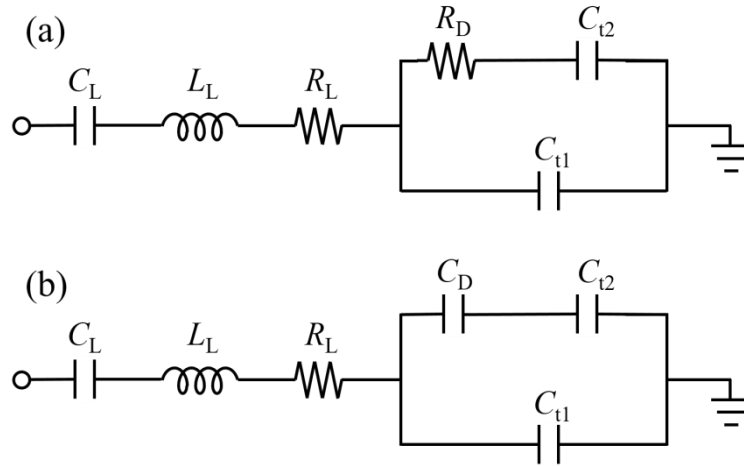


Figure 3-23 Equivalent circuit for PIN-diode switched coil using a capacitor with applied DC bias. When the PIN-diode is (a) forward biased it presents resistance R_D , which is controlled by the DC forward current, while when (b) reverse biased it appears as a capacitance C_D .

3.3.2.2 PIN-diode switched coil using an inductor

Equivalent circuits for a PIN-diode switched coil using an inductor with bias conditions are shown in 3-25.

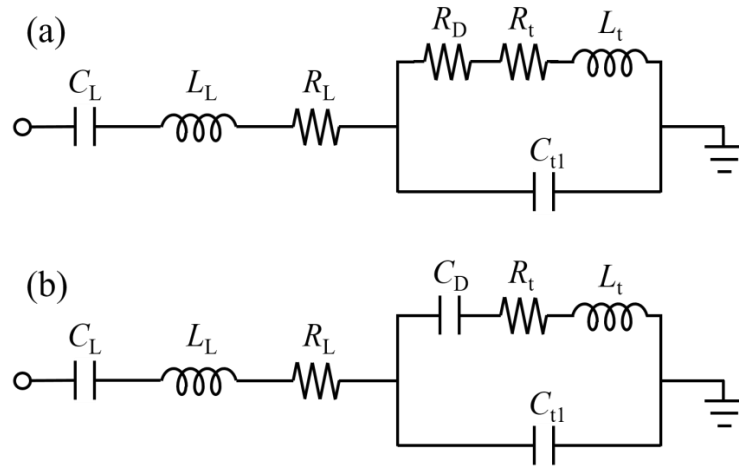


Figure 3-24 Equivalent circuit for PIN-diode switched coil using an inductor with (a) with forward bias, and (b) with reverse bias.

The equivalent circuits using a PIN-diode controlled inductor switching network with forward and reverse bias are the same as LC and LCC trapped coil, respectively. When reverse bias is applied to the PIN-diode, this coil is operating similarly to the single-tuned coil with two capacitors (C_L , C_{t1}) due to the small C_D . Since the coil can be operated similarly to the LC trapped coil with forward bias, the resonance frequency of the coil can be split to both lower and higher frequencies.

3.3.3 Double-tuned birdcage coil

A double-tuned birdcage coil may be designed using either a trap circuit [17, 24] or a PIN-diode [22] as shown in Figure 3-25 (a) and (b), respectively. It is also possible to use a mechanical actuator instead of the PIN-diode to avoid losses from the PIN-diode [94]. Other methods of designing a double-tuned birdcage coil is adding their rungs or end-rings to tune to another frequency. This can be done either by individually tuning alternating rungs to one frequency and the others to the other frequency (Figure 3-25 (c)) [95, 96] or by tuning the end-rings into four-ring birdcage coils (Figure 3-25 (d)) [27]. The length of homogenous region along the main magnetic field axis of optimized four end-ring birdcage coil was compared with single tuned birdcage coil at 3T by Duan et al. [29]. Length of homogenous region along the main magnetic axis reached 96.7% at ^1H and 96.8% at ^{31}P .

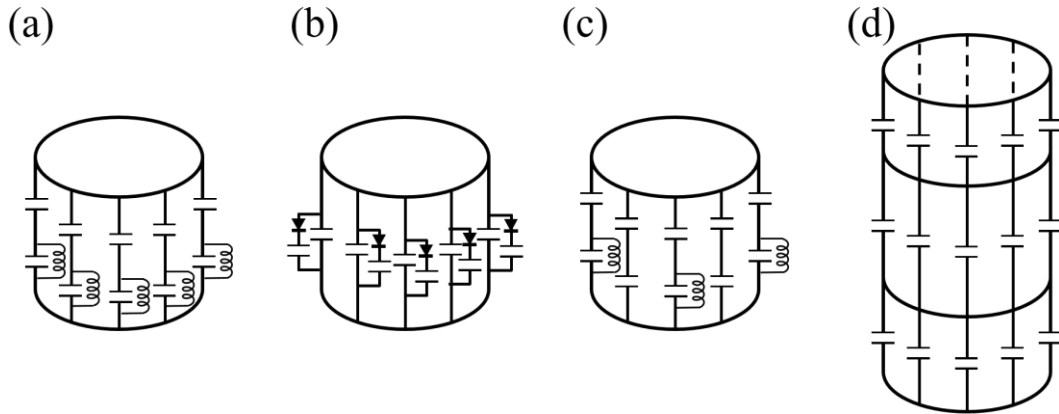


Figure 3-25 Layout of (a) trapped, (b) PIN-diode switched, (c) alternate rungs, and (d) four-ring double-tuned birdcage coil.

However, those double-tuned birdcage coils are not a good solution when two resonant frequencies are very close to each other. There are some other methods to create double-tuned birdcage coil which are operate at ^{19}F (40.052 MHz/T) and ^1H (42.576 MHz/T). One of the popular methods is to split the two degenerate modes by non-uniform distribution of capacitance in the columns of birdcage coil [97]. Another method is inserting an additional shield with a smaller radius on the coil which switches the resonance frequency to ^{19}F frequency [98]. Muftuler et al. suggested an automatically tuned birdcage coil by using variable capacitance (varactor) diodes [99]. And a double-tuned birdcage coil using coupled resonator method was developed for small animal imaging at 4.7T by Hu et al. [100]. It may also be possible to use PIN-diode with inductor to dual-tune the birdcage at ^{19}F and ^1H frequencies [90].

3.3.4 Dual-coil approach

3.3.4.1 Coupling between two separate coils

A double- or triple-tuned coil can be also constructed using separate coils such as two birdcages in co-axial configuration [101-104] and concentric loops [20, 105]. One coil is tuned to ^1H while the other is tuned to the X-nuclei frequency. The coupling effect between the two coils has been explained in section 3.1.3.1. Although each coil resonates at a different frequency, mutual coupling can cause interference between the coils as shown in Eq. 3-14 [102]. When Loop2 shown in Figure

3-5 is resonating lower than the resonance frequency of Loop1, the denominator of I_2 in Eq 3-14 is negative and induces a current in Loop2, being in phase with the current in Loop1 (capacitive reactance is added into Loop1).

$$I_2 = \left| \frac{X_M}{R_2 + jX_{C2} + jX_{L2}} \right| I_1. \quad 3-45$$

When Loop2 is tuned at a higher frequency compared to Loop1, the denominator of I_2 is positive and the current induced in Loop2 is 180 degrees out of phase with the current in the Loop1 (inductive reactance is added into Loop1).

$$I_2 = - \left| \frac{X_M}{R_2 + jX_{C2} + jX_{L2}} \right| I_1. \quad 3-46$$

The counter-rotating currents can cause significant loss in the magnetic field intensity [102]. Thus, it is also important to decouple the signal (especially at higher resonance frequency) for the double-tuned coil. The geometric decoupling which has described in section 3.1.3.1.2 can also be applied for double-tuned coils [54, 106-110].

3.3.4.2 Decoupling using lumped element traps

The characteristic of the trapped coil has been explained in section 3.3.1. Although the performance of the circuit with trap is lower than that of the circuit without trap, as shown above, traps can be used on the lower frequency coil to improve performance at the higher frequency when separate coils are used [20, 105, 111]. However the traps are not necessary for the higher frequency coil since the mutual coupling is much less at lower frequency and induced current is in phase with that of self-current [112]. For the purpose of decoupling, the trapped coil is tuned to X-nuclei Larmor frequency with low impedance and tuned to ^1H Larmor frequency with high impedance while the other coil is tuned to ^1H without trap. Figure 3-26 shows two double-tuned coils using two concentric loops which

are decoupled using *LC* trap and *LCC* trap.

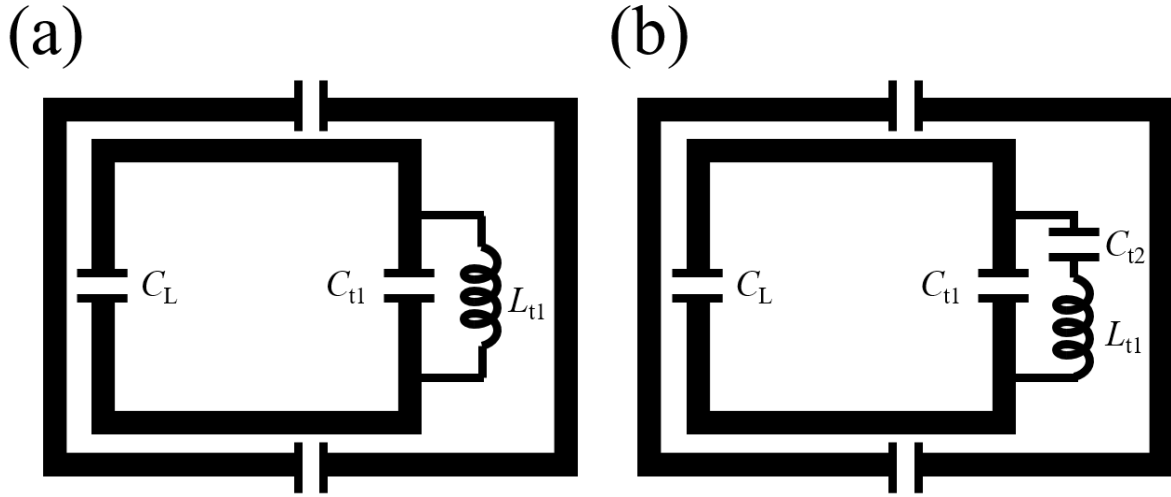


Figure 3-26 Double-tuned coils using concentric loops with (a) *LC* trap and (b) *LCC* trap.

The outer coil is a higher frequency single-tuned coil while the inner coil is tuned at lower frequency with high impedance at higher frequency. For the *LC* trapped coil, the required value of C_L and C_{t1} can be written as function of ω_1 , ω_2 , L_{t1} , and inductance of the loop L_L .

$$C_L = \frac{L_{t1}C_{t1}\omega_1^2 - 1}{\omega_1^2(L_L L_{t1}C_{t1}\omega_1^2 - L_L - L_{t1})}, \quad 3-47$$

$$C_{t1} = \frac{1}{\omega_2^2 L_{t1}}. \quad 3-48$$

For the *LCC* trapped coil, the required value of C_L and C_{t1} can be calculated using ω_1 , ω_2 , L_{t1} , L_L , and C_{t2} .

$$C_L = \frac{L_{t1}C_{t1}C_{t2}\omega_1^2 - C_{t1} - C_{t2}}{\omega_1^2(L_L L_{t1}C_{t1}C_{t2}\omega_1^2 - L_L C_{t1} - L_{t1}C_{t2} - L_{t1}C_{t2}) + 1}, \quad 3-49$$

$$C_{t1} = \frac{C_{t2}}{\omega_2^2 L_{t1} C_{t2} - 1}.$$

3-50

Figure 3-27 shows the total resistance of the *LC* and *LCC* trapped coil at two different resonance frequencies.

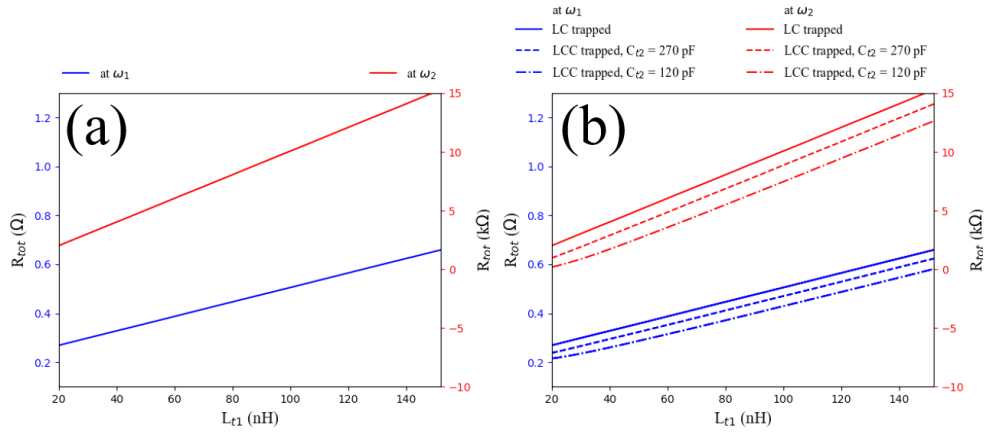


Figure 3-27 Resistance of (a) *LC* trapped coil and (b) *LCC* trapped coil at two different frequencies as a function of trap inductance when coil traps are used for blocking.

The total resistance of the *LCC* trapped coils at both ω_1 and ω_2 is lower than that of *LC* trapped coil. However, the total resistance of those coils at ω_2 is usually high enough to decouple the higher frequency signal. Therefore, *LCC* trapped coils may be a better choice to be used instead of the *LC* traps [20, 48].

3.4 RF chain

3.4.1 Preamplifier

The total noise figure of the receive system N_F is calculated by Friis formula for noise factor [113].

$$N_F = N_1 + \frac{N_2 - 1}{G_1} + \frac{N_3 - 1}{G_1 G_2} + \frac{N_4 - 1}{G_1 G_2 G_3} + \cdots + \frac{N_n - 1}{G_1 G_2 \cdots G_{n-1}}, \quad 3-51$$

where N_i and G_i are the noise figure and power gain, respectively, of the i -th stage, and n is the number of stages. The MR signal received by RF coil is at microvolt order of magnitude. If the received MR signal is transported directly out of the magnet room through a long cable without pre-amplification, the signal is largely attenuated, leading to a serious loss of SNR. A preamplifier is generally placed close to the receive coil (e.g. G_1 and N_1). When the preamplifier has high gain and low noise figure, N_F is lower. The input impedance of the preamplifiers used in MRI systems is usually 50 Ω . However, preamplifiers with either high or low input impedance are used with array coils for decoupling purposes, as described in section 3.1.3.1.4.

3.4.2 T/R switch

The RF coil can be used not only for Tx-only or Rx-only, but also for transceiver. The primary purpose of a T/R switch is to physically connect the RF coil either with the power amplifier in Tx or with the preamplifier in Rx. When one RF coil is used for both Tx and Rx, the preamplifier can be permanently damaged by the transmit signal which is (typically eight orders of magnitude) higher than the receive signal. A transmit and receive (T/R) switch is usually designed to protect the preamplifier from the RF transmit power [89, 114]. The PIN-diodes are typically used for T/R switch employed in MRI systems. Figure 3-5 shows an example of the T/R switch.

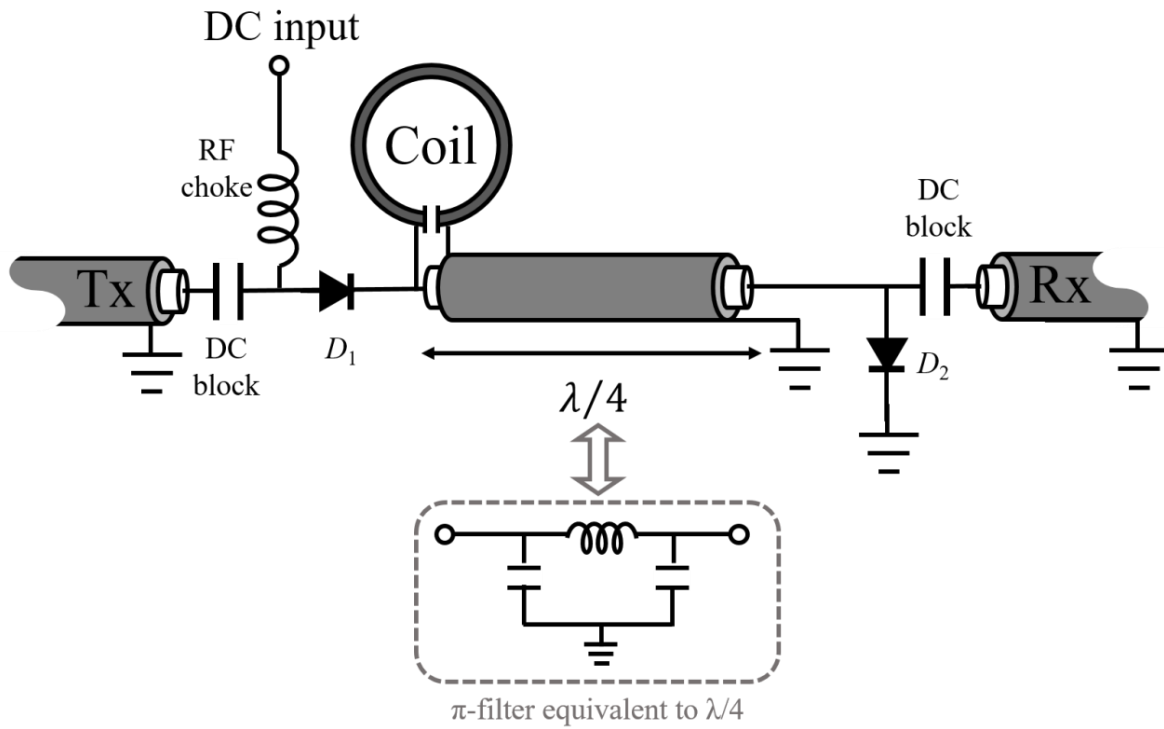


Figure 3-28 $\lambda/4$ T/R-switch using PIN-diodes

During transmission, the PIN-diodes are forward biased to close PIN-diode switch. Closing D_1 connects the Tx port to the coil, while the Rx port is isolated via a quarter-wavelength transmission line by closing D_2 . Otherwise, both PIN-diode switches are opened by applying direct current (DC) reverse voltage. In this way, the MR signal detected by the coil can pass to the preamplifier. The quarter-wavelength transmission line can be replaced by a lumped element π -filter which provides phase shift of $\lambda/4$.

3.4.3 Hybrid coupler

The hybrid coupler is widely used for power dividing and combining. This coupler can be designed using not only transmission line (micro-strip / strip-line) but also a lumped element as shown in Figure 3-29 [115].

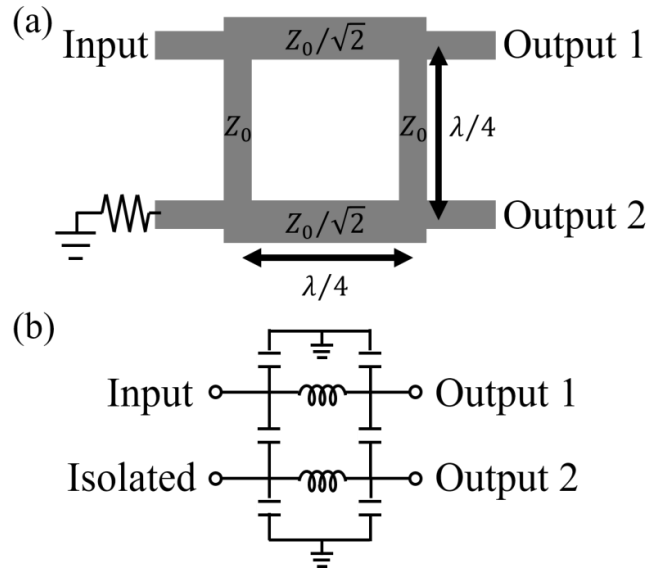


Figure 3-29 (a) Geometry of the hybrid coupler and (b) its example schematic using lumped elements.

The hybrid coupler is widely used for Tx quadrature coils since the input power is split equally (3dB coupler) between two output ports, with a 90° phase difference. The isolated should be terminated by a 50Ω load to absorb power reflected from the coil terminals. This hybrid coupler is also commonly used for Tx/Rx quadrature coil. During Rx, MR signals come from the two output ports and a combined signal can be detected at the isolated port. In this case the input port in Figure 3-29 should be terminated by a 50Ω load. To make this possible, the hybrid coupler is combined with a T/R switch, as shown in Figure 3-29 [89, 106].

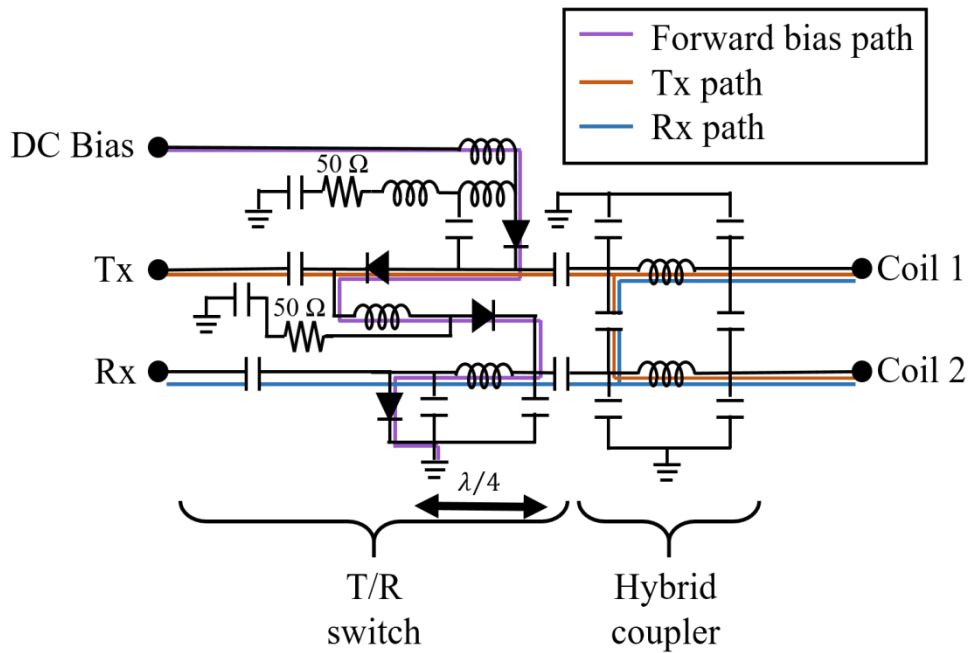


Figure 3-30 Schematic diagram of quadrature T/R switch.

When all the PIN-diodes are conducting by applying a forward DC bias current, the Rx port is isolated via a quarter-wavelength transmission. Signal input from the Tx port is then divided into two coil ports (coil 1, coil 2) with 90° phase difference. During Rx, all the PIN-diodes are reverse biased, input signals from both coil ports are combined and transmitted to the Rx port.

3.5 Evaluation of the coil and the RF chain

3.5.1 S-parameter

Scattering parameters (S-parameters) describe reflection and transmission characteristics (amplitude and phase) in frequency domain are arranged in a complex valued matrix. For high frequencies, it is convenient to describe a given network using S-parameters rather than voltages, currents and impedances [116]. Measuring voltages and currents is often problematic because the measurement probe affects the measurement itself. Figure 3-31 shows an example of a two-port network.

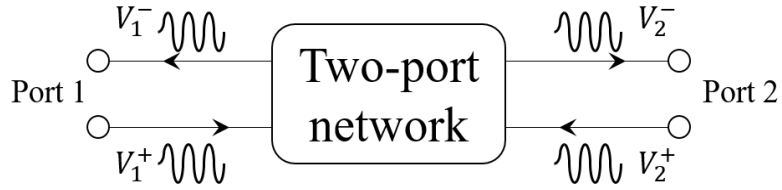


Figure 3-31 An example of a two-port network.

V_1^- , V_1^+ , V_2^- , and V_2^+ in Figure 3-31 are voltage waves which are traveling waves normalised by the characteristic impedance Z_0 of the transmission line. The square of the magnitude of these variables has the dimension of voltage. The relationship between the reflected, incident voltage waves and S-parameter is given by [117]

$$\begin{pmatrix} V_1^- \\ V_2^- \end{pmatrix} = \begin{pmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{pmatrix} \begin{pmatrix} V_1^+ \\ V_2^+ \end{pmatrix}. \quad 3-52$$

If the voltage wave applied from port 1 () is considered, it waves exiting from either port 1 itself () or port 2 (). However, according to the definition of S- parameters, port 2 is terminated in a load identical to the system impedance. Therefore, will be absorbed making equal to zero as

$$S_{11} = \frac{V_1^-}{V_1^+}, S_{21} = \frac{V_2^-}{V_1^+}. \quad 3-53$$

Similarly, if port 1 is terminated in the system impedance then becomes zero, giving

$$S_{12} = \frac{V_1^-}{V_2^+}, S_{22} = \frac{V_2^-}{V_2^+}. \quad 3-54$$

The two-port S-parameters have the following generic descriptions:

S_{11} and S_{22} are the input and output reflection coefficient of the network circuit, respectively.

S_{21} and S_{12} are the forward and reverse voltage gain, respectively.

3.5.1.1 Detection of the resonance frequency of RF coil

Figure 3-32 shows the detection of the resonance frequency of an RF coil using two decoupled “sniffer loops” [118].

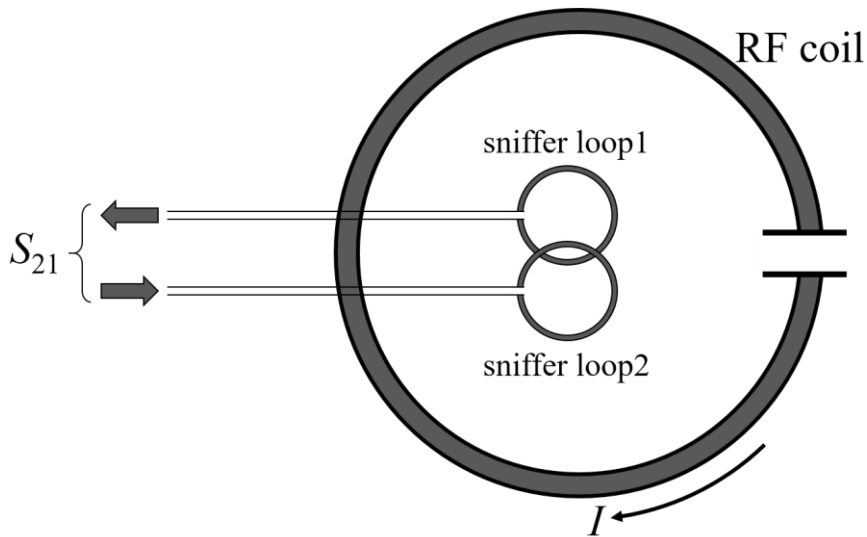


Figure 3-32 Detection of resonance frequency using two decoupled sniffer loops.

The RF coil absorbs most of the power radiated from sniffer loop2, at its resonance frequency. An output voltage is then transmitted to sniffer loop1 due to the current I in the RF coil. Thus, S_{21} measured with two sniffer loops has a peak at the resonance frequency of the RF coil [52, 119].

3.5.1.2 Q-factor measurement

The Q -factor relates the total magnetic field energy in the RF coil (the reactance) to the energy dissipated (the resistance) during each cycle of oscillation, meaning that it is a ratio of resonance

frequency f ($f = \omega/2\pi$) to bandwidth BW [120].

$$Q = \frac{f}{BW}. \quad 3-55$$

The Q-factor can also be measured using two sniffer loops with a S_{21} measurement. On the display of the network analyser, the typical Lorentzian shape is visible, where the frequency is on the horizontal axis, and the log of the transmission sensitivity on the vertical axis. To get the Q-factor, the centre frequency of the peak (resonance frequency) is divided by -3dB bandwidth [120].

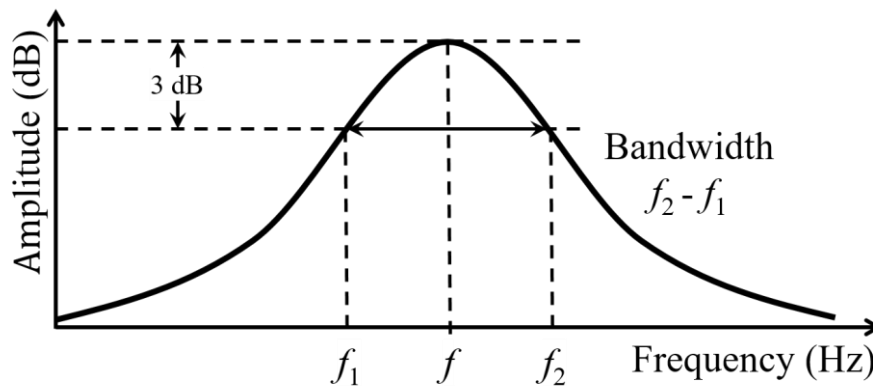


Figure 3-33 Q-factor measurement points in the transmission coefficient (S_{21}).

As the bandwidth is taken between two -3dB point, it is given as

$$BW = \frac{R}{2\pi L}. \quad 3-56$$

Thus, the Q-factor can be expressed in terms of circuit parameters as [121]

$$Q = \frac{\omega L}{R}. \quad 3-57$$

3.5.2 SNR measurement

Signal to noise ratio (SNR) is in general a power ratio. In imaging it conveys an intensity ratio [122], expressed as

$$\text{SNR} = \frac{\bar{S}}{\sigma}. \quad 3-58$$

where $\bar{S}$ is mean signal intensity from the phantom ROI and σ is the noise which is estimated from the standard deviation from an artefact-free background ROI [123]. If a magnitude image is evaluated as shown in Figure 3-34, the image noise will not be Gaussian distributed, but rectified to a Rician distribution.

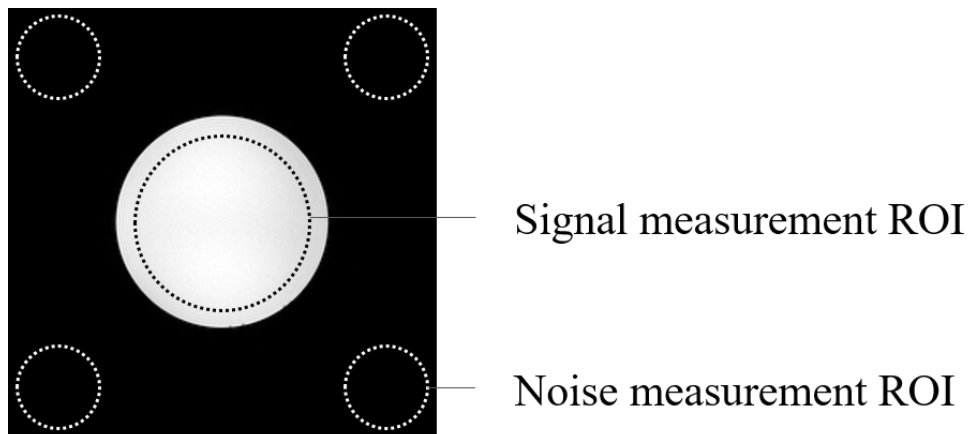


Figure 3-34 A schematic showing signal and background noise measurement ROI.

The change of noise distribution has to be compensated for as the SNR measure assumes the noise is

Gaussian distributed. For a single-channel system the correction factor is [122-124]

$$\sigma = \frac{SD}{0.66}. \quad 3-59$$

When computing the noise standard deviation from multi-channel magnitude reconstructed images, the correction factor is higher than that of single-channel system (e.g. 0.70 for 12-channel system and 0.71 for 32-channel system) [125]. Note that the above definitions hold for non-accelerated imaging in the context of MR. If receive acceleration (e.g. subsampling of the receive k -space) is employed the reconstructed images display an inhomogeneous noise distribution. In this case voxel based SNR calculation have to be carried out [126].

Chapter 4

DEVELOPMENT OF A PIN-DIODE CONTROLLED, QUADRATURE-ENHANCED, DOUBLE-TUNED RF COIL FOR SODIUM MRI

4.1 Introduction

For the heterogeneous resonator, a double-tuned coil can be designed with the use of two geometrically isolated coils whose B_1 field directions are perpendicular to each other as described in the section 3.1.3.1.3 and 3.3.4.1. For example, one coil can be a loop tuned to ^1H and the other tuned to the other nucleus, can be a butterfly [106, 107], a double-loop [108], a monopole antenna [109], or micro-strip transmission line [110, 127-129]. The advantage of this concept is that it does not require additional lossy components for decoupling, such as traps or PIN-diodes; therefore, there is no SNR degradation. However, the double-tuned coil with this concept can only work as a linear mode and not as a quadrature mode.

In this chapter, a quadrature-enhanced, double-tuned coil is designed and implemented. The proposed novel concept uses PIN-diode switches, inserted only in the loop coil, to shift the resonance frequency between ^1H and ^{23}Na so that ^{23}Na signal can be acquired in quadrature while the capability of using ^1H scan remains.

4.2 Concepts

4.2.1 A quadrature-enhanced dual-tuned coil

Figure 4-1 (a) shows a double-tuned coil which consists of a frequency-switchable $^1\text{H}/^{23}\text{Na}$ loop

coil (inner dimension: 28 mm $\times$ 22 mm, track width = 3 mm) and single-tuned ^{23}Na butterfly coil (inner dimension: 49 mm $\times$ 22 mm, coil path width = 3 mm). These two coils were carefully overlapped to isolate B_1 fields in a geometrically decoupled solution. Three non-magnetic PIN-diodes (MA4P7418NM-1072, MACOM, USA), two for tuning and one for matching, together with a capacitor in series were inserted only in the loop coil, in order to enable the loop coil to function at dual frequencies.

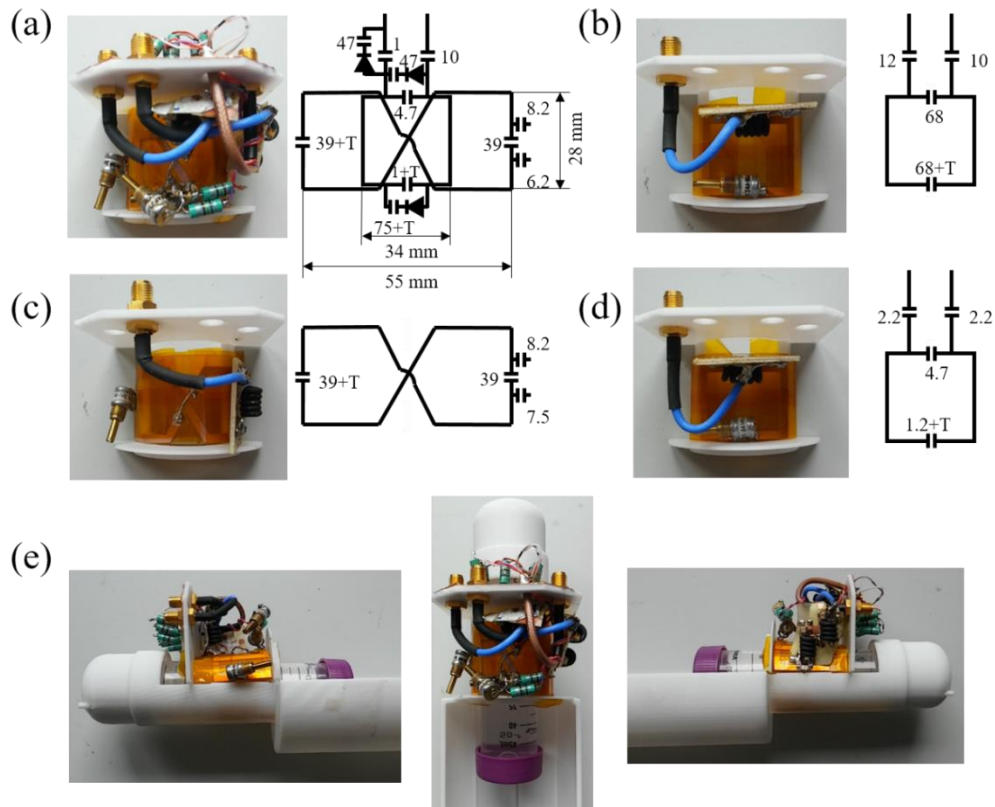


Figure 4-1 A photo and schematic diagram containing component values of (a) a quadrature-enhanced dual-tuned coil, (b) ^{23}Na single-tuned loop coil, (c) ^{23}Na single-tuned butterfly coil, and (d) ^1H single-tuned loop coil. (e) A photo of the double-tuned coil assembled with an animal bed and a phantom. Capacitor values are given in picofarads, with the suffix +T indicating the addition of a trimmer capacitor in parallel with the fixed capacitor [130].

Three PIN-diodes on the loop coil were connected to a PIN-diode control unit shown in Figure 4-3 and described in section 4.2.3 via $2.7 \mu\text{H}$ RF chokes. The butterfly ^{23}Na coil maintained its single-tuned formation without any loops. Using this configuration, the ^{23}Na coil can be driven in quadrature but in a linear manner for the ^1H . Figure 4-1 (e) shows the double-tuned coil photo assembled with

the animal bed and with a 45 mM NaCl doped water phantom. In order to compare the performance of the double-tuned coil, single-tuned reference coils (Figure 4-1 (b), (c), (d)) were also built with identical type (loop or butterfly) and geometry on a precisely duplicated 3D-printed former.

4.2.2 PIN-diode control unit

PIN-diodes are often used with arranged in series or parallel, when many of PIN-diodes need to be driven from a limited number of DC bias outputs [88, 131]. When PIN-diodes are connected in parallel (Figure 4-2 (a)), same reverse voltage is applied to the each diode but forward current divides into each diode due to the characteristics of a parallel circuit. For the many number of channels, phased array, receive-only coil, detuning diodes are often connected in parallel due to the limited bias source from the scanner [88]. Small forward bias current is still enough to detune the loop.

When the PIN-diodes are connected in series (Figure 4-2 (b)), the same forward current is applied to each diode. The PIN-diodes are often used in its series connection to tune the transmit-only coil [131] since the PIN-diode resistance is inversely proportion to the applied forward current [23]. However, the reverse voltage to each PIN-diode is not applied to the each PIN-diode properly. One of the solutions to apply same reverse voltage to each PIN-diode is adding resistors with high resistance (e.g. 10 k Ω) in parallel to the PIN-diodes as shown in Figure 4-2 (c). Applied reverse voltage then divides equally across each PIN-diode.

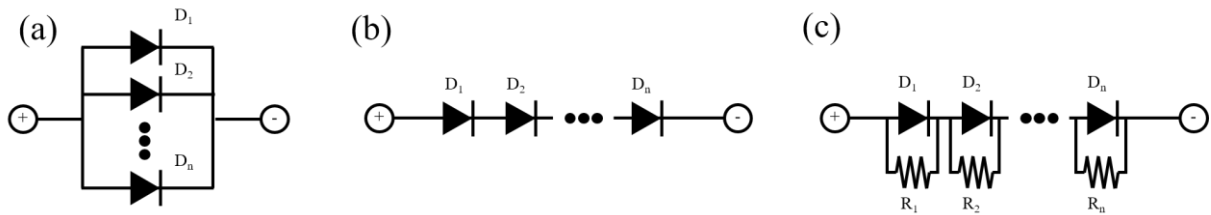


Figure 4-2 Schematic diagram of PIN-diodes connection. (a) Parallel, (b) series connection, and (c) series connection with parallel resistors.

Thus, the PIN-diode control unit (Figure 4-3) was also designed and attached on the coil former, which enables a DC bias either current (100 mA) or voltage (-30V) to be contributed equally to the individual PIN-diode in the double-tuned coil [132]. As described in Figure 4-3 (a) and (b), the

control unit includes two NPN silicon transistors ($TR_{1,2}$, BC847B, Infineon, Germany), four silicon Schottky diodes (D_{4-7} , BAS70-04, Infineon, Germany), and three non-magnetic chip resistors (RCWP1206-1C, Vishay, USA). Figure 4-3 (b) and (c) shows the operation of the unit. When forward biased, the transistors are turned on and allow the applied forward current (100 mA) to flow to the serially connected PIN-diodes but not through the Schottky diodes. In contrast, when reverse biased, the transistors are not activated, but all Schottky diodes are turned on, allowing all PIN-diodes to be connected in parallel with reverse voltage. A 47 k Ω resistor is added between the Schottky diodes because the Schottky diodes are turned on by small current (1 mA). By measuring applied forward current and reverse voltage using a multimeter (87-V, Fluke, USA) and a bench-top power supply (72-6905, Tenma, OH, USA), correct operation of the unit was confirmed. When 100 mA of forward current is applied to the PIN-diodes through a PIN-diode control unit, 100 mA of current flow is observed from each diode. When -30 V of reverse voltage is applied from the power supply, 28.7 V of reverse voltage is applied on each PIN-diode. Although the transistors are slightly magnetic, the PIN-diode control unit was placed at some distance away from the coil and object. Nevertheless, the unit can be relocated further away from the coil or outside the magnet bore since it functions to distribute DC for the PIN-diodes. Importantly, there was no sign of any interference on the images caused by these transistors.

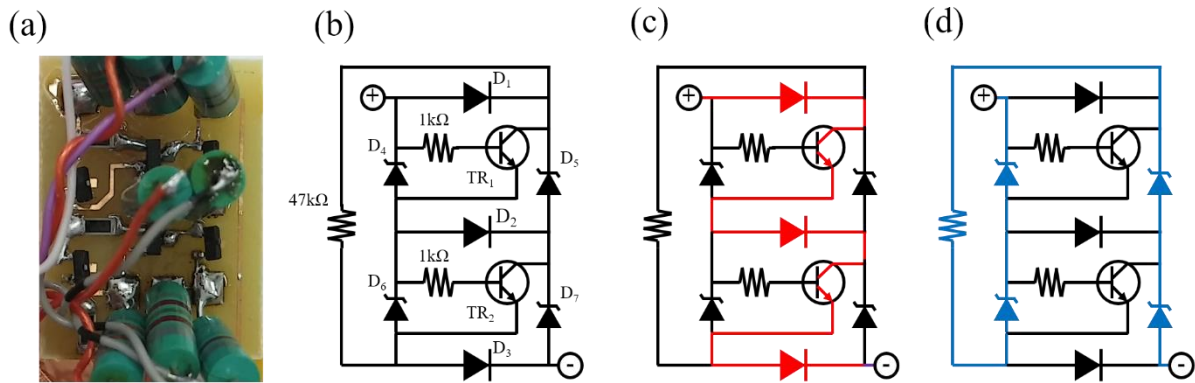


Figure 4-3 (a) A photograph and (b) schematic circuit diagram of the PIN-diode control unit for three PIN-diodes and (c) its operation with forward current and (d) with reverse voltage [130].

4.2.3 T/R switch for the quadrature-enhanced coil

Since sodium works in quadrature while proton works in linear, it was necessary to design an RF interface capable of providing the correct signals to the individual coil elements. Figure 4-4 shows a

circuit diagram including component values (Figure 4-4 (b)) and a photo (Figure 4-4 (c)) of the home-built RF interface, which was used in all MR measurements. This unit was designed using Eagle software (Autodesk, USA) and was etched on a double-sided printed circuit board (FR4, copper thickness of 35 μm). Since the $^1\text{H}/^{23}\text{Na}$ double-resonant loop coil was connected only to a single feed point, a ^{23}Na trap for ^1H path was added at the input port of the T/R switch so that the ^{23}Na trap filters ^{23}Na signal out and only allows ^1H signal pass. Two ^1H traps were attached at both inputs of sodium quadrature hybrid. One was used for the same purpose, to block ^1H signal allowing only ^{23}Na signal to pass through, and the other was used to keep the same phase delay between ^{23}Na butterfly coil and hybrid coupler. Tx signal can be routed to either the ^1H or ^{23}Na units by switching the PIN-diode in the both ^1H and ^{23}Na T/R switch unit. A ^{23}Na hybrid coupler, pre-amplifiers for both nuclei (WanTcom Inc., USA) and pre-amplifier protection circuits were also included in the RF interface. This unit was then evaluated by measuring Tx and Rx isolation of ^1H and ^{23}Na , and the phase difference between ports of ^{23}Na using the network analyser.

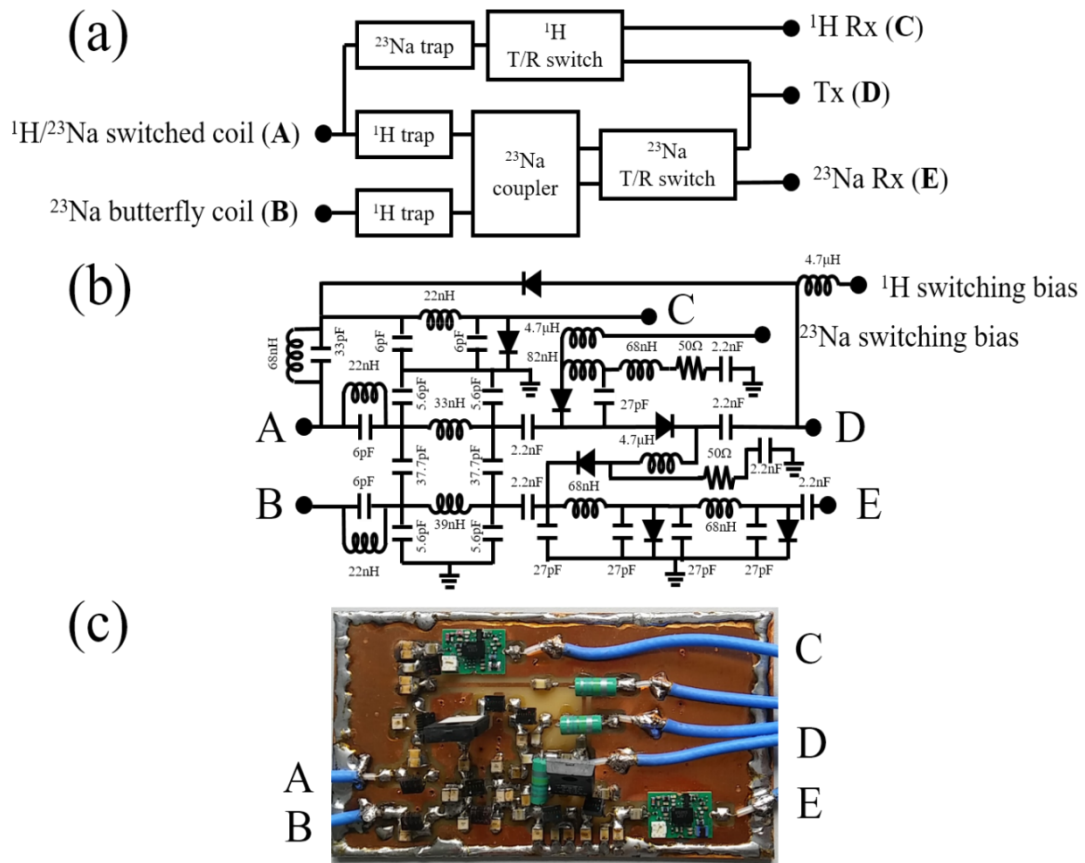


Figure 4-4 (a) Block diagram, (b) schematic diagram, and (c) photo of the RF interface [130].

4.3 Methods

The return losses (S_{11}) of these coils and the loaded/unloaded Q -factors were measured using a vector network analyser (ZNB4, Rohde & Schwarz, Germany) on the bench for comparison. S-parameters and phase differences of the suggested T/R switch with ^{23}Na hybrid coupler are also measured using a network analyser.

4.4 Result

Table 4-1 is summary of measured S_{11} , unloaded and loaded Q -factors, and Q -ratio of all coils evaluated in this work.

Table 4-1 Return loss (S_{11}) and Q -factor (Unloaded/Loaded/Ratio) of the coils [130].

			S_{11} (dB)	Q_{Un}	Q_L	Q_{Ratio}
^1H	Single-tuned	Loop	-18.19	111	75	1.48
	Double-tuned	Loop	-15.75	78	59	1.32
^{23}Na	Single-tuned	Loop	-27.94	175	148	1.18
		Butterfly	-22.48	186	162	1.15
	Double-tuned	Loop	-20.09	117	105	1.11
		Butterfly	-19.40	159	139	1.14

The captured network analyser screens of measured return losses (S_{11}) are shown in Figure 4-5. It was found that the return losses of all coils were below -15 dB, indicating excellent tuning condition of all coils to their corresponding Larmor frequencies (400.0 MHz for ^1H and 105.7 MHz for ^{23}Na). Q -factors of the double-tuned coil were found to be comparable with those of single-tuned coils, although there seem to be a small reduction of the Q -factor in the double-tuned loop coil.

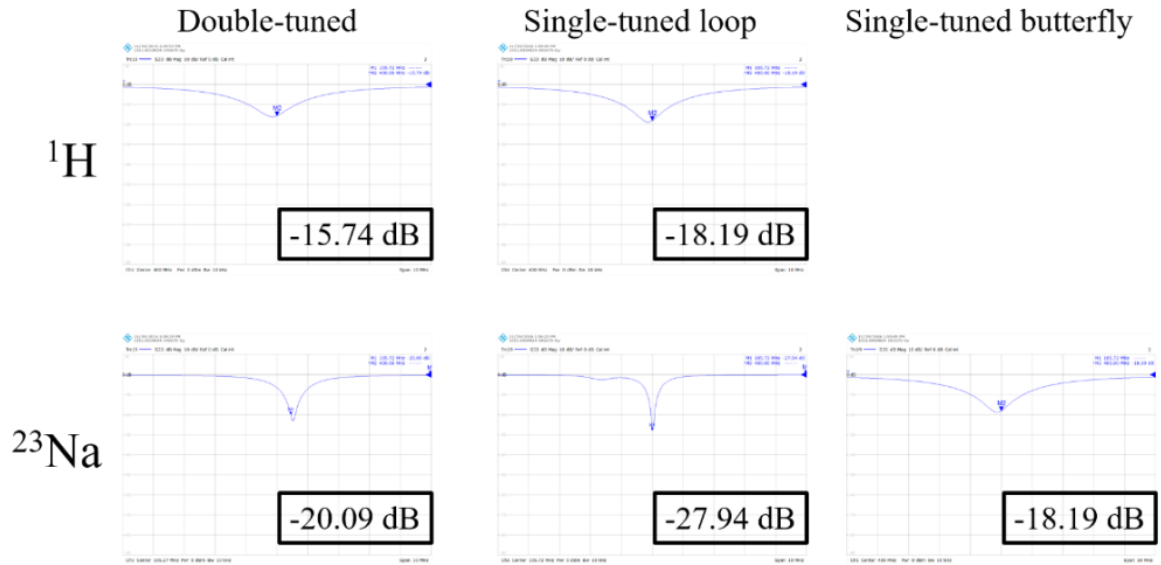


Figure 4-5 Measured return losses of the double-tuned coil (left column), single-tuned loop coil (middle column) and single-tuned butterfly coil (right column) at 400 MHz (top row) and 105.7 MHz (bottom row).

The characteristics of the RF interface unit were evaluated in Figure 4-6, captured on the network analyser screen, displays Tx and Rx isolation between the two ports labelled as A-E in Figure 4-4. S_{AD} , for example, denotes the signal loss detected on port A, coming from port D. Phase differences between two ^{23}Na ports were also provided.

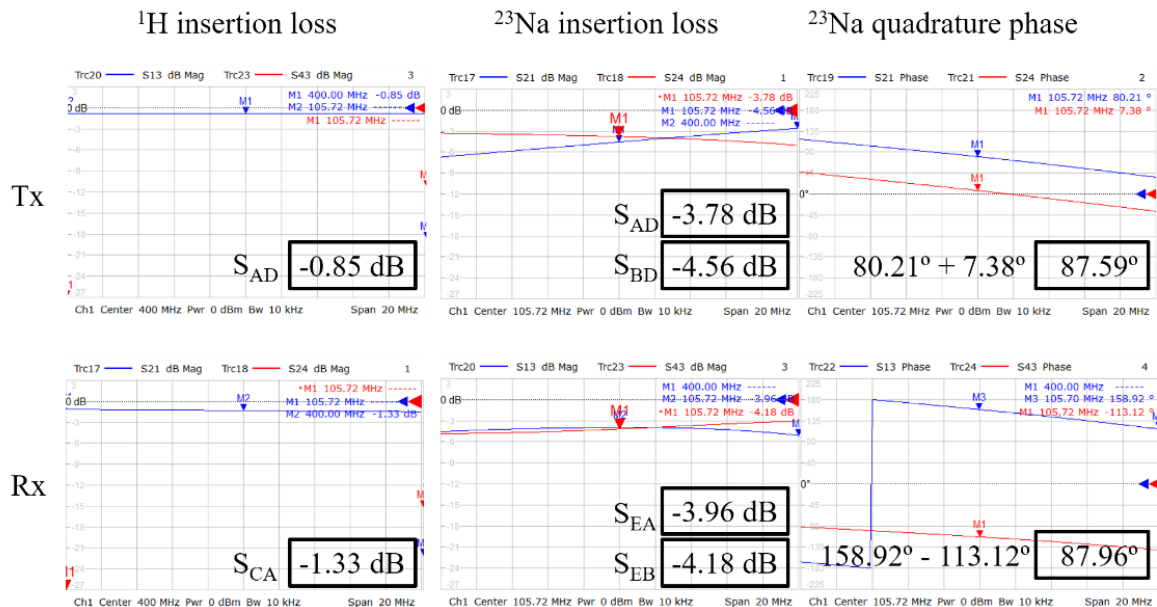


Figure 4-6 Measured isolation, and phase differences between two ^{23}Na ports.

4.5 Discussion and conclusion

The quadrature-enhanced, double-tuned coil design concept was proposed and constructed. The proposed coil comprises of a ^{23}Na single-tuned butterfly coil and a PIN-diodes controlled double-tuned loop coil, carefully overlaid to minimise coupling. Using the RF interface shown in Figure 4-3, ^{23}Na signals can be acquired in quadrature while ^1H signal is acquired in linear mode.

Conventionally, an arrangement of loop/butterfly coil is designed for quadrature coil to gain up to approximately 40% SNR [71, 133] or a double-frequency coil to gain ^1H function [107]. However, the efficiency of the PIN-diode switched coil at lower frequency (^{23}Na resonance frequency) is usually less than geometrically identical single-tuned coil due to the resistance of the diode [22, 23]. In the proposed design, the sodium coil can be driven in a quadrature mode using PIN-diode switched coil and butterfly single-tuned coil, so that the efficiency of this double-tuned coil at ^{23}Na frequency can be improved.

Chapter 5

DESIGN OF A FOLDED FOUR-RING DOUBLE-TUNED BIRDCAGE COIL FOR RAT BRAIN SODIUM IMAGING AT 9.4 T

5.1 Introduction and overview

In this Chapter, a double-tuned folded four-ring birdcage coil for brain imaging with high efficiency at ^{23}Na Larmor frequency is introduced. Since generating homogeneous B_1 field is important for transmit coils, birdcage coils are widely used for the single-tuned transmit coil. The birdcage coil can be also used as a double-tuned coil. There are several methods to double-tune birdcages and some were introduced in section 3.3.3. Among them, the four-ring birdcage coil design is preferred since it does not contain additional lossy components, such as trap circuit or PIN-diodes [88, 95], which lead to decrease of power efficiency and degradation of image quality. This four-ring approach, adding two additional end-rings on a conventional birdcage coil employs tuning at one resonance frequency using the inner end-rings structure and at the other frequency using the outer structure [27]. Several configurations of the four-ring birdcage can be used, such as low-pass/high-pass for both inner and outer, or low-pass for inner and high-pass for outer [28] (Figure 5-2). In most cases, the inner part tends to be set at the X-nuclei and with a low-pass configuration [30].

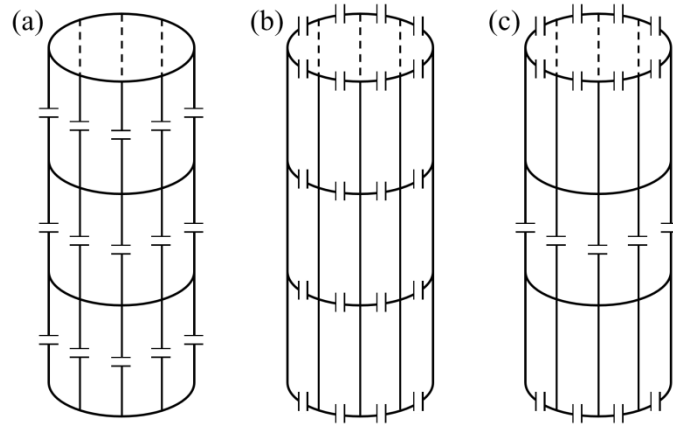


Figure 5-1 Layout of the (a) low-pass, (b) high-pass for both inner and outer and (c) low-pass for inner and high-pass for outer four-ring birdcage coil.

However, in order for the four-ring implementation to maintain efficiency and sensitivity at both frequencies an optimised inner- to outer rung length ratio is required [29]. This ratio is substantial and thus causes a substantial increase in overall coil length [27]. This requirement of additional length for the outer end-rings restricts the accessible space which can be problematic, particularly for in vivo brain studies due to the shoulders.

When the performance of the ^1H channel is less crucial, the outer structure can be modified using a different configuration, for example an Alderman-Grant coil (“half-Helmholtz”) [134] or a split birdcage [135] (Figure 5-2).

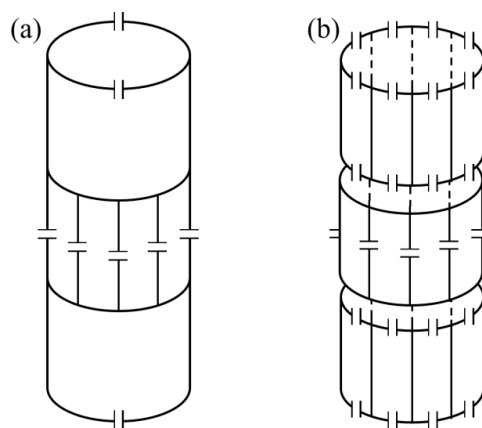


Figure 5-2 Layout of the (a) inner birdcage and outer Alderman-Grant coil (b) and split birdcage which is consisted of a low-pass inner birdcage and two high-pass outer coils.

In this chapter, a double-tuned four-ring birdcage coil with a modification that entails folding the outer end-rings by 90 degree to remove the length restriction issue is introduced. The performance of this design is compared with a single-tuned and a larger four-ring birdcage coil on a phantom and in an in vivo healthy rat brain at 9.4 T.

5.2 Concepts

A folded four-ring quadrature birdcage coil was designed and constructed with a double-tuned configuration as shown in Figure 5-3(a); an outer high-pass coil for ^1H (400 MHz) and inner low-pass coil for ^{23}Na (105.72 MHz) corresponding to the respective Larmor frequencies at 9.4 T.

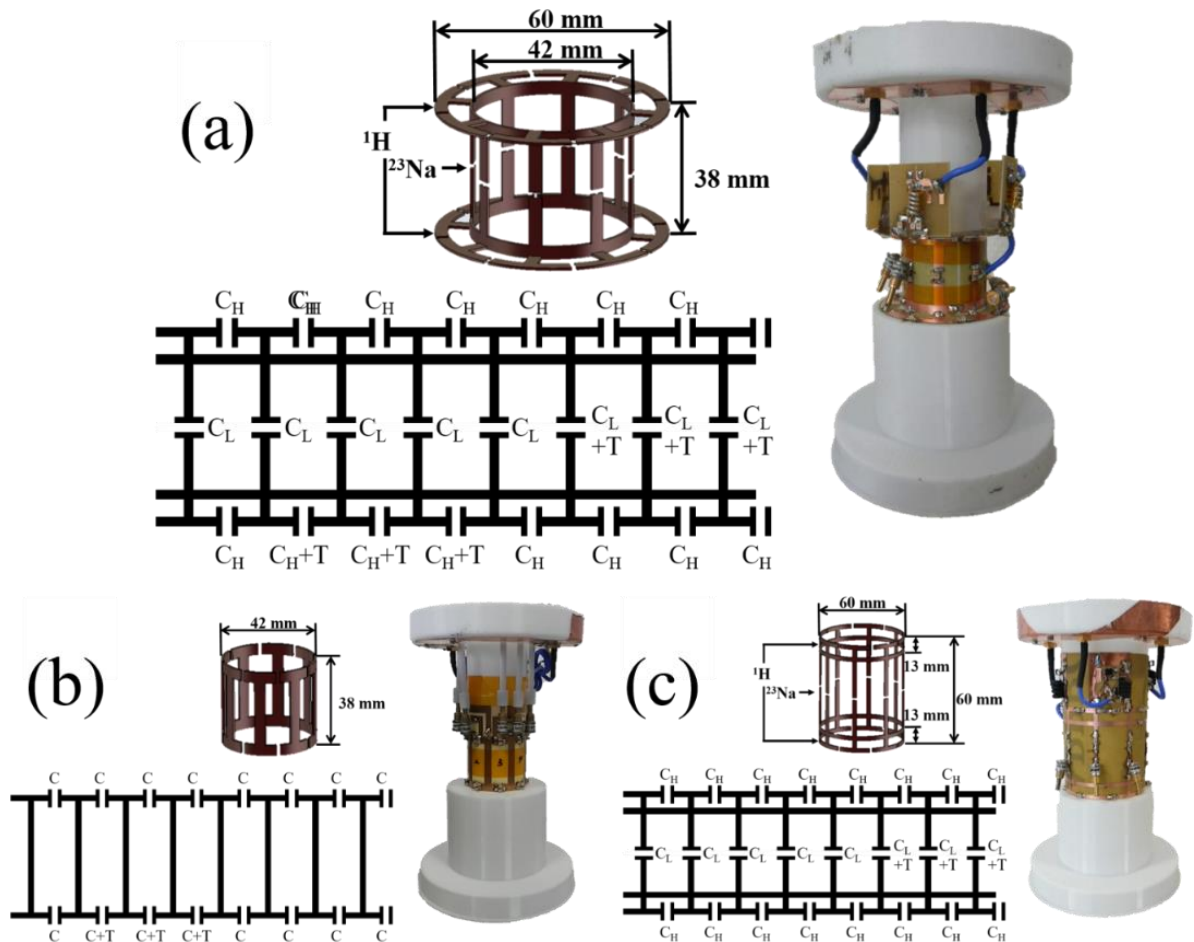


Figure 5-3 Photographs, dimensions and schematic diagram of (a) the proposed 90 degree folded four-ring double-tuned birdcage coil, (b) the single-tuned ^1H and ^{23}Na birdcage coil, (c) the large four-ring birdcage coil.

The coil with inner/outer rings were tuned using 47/12 pF non-magnetic fixed capacitors (C_L/C_H , 100B, American Technical Ceramics, USA), respectively, and fine adjustment of their frequencies and of 50 Ω matching condition was carried out using 1~10 pF trimmers (NMAP10HVE, Voltronics, USA). Both ^1H and ^{23}Na coaxial-cable-wound cable traps were built and inserted in each channel. As references, geometrically identical single-tuned birdcage coils [6] (Figure 5-3(b)) tuned to each nucleus (^1H and ^{23}Na) and a large four-ring birdcage coil (Figure 5-3(c)) were constructed for comparison.

All coils were driven in quadrature. The T/R RF chain included home-built quadrature hybrids, phase shifters, T/R switches and commercial low noise pre-amplifiers (WanTcom INC., USA) for each frequency. The ^{23}Na T/R switch with quadrature hybrid is show in Figure 5-4.

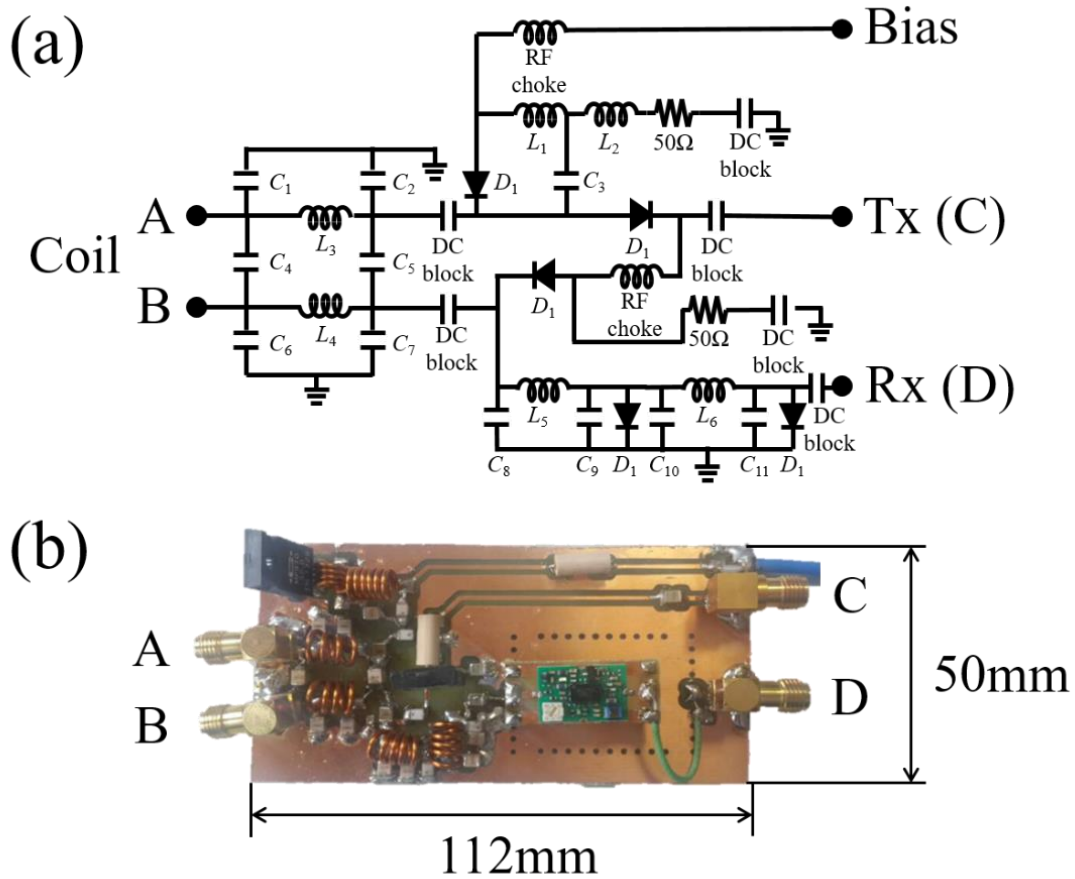


Figure 5-4 (a) Schematic diagram, and (b) photo of the ^{23}Na T/R switch with hybrid coupler.

5.3 Bench measurements

A uniform phantom (100 mm length and 28 mm diameter) containing 45 mM NaCl – which is equal to the total sodium concentration in rat brains [136] (Figure 5-5) was prepared in order to measure the coil quality.

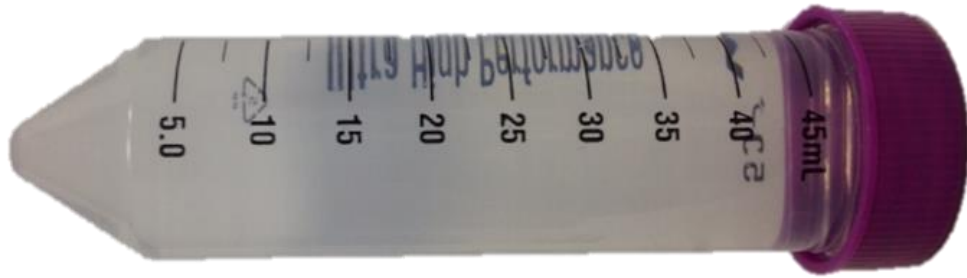


Figure 5-5 A uniform phantom used for bench measurement and phantom experiment.

The matching of the folded four-ring birdcage coil was measured using a network analyser (ZNB, Rohde & Schwarz, Germany). S-parameters in the loaded condition were obtained by directly connecting the coil to network analyser (which is different to the measurements reported so far where decoupled sniffer loops have been employed). The loaded and unloaded Q -factors (Q_{Un} , Q_L) were also measured using decoupled sniffer loops [118] and the ratio (Q_{Ratio}) was calculated. S-parameters and phase differences of the ^{23}Na T/R switch with hybrid coupler are also measured using a network analyser.

5.4 Results

It was found that all coils were tuned and matched well to 105.72 MHz (^{23}Na) or 400 MHz (^1H), respectively, as return losses of each channel of the coils were less than -22dB and the isolation between two quadrature ports were better than -13.7dB for ^1H and -27 dB for ^{23}Na . Figure 5-6 shows the measured S-parameters of the folded four-ring birdcage coil at the both frequencies.

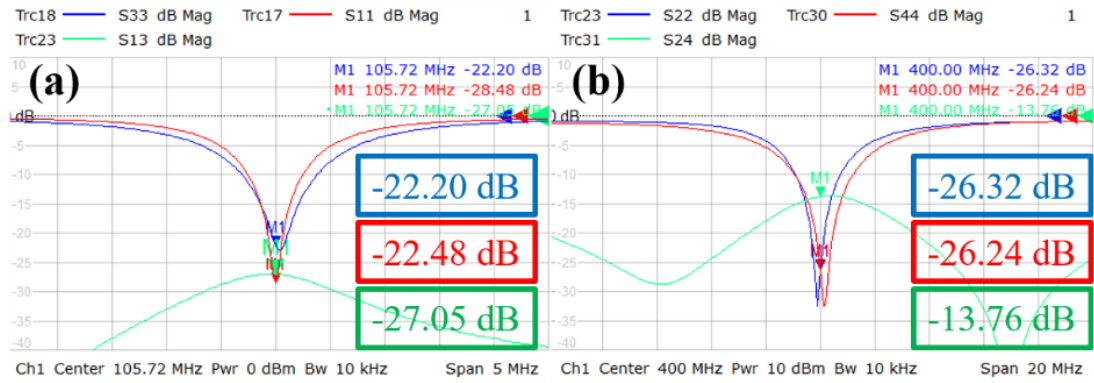


Figure 5-6 Measured S-parameters (return loss at coil port A: blue, return loss at coil port B: red, and isolation between coil port A and port B: green) at (a) ^{23}Na at 105.72 MHz and (b) ^1H at 400 MHz of the folded four-ring birdcage coil.

Measured isolation and phases of the ^{23}Na T/R switch are shown in Figure 5-7. isolation of both Tx and Rx paths at all ports are better than -1.0 dB and Phase differences of Tx and Rx paths are 90.4° and 85.3° , respectively.

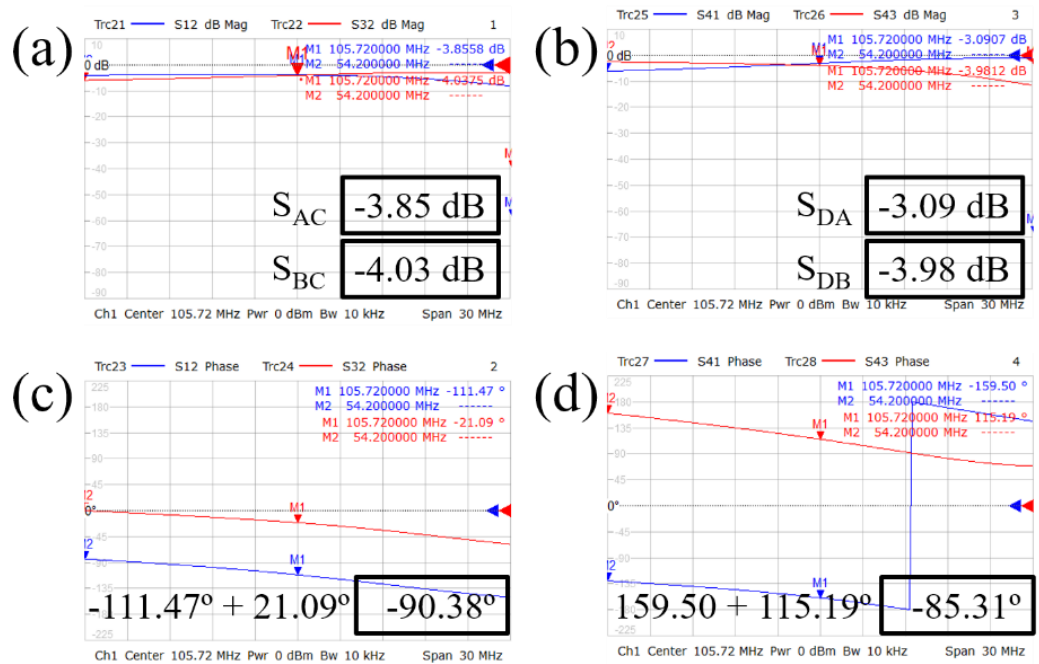


Figure 5-7 Measured isolation of ^{23}Na T/R switch on the (a) Tx (A: port A, B: port B, C: Tx port), (b) Rx (D: Rx port) path and phase differences on the (c) Tx, (d) Rx path.

Table 5-1 shows the Q -factors measured for all coils.

Table 5-1 Measured Q -factors of the folded four-ring, single-tuned and large four-ring birdcage coil.

	^1H			^{23}Na		
	Q_{Un}	Q_{L}	Q_{Latio}	Q_{Un}	Q_{L}	Q_{Latio}
Folded four-ring	95	69	1.4	155	130	1.2
Single-tuned	93	60	1.5	157	125	1.3
Large four-ring	70	46	1.5	166	146	1.1

5.5 Discussion and conclusion

Conventional four-ring birdcage coils are elongated along the z -axis and can be restricted in position due to the shoulders, which may lead to limitations in in vivo brain studies of humans and animals. Although the length of four-ring coil can be shortened by the optimisation presented in [29], this still requires extension of the coil diameter to be larger for the in vivo animal brain studies, which in turn results in degradation of the quality of the coil and image due to a reduced filling factor. In comparison, the proposed four-ring coil provides one important benefit in that it can avoid length restriction by folding the outer rings vertically. Since MRI scanners are usually designed for whole body application, the increased diameter in radial direction will not be an issue for head imaging. This folded birdcage coil therefore helps to build double-tuned coils with a smaller diameter, leading to higher filling factors and the associated increase in ^{23}Na SNR.

Chapter 6

EXPERIMENTAL RESULTS AND ANALYSIS

All MR imaging experiments were carried out on a 9.4 T small animal scanner (Figure 6-1) [137] and served to evaluate the performance of the proposed double-tuned coil.



Figure 6-1 A home-integrated 9.4 T small animal scanner.

The MR system consists of an actively-shielded, horizontal-bore superconducting magnet and gradient/shim coil (inner diameter of 120 mm, 600 mT/m, up to 2nd order shim) from Varian/Agilent Technologies, USA. Major system electronics and clinical software/user interface (SYNGO) are from Siemens Healthcare, Germany [90, 137].

6.1 Phantom experiments

6.1.1 Comparison of the surface coils

A basic adjustment including B_0 shimming and Tx power calibration was completed with the use of the system's standard adjustments. Structural images were acquired using either a single-tuned ^1H coil or the ^1H channel of the dual-tuned coil. ^1H and ^{23}Na axial images were obtained using a GRE (TR = 95 ms, TE = 4.11 ms, $0.1\text{ mm} \times 0.1\text{ mm} \times 1\text{ mm}$ resolution, 2 averages, and scan time = 1.14 minutes) and a 3D FLASH (TR = 40 ms, TE = 2.73 ms, $1\text{ mm} \times 1\text{ mm} \times 2\text{ mm}$ resolution, 64 averages, and scan time = 10:15 minutes) sequence, respectively. Noise images (Tx power = 0) were additionally acquired for SNR calculation. A region of interest (ROI) was drawn in which a rat brain would be located and SNR was calculated. Two SNR profiles along the y-axes (one centre, one side) on the ^{23}Na images were plotted. These SNR values were compared with single-tuned coils under the same condition. The acquired MR data were analysed using Python (Python Software Foundation).

6.1.2 Comparison of the birdcage coils

All the imaging parameters used in phantom experiments are described in the Table 6-1.

Table 6-1 MR image acquisition parameters used in phantom experiments.

Image	Sequence	TR (ms)	TE (ms)	No. Avgs.	FA(°)	Resolution (mm ³)	Scan time (minutes)
^1H loc.	GRE	9	4.27	2	20	$0.2 \times 0.2 \times 1$	0:08
^1H	GRE	150	3.94	4	30	$0.2 \times 0.2 \times 1$	2:34
^{23}Na	3D FLASH	30	3.28	72	60	$1 \times 1 \times 1$	9:48
^{23}Na FA map	3D FLASH	200	2.29	16	60, 120	$2 \times 2 \times 2$	10:15

In order to acquire FA maps of the ^{23}Na coils, a double angle method [138] was used and B_1 homogeneities of all ^{23}Na coils were compared. For this purpose a 3D FLASH ^{23}Na sequence was utilised with two flip angles, 60 and 120°. The flip angle maps were calculated using the Eq. 6-1.

$$\alpha(\mathbf{r}) = \arccos\left(\frac{I_2(\mathbf{r})}{2I_1(\mathbf{r})}\right), \quad 6-1$$

where I_1 , and I_2 are the two images acquired with tip angle α_1 and $\alpha_2 = 2\alpha_1$, respectively, and $\mathbf{r}$ represents spatial position. Relative uniformity (RU) was then calculated using the Eq. 6-2 by dividing the number of pixels within the phantom area whose relative deviation is smaller than 10% of the mean α through the overall number of pixels [14, 73]:

$$\text{RU} = \frac{\text{Number of pixels with } \left(\left|\frac{\alpha - \bar{\alpha}}{\bar{\alpha}}\right| \times 100 < 10\right)}{\text{Total number of pixels}} \times 100. \quad 6-2$$

The SNR of the proposed coil was also compared with that of the reference coils under the same conditions. In order to calculate SNR, noise images with zero transmit power ($P_{\text{Tx}}=0\text{W}$) were additionally obtained. The SNR values were calculated using NEMA method number 2 [123]; signal mean of the selected ROI from the magnitude image divided by noise standard deviation from the noise image of the identical ROI chosen for the magnitude image. The acquired MR data were processed and analysed using Python (Python Software Foundation).

6.2 In vivo experiments of the folded four-ring birdcage

In vivo ^1H and ^{23}Na MR images were then obtained with the folded four-ring birdcage coil using female Wistar rats. All animal experiments were approved by the Animal Protection Committee of the local government according to the German Protection of Animals Act. The rats (~346 g) were housed under standard housing conditions in the animal facility. Rats were anaesthetised with 2 to 5 %

isoflurane in oxygen, and during the measurement breathing rate and body temperature were continuously controlled. All the imaging parameters used in the in vivo experiments are described in the Table 6-2.

Table 6-2 MR image acquisition parameters used in in vivo experiments.

Image	Sequence	TR (ms)	TE (ms)	No. Avgs.	FA	Resolution (mm ³)	Scan time (minutes)
¹ H loc.	GRE	9	4.27	2	20	0.18 × 0.18 × 1	0:08
¹ H	TSE	3000	35	4	90	0.2 × 0.2 × 1	4:13
²³ Na	3D FLASH	40	2.26	128	60	1.2 × 1.2 × 2	30:50

6.3 Result

6.3.1 Comparison of the surface coils

Figure 6-2 shows the ¹H and ²³Na axial images obtained using single- and double-tuned coils including associated SNR values of the ROI and SNR profiles along the y-axes. Adjusted Tx reference voltages for single- and double-tuned coils were 10.8 V and 14.1 V, respectively. The ²³Na SNR in the selected ROI, acquired by the double-tuned coil, is nearly 33% and 17% higher in comparison with loop and butterfly single-tuned coil, respectively. SNR plots also show the superiority of double-tuned coil in ²³Na. However, the SNR of ¹H is about 30% lower than that of the single-tuned coil.

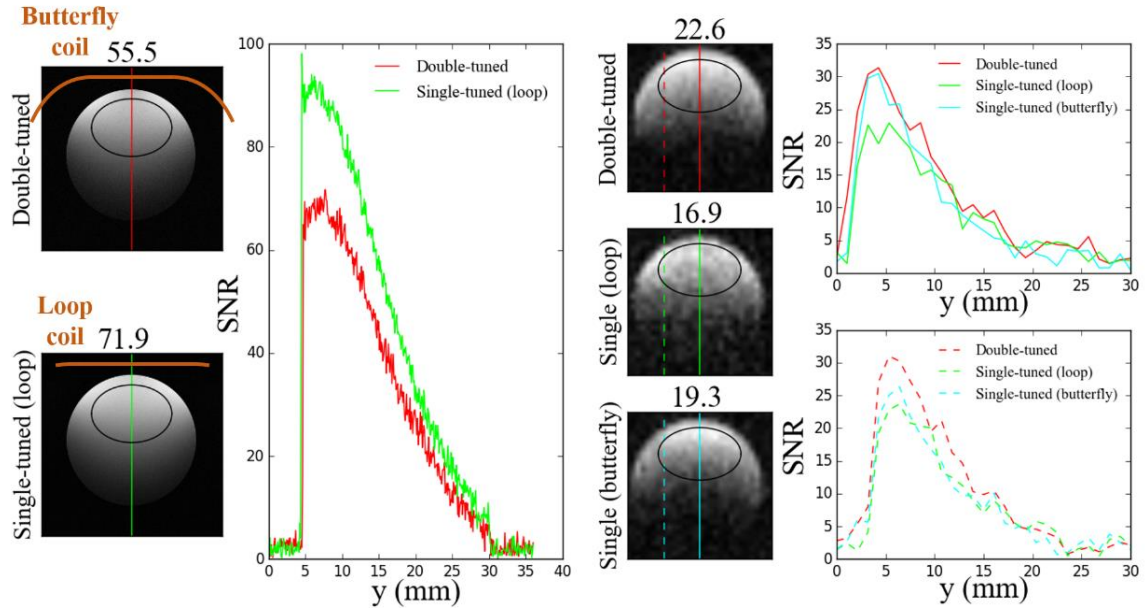


Figure 6-2 ^1H (left) and ^{23}Na (right) images and SNR profile plots of double-tuned coil and single-tuned reference coils [130].

6.3.2 Comparison of the birdcage coils

Figure 6-3 shows the uniform phantom ^1H (left column) and ^{23}Na (middle column) images obtained using folded four-ring (top row), single-tuned (middle row) and large four-ring (bottom row) birdcage coils. ^{23}Na flip angle maps acquired with the described coils are also shown in Figure 6-3 (right column). Table 6-3 summarizes the SNR values and the B_1 uniformity of the ^{23}Na coils.

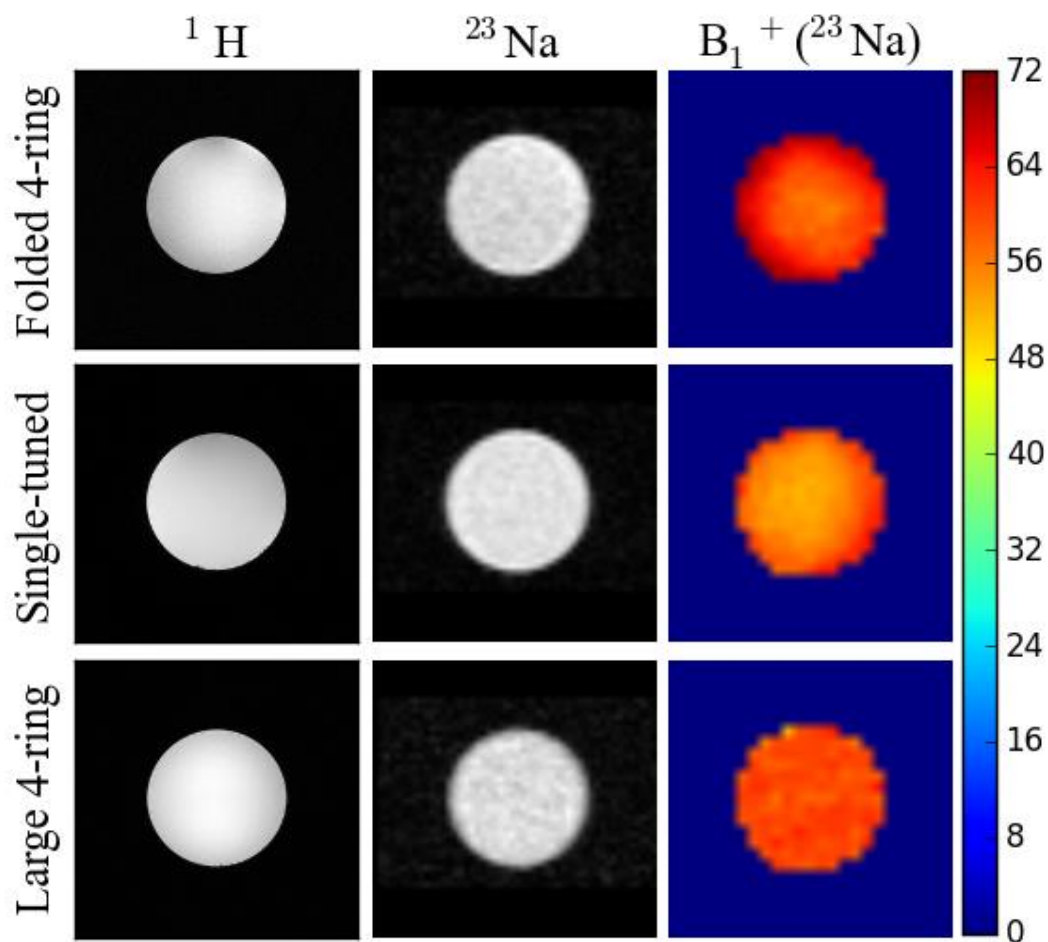


Figure 6-3 ^1H image (left column), ^{23}Na image (middle column) and associated ^{23}Na flip angle map (right column) of folded four-ring (top row), single-tuned (middle row), and larger four-ring (bottom row) birdcage coils [139].

Table 6-3 SNR and ^{23}Na relative B_1 uniformity (RU) values measured by folded four-ring, single-tuned and large four-ring birdcage coils [139].

	SNR		^{23}Na RU (%)
	^1H	^{23}Na	
Folded 4-ring	68.85	25.43	91.3
Single-tuned	369.48	27.29	93.5
Large 4-ring	147.36	14.03	99.3

Figure 6-4 shows the ^1H localiser image obtained with test phantom using folded four-ring birdcage coil.

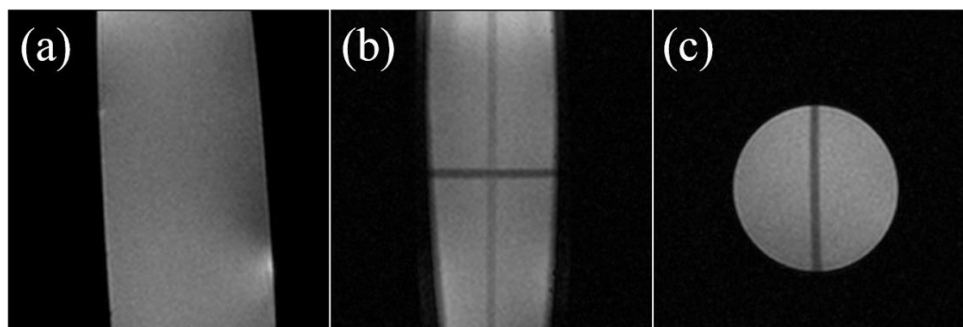


Figure 6-4 ^1H localiser image of the phantom [139]. (a) Sagittal. (b) Coronal. (c) Axial.

6.3.3 In vivo experiments

Figure 6-5 shows the ^1H scout image of the in vivo rat brain viewed in three orthogonal (Sagittal, Coronal, Axial) planes.



Figure 6-5 ^1H localiser image of the in vivo rat [139]. (a) Sagittal. (b) Coronal. (c) Axial.

A set of in vivo ^1H and ^{23}Na axial images are shown in Figure 6-6, which covers the whole rat brain and shows expected anatomy and adequate image quality. ^{23}Na images are weighted towards total tissue sodium concentration and compare well to the anatomic ^1H images. Vitreous humour of the eyes possesses expected high contrast due to its large sodium concentration.

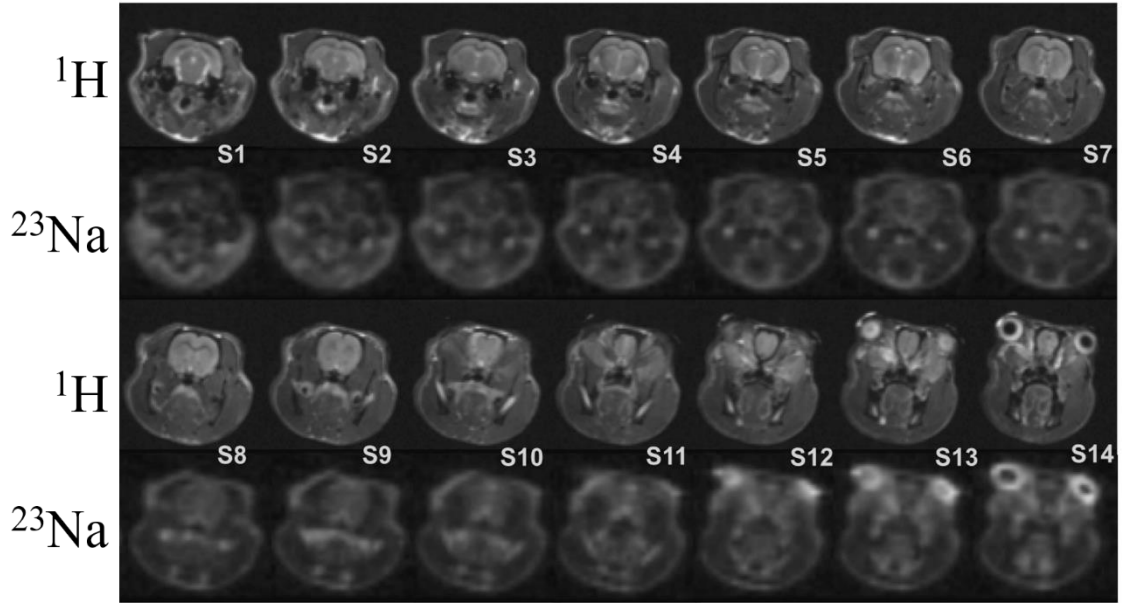


Figure 6-6 In vivo ^1H (top) and ^{23}Na (bottom) images with whole rat brain coverage [139]. S1~S14 denote slice numbers.

6.4 Discussion and conclusion

The quadrature-enhanced, double-tuned coil design concept was successfully demonstrated on a 9.4 T animal scanner and was shown to outperform the geometrically identical single-tuned coils used for comparison.

Although lower than system supplied voltage (-30 V) was applied during the ^1H imaging as the RF transmit voltage ($V_{RF} < 15V$), approximately 30% higher RF transmit voltage was required than that of single-tuned coil. When the PIN-diodes are used for the switched tuned coil, higher than applied RF transmit voltage is applied to the PIN-diode. The requirement of higher RF transmit voltage for double-tuned coil might be due to insufficient reverse voltage being applied to each PIN-diode during the acquisition [140] and more details described in Appendix B. Nevertheless the transmit signal is compressed because of the insufficient reverse voltage, it still can be used for B_0 field shimming and scout imaging. Lower than system supplied voltage can be also applied using power supply to prevent Tx signal compressing.

Instead of using the PIN-diode switch, one can add LC or LCC trap circuits to achieve the same purpose [18]. Depending on the purpose, sensitivity of the coil at two resonance frequencies can be

adjusted by choosing an inductance of traps on the coil as shown in the section 3.3.1 [19]. These traps, or PIN-diodes, can also be inserted into the butterfly coil.

Further improvement can be also achieved with the optimisation of a single-tuned butterfly coil [53] and combination between coils. Moreover, since the B_1 field is distorted at high magnetic field, there may be a chance that 90 degrees is not the optimised phase difference [141]. An adjustment of the phase difference of the ^{23}Na hybrid coupler may offer additional improvement in quadrature operation [142]. In addition, this concept can be applied to build multi-channel array coils [143-145]. Here, the PIN-diode control unit may be useful because of the limited bias source from the scanner [88].

In terms of birdcage coil experiment, the relative uniformity of the large coil is slightly higher than that of the other coils because of the coil's physical diameter. However, the ^1H and ^{23}Na SNR values of the large coil in phantom measurements reached only approximately 40% and 51%, respectively, to those of the single-tuned coils due to, for example, the increased filling factor. In case of the folded four-ring coil, since the outer end-rings were folded vertically and used to tune at ^1H Larmor frequency, the outer birdcages work similar to the butterfly surface coil. As the result, signal intensity of the ^1H higher at near to the end-rings as shown in Figure 6-4. Although ^1H SNR of the folded four-ring coil was reduced by folding the outer end-rings, the image quality of ^1H on the image in Figure 6-3 appeared to be sufficient. Importantly, the ^{23}Na image quality was found to be nearly as good as that of the single-tuned coil (~93% in SNR) and much superior to that of the large four-ring coil. The loss in RU of the folded four-ring coil is also only 2% compared to the single-tuned coil.

The performance of the folded four-ring design was demonstrated and this new design shows a benefit for ^{23}Na imaging in terms of a better SNR than larger diameter implementations. This coil concept can also be applicable for other nuclei, such as ^{13}C or ^{31}P where a double-resonant coil is essential for ^1H decoupling and nuclear Overhauser effect [146].

Chapter 7

DISCUSSION AND CONCLUSIONS

The main goal of this thesis was to develop double-tuned RF coils for ^{23}Na brain imaging with high efficiency at ^{23}Na Larmor frequency. For the Homogeneous resonator, a four-ring birdcage coil was modified by folding the outer end-rings by 90 degrees to cope with the available space limitation. Regarding the heterogeneous resonator, a PIN-diode controlled double-tuned coil was modified by adding a butterfly loop coil to use circularly polarised magnetic field. Since it was found that higher RF transmit power is required for the PIN-diode controlled coil compared to that of single-tuned coil, transmit and receive performance of the PIN-diode coil was also discussed.

B_1 field homogeneity is an important requirement for the transmit coil. So that, for the single-tuned case, a birdcage coil is commonly used because it generates homogeneous B_1 field. For the double-tuned birdcage, four-ring, double-tuned, birdcage coil is preferred since the efficiency and the homogeneity of the four-ring birdcage coil at two resonance frequencies are comparable to those of the single-tuned birdcage coils. However, the additional leg length required for the outer two end-rings is limited by the accessible space, in particular for brain studies of humans and animals. To overcome this limitation, a folded four-ring birdcage coil was suggested and constructed. In spite of the lower ^1H SNR and B_1 field homogeneity, when compared to that of the single-tuned birdcage coil, and larger size, both the SNR and B_1 field homogeneity of ^{23}Na of the four-ring birdcage coil are comparable to that of the single-tuned birdcage coil. In vivo healthy rat brain images at 9.4 T using the folded four-ring birdcage coil are demonstrate this.

When the PIN-diode is added to the single-tuned RF coil together with series capacitor, the resonance frequency is actively switched by adding or removing a capacitor. The RF coil can be tuned at lower resonance frequency when the PIN-diodes are forward biased while it is initially tuned at higher resonance frequency with applied reverse bias for the PIN-diodes. However, the sensitivity at lower frequency is lower than that of single-tuned coil because of the resistance of the PIN-diode. To enhance the transmit sensitivity and SNR for the ^{23}Na , a single-tuned ^{23}Na butterfly coil was added to the PIN-diode controlled coil for quadrature driving. Although the ^1H SNR was lower than that of the single-tuned loop coil, ^{23}Na SNR was higher than that of single-tuned both the loop and the butterfly coil.

A special PIN-diode control unit and a T/R switch for operating this unique coil were also introduced. When the coil is used without this PIN-diode control unit, both forward current and reverse voltage are divided between each PIN-diode. For the coil introduced in Chapter 4, a single bias channel can also be used to each PIN-diode for the same purpose since there are more than enough channels on the scanner. However, reduction of number of bias cables between the coil and the scanner can be helpful to minimise potential RF noise because these cables are usually far longer than the wavelength of the RF signal [89]. If a lower forward current is applied to the PIN-diode, resistance of the PIN-diode is higher compared to that with sufficient forward current, reducing the ^{23}Na SNR, due to the higher resistance. The reverse voltage for the PIN-diode should be higher than the peak voltage of the RF signal. When insufficient reverse bias is applied to the PIN-diode, the RF transmit signal can be compressed. When the suggested PIN-diode control unit is used to apply both forward current and reverse voltage to the several PIN-diodes, equal and undivided DC bias is applied to each PIN-diode.

Although applied RF transmit voltage was lower than system supplied reverse voltage for the PIN-diode controlled, quadrature enhanced, double-tuned coil during the ^1H imaging, approximately 30% higher RF transmit voltage was required than that of single tuned coil. To address this effect, minimum required reverse voltage for the PIN-diode was calculated using circuit analysis. It was demonstrated that the RF voltage across the PIN-diode for the switch-tuned coil is much higher than the applied RF transmit signal because the RF coil is a resonance circuit. Therefore, a very large reverse bias voltage should be applied to the PIN-diode to avoid signal compressing. It was demonstrated that applying sufficient reverse voltage to the PIN-diode controlled coil for human use is impractical for two reasons. One reason is that the PIN-diode may be damaged by exceeding the reverse breakdown voltage.

Transmit and receive performance of the PIN-diode switch-tuned coil was also investigated using a simple loop coil tuned to ^{23}Na and ^1H Larmor frequency at 4 T. It was demonstrated that the RF transmit signal is compressed when the PIN-diodes are insufficiently reverse biased. As the result, the flip angle acquired using PIN-diode switch-tuned coil was not linearly increased with applied RF voltage due to the insufficient reverse voltage. Hence, when the flip angles are controlled by RF voltage, pulse sequences using more than two different flip angles cannot perform well such as spin echo, inversion recovery and so on. Adjusting the flip angle by changing the RF pulse duration may be helpful to avoid this effect. However, PIN-diode switches remain an interesting option for the receive-only coil and MEMS switches can be used instead of PIN-diodes to avoid insufficient reverse voltage problem.

Summarising, the performance of the quadrature-enhanced surface coil and folded four-ring

birdcage coil was demonstrated and these new designs have a distinct benefit for ^{23}Na imaging experiments. It is also possible to combine both coils as a Transmit-only/Receive-only RF system. The folded four-ring birdcage can then be used as Transmit-only coil with homogeneous B_1 field for ^{23}Na excitation. Then combine a ^{23}Na CP receive array from quadrature-enhanced surface coil for the higher SNR. These coil concepts can also apply for other X-nuclei.

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ACKNOWLEDGEMENTS

I am very thankful for all these years that I have worked at the Institute of Neuroscience and Medicine – 4 at Forschungszentrum Jülich. I want to give my appreciation to all that lab members that worked with me, helped me, taught me and guided me. I would not be able to go this far without the help from so many patient people.

I would like to show my most sincere gratitude to my professor and advisor Prof. N. Jon Shah. Everything I have learned and done is based on your guidance and patience. I learned quite a lot not only from your advice for my research, but also your surprising ability to keep this large institute both capable and cordial.

I am also greatly indebted to my supervisors, Dr. Arthur W. Magill, Dr. Chang-Hoon Choi, and Dr. Jörg Felder. Their extensive scientific knowledge and excellent advice have been of vital importance to the success of this work, as well as for my personal scientific development. I thoroughly enjoyed our many brainstorming sessions and discussions about new ideas.

I also want to thank my colleagues for the help during my work:
YunKyoung, Tim, Nadine, and Annette for the help on 3D printing and PCB etching.
Avdo and Stefan for their technical assistant on the system.
Dr. Wieland A. Worthoff and Aliaksandra for their technical assistant on the ^{23}Na MR imaging.
Dr. Antje Willuweit and Mr. Schöneck for their assistant to acquire rat brain image.
Markus for the help to set up Python for MR image processing.
Dr. Seong Dae Yun, Dr. Suk-Min Hong, Ms. Rick, and Ricardo (a.k.a. 철수) for the fruitful discussions and proofreading parts of this dissertation.
Anna (a.k.a. Matilde) for translating to German the abstract of this thesis and being the best office mate.
Arne for the help and valuable discussions about RF coils and making lovely memories in Jülich.

I am very grateful to my parents, younger brother, and parents in law. They were always supporting and encouraging me with their best wishes.

Last but not the least, I am truly indebted to my devoted wife Youjung, my son Seung, and my daughter Yul. Their love and support without any complaints or regrets have allowed me complete this doctoral study.

LIST OF PUBLICATIONS

Journal articles

1. Y. Ko, S. D. Yun, S.-M. Hong, Y. Ha, C.-H. Choi, N. J. Shah, J. Felder “MR-compatible, 3.8 inch dual organic light-emitting diode (OLED) in-bore display for functional MRI.” PloS one, 13(10), (2018)
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Title: Design and implementation of a simple multinuclear MRI system for ultra high-field imaging of animals

Author: Chang-Hoon Choi, YongHyun Ha, Pandichelvam Veeraiah, Jörg Felder, Klaus Möllenhoff, N. Jon Shah

Publication: Journal of Magnetic Resonance

Publisher: Elsevier

Date: December 2016

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Title: Design and use of a folded four-ring double-tuned birdcage coil for rat brain sodium imaging at 9.4 T

Author: YongHyun Ha, Chang-Hoon Choi, Wieland A. Worthoff, Aliaksandra Shymanskaya, Michael Schöneck, Antje Willuweit, Jörg Felder, N. Jon Shah

Publication: Journal of Magnetic Resonance

Publisher: Elsevier

Date: January 2018

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Title: Development and implementation of a PINdiode controlled, quadrature-enhanced, doubletuned RF coil for sodium MRI

Author: Yong Hyun Ha

Publication : Medical Imaging, IEEE Transactions on

Publisher: IEEE

Date: Dec 31, 1969

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Appendix A

AN ANALYSIS OF DOUBLE-TUNED, MRI RF COIL USING AN LC TRAP

C_L and C_{tl} values were calculated with L_{tl} of 22 nH and L_L of 150 nH at 105.72 MHz and 400.0 MHz ($^{23}\text{Na}/^1\text{H}$ Larmor frequencies at 9.4 T) using Eq. 3-33 and 3-34. Table A-1 shows the calculated values using both plus and minus sign before the square root in Eq. 3-34.

Table A-1 Calculated components for double-tuning the LC trapped coil.

	Calculated using minus sign before the sqrt in Eq. xx	Calculated using plus sign before the sqrt in Eq. xx
C_L	13.02 pF	1.07 pF
C_{tl}	8.35 pF	101.87 pF

To identify the effect of the Eq. 3-34, S-parameters were simulated with calculated value using RF simulation software RFSim99 (originally from HyDesign Ltd. UK). When C_{tl} was calculated using Eq. 3-34 with the plus sign before the square root, the third resonance, which is the resonance described by an impedance maximum, is centred at 105.91 MHz. It almost coincidences with the lower resonance frequency of 105.72 MHz desired for acquiring sodium signal. The result is that the coil does not reach an impedance minimum at the sodium frequency and cannot be used to pick up signal from the ^{23}Na nuclei. Instead the coil almost behaves like a single tuned coil as shown in Figure A-1 (right).

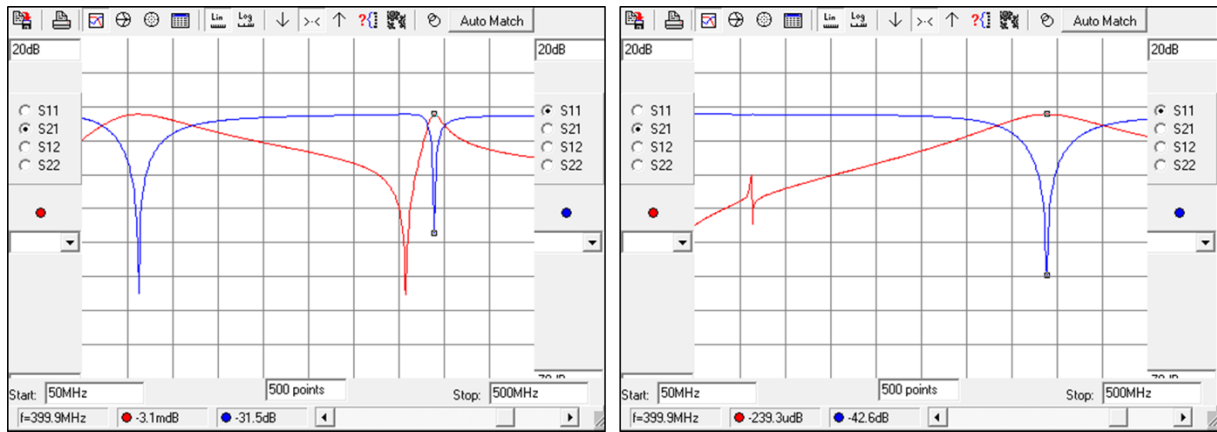


Figure A-1 S-parameter graph of the LC trapped coil with L_{tl} of 22nH, C_L and C_{tl} values which were calculated using minus (left) and plus (right) sign in front of square root in Eq. 3-34.

Appendix B

MINIMUM REVERSE BIAS CALCULATION FOR PIN-DIODE SWITCHED DUAL-TUNED RF COILS

Figure B-1 shows a schematic diagram of the PIN-diode switched dual-tuned coil which has one PIN-diode controlled capacitor switching network. For PIN-diodes used on MR coils, C_d is typically in the range 0.5-1.5pF.

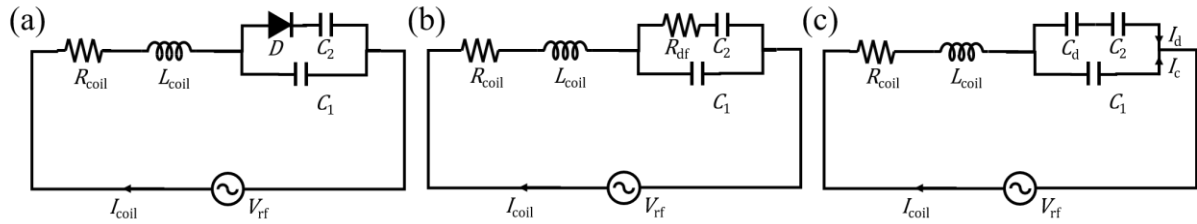


Figure B-1 Circuit diagram showing (a) loop coil with a PIN-diodes switching unit consisting of capacitor C_1 in parallel with a PIN-diode D and a second capacitor C_2 . If the diode were ideal, forward or reverse biasing it would add or remove C_2 from the circuit. However, when the diode is (b) forward biased it presents resistance R_{df} , which is controlled by the DC forward current, while when (c) reverse biased it appears as a capacitance C_d .

When using PIN-diodes in high power RF circuits, is that sufficient reverse bias voltage is provided, such that the applied RF signal does not bring the diode into conduction, which can be occurred when the RF voltage exceeds the DC reverse bias voltage. When switched to the higher frequency (ω_h), PIN-diodes will be operated in reverse bias mode during high power transmission and presents capacitance C_d . The impedance of the coil is then

$$Z_{coil} = R_{coil} - j \frac{C_2 + C_d}{\omega_h (C_1 C_2 + C_1 C_d + C_2 C_d)} \quad \text{B-1}$$

When RF transmit signal is applied with peak voltage ($|V_{rf}|$), peak voltage across diode ($|V_{drf}|$) is

$$|V_{drf}| = \frac{1}{\omega_h C_d} \left(\frac{C_2 C_d}{C_1 C_2 + C_1 C_d + C_2 C_d} \right) \frac{|V_{rf}|}{R_{coil}} \quad \text{B-2}$$

$|V_{drf}|$ is much higher than $|V_{rf}|$. The minimum reverse bias for the diode ($|V_{dc}|$) can be calculated using below equation [147].

$$|V_{dc}| = \frac{|V_{drf}|}{\sqrt{1 + \left(\frac{\omega_h W}{2v_{sat}} \right)^2}} \quad \text{B-3}$$

where W is intrinsic region width of the diode and v_{sat} is saturation velocity of the charged carrier in the intrinsic region. This velocity varies with carrier type, semiconductor material, and temperature, but is approximately 10^7 cm/s for electrons in silicon at 290 K [147]. W of the used PIN-diode (MA4P4002B-402, MACOM, USA) is $175 \mu\text{m}$. When reverse voltage is smaller than $|V_{dc}|$, the RF peak voltage is compressed [148].